12-Sep-2025

SEC-T 0x10sion Music Synthesizer Badge

Assembly Instructions

(with re-programming how-to)

V 0.2

open source hardware

0X10SION

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CORNFIELD ELECTRONICS

Learn To Solder

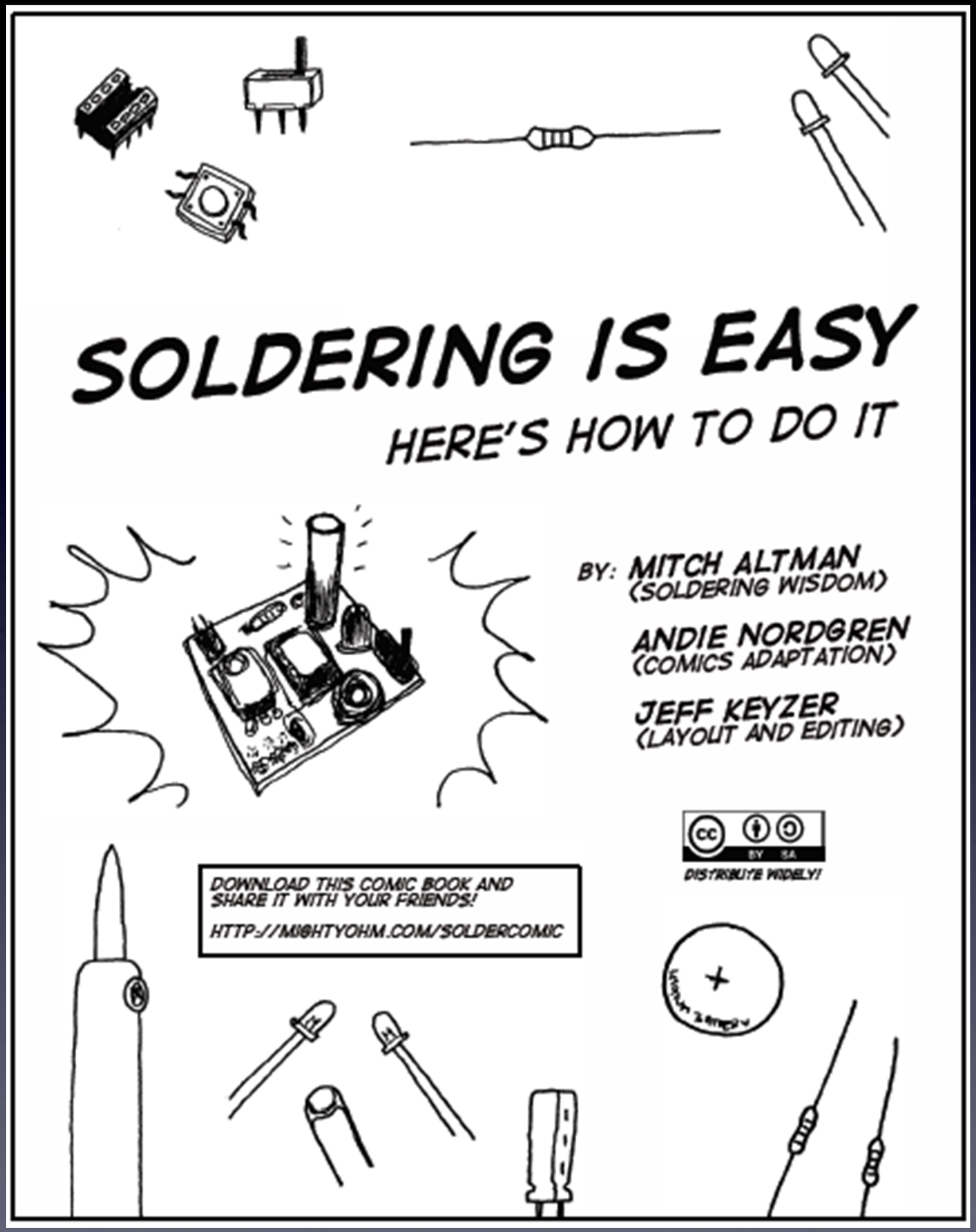
The following photos will
show you how to solder.

But feel free to download
the “Soldering Is Easy” comic book
for free!

(In many different languages.)

download for free at:
http://mightyohm.com/soldercomic

Learn To Solder



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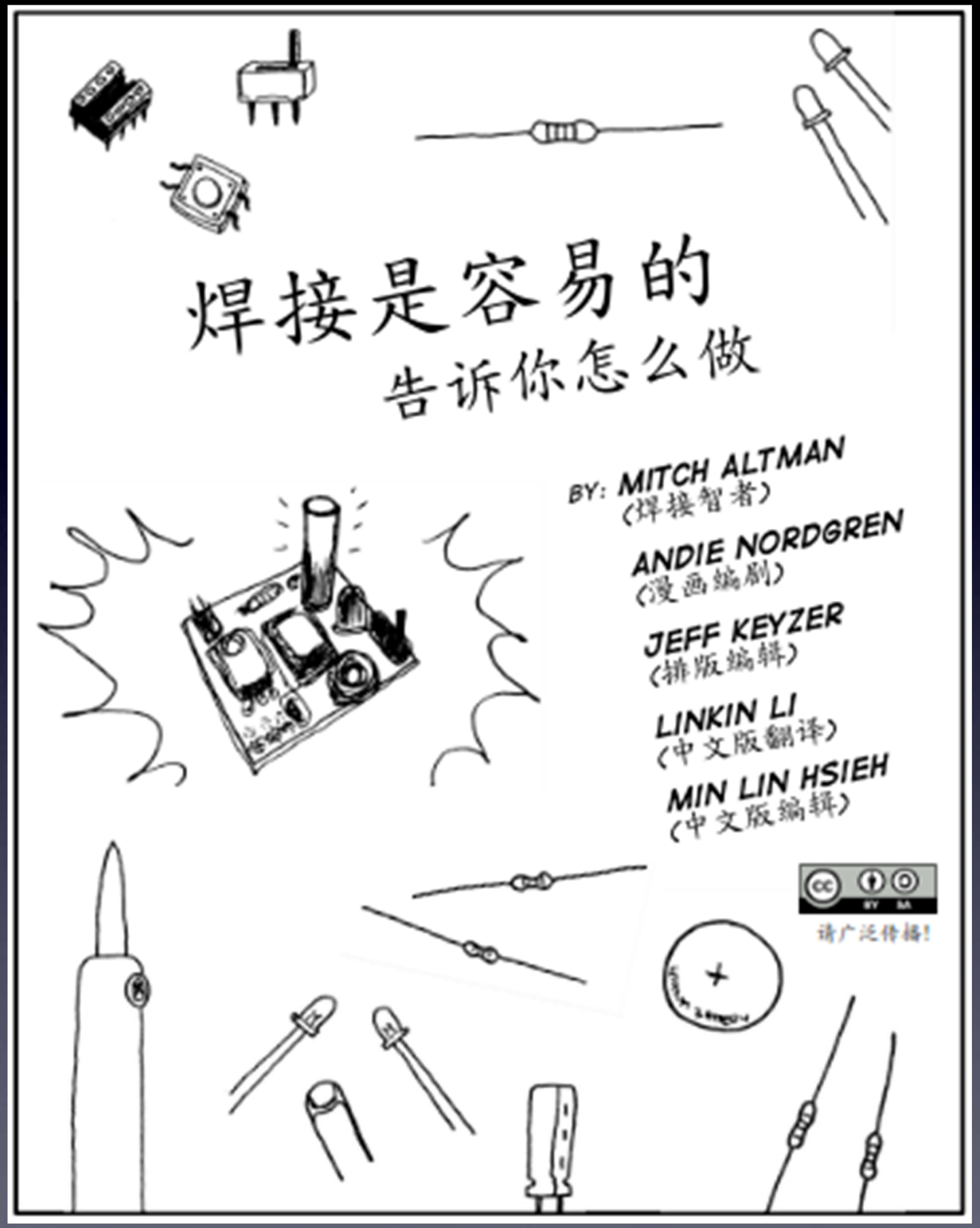
Learn To Solder



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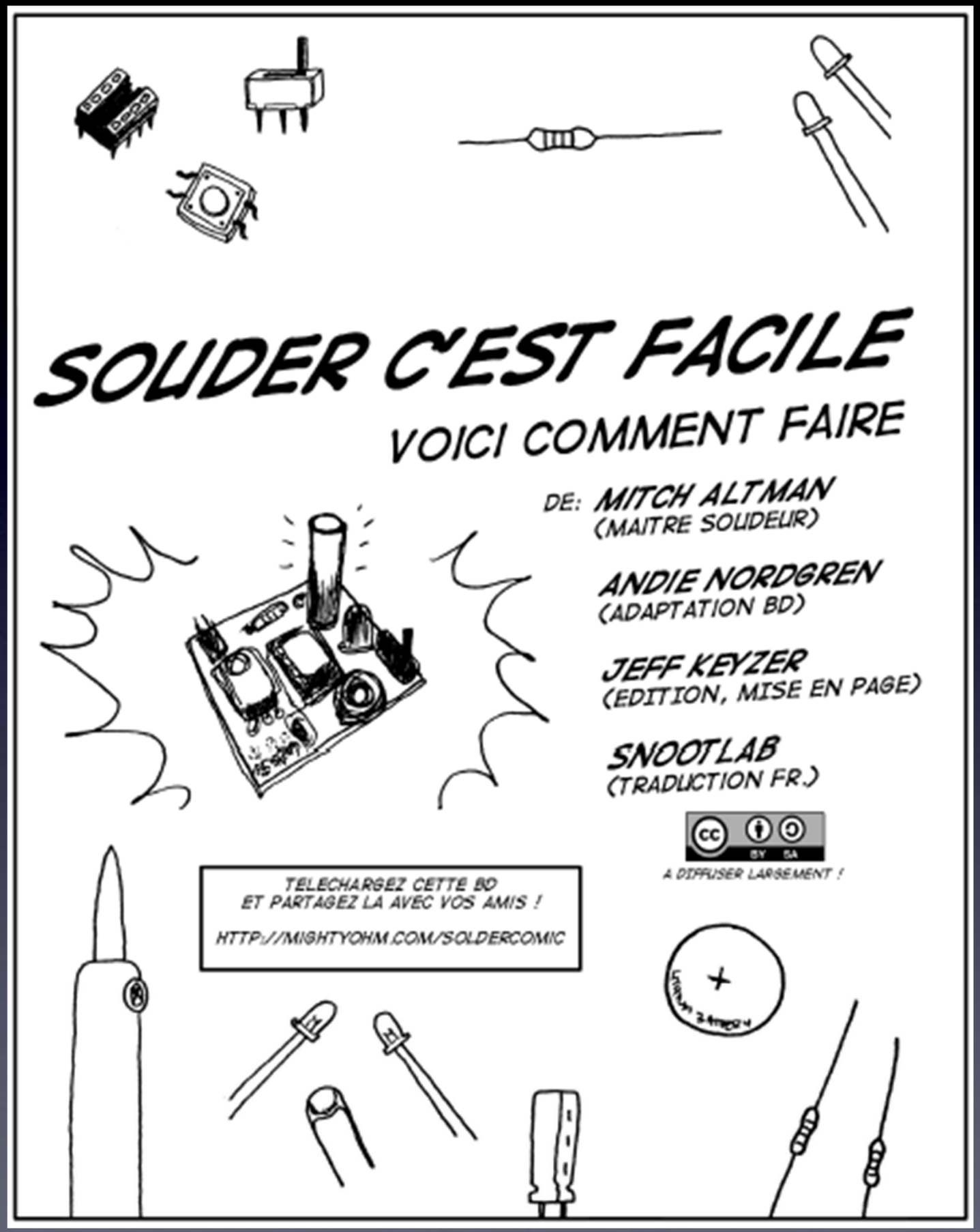

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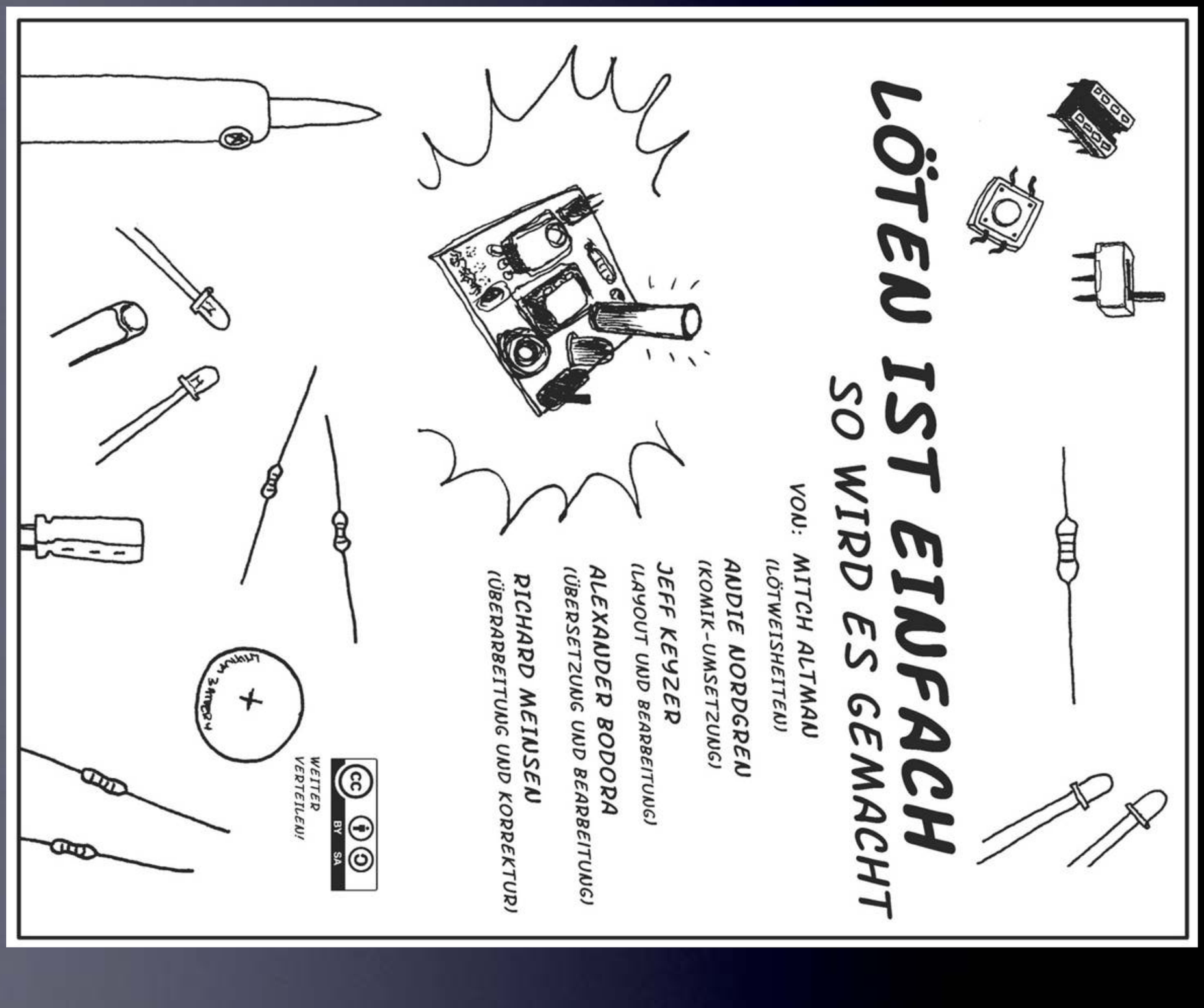

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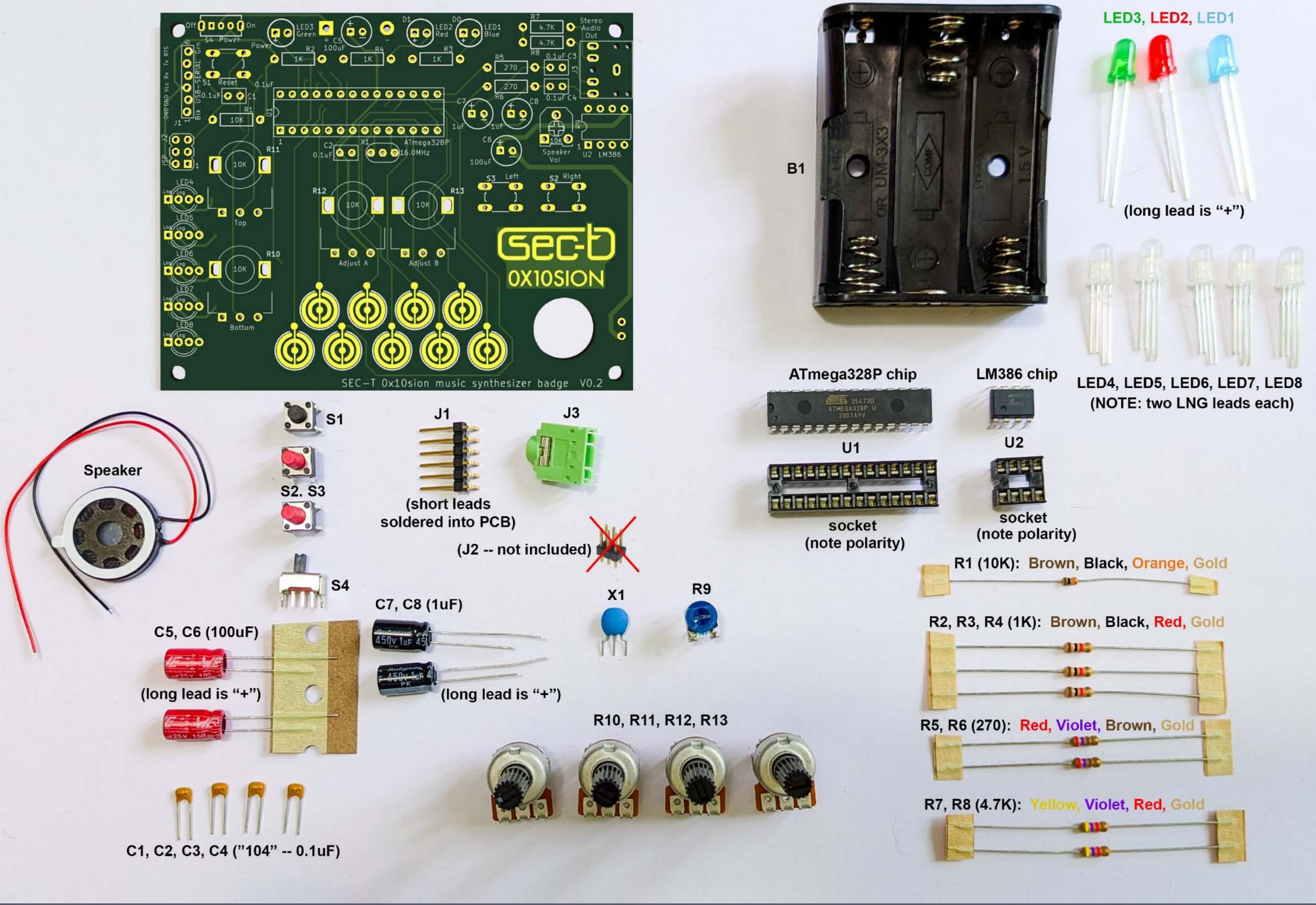
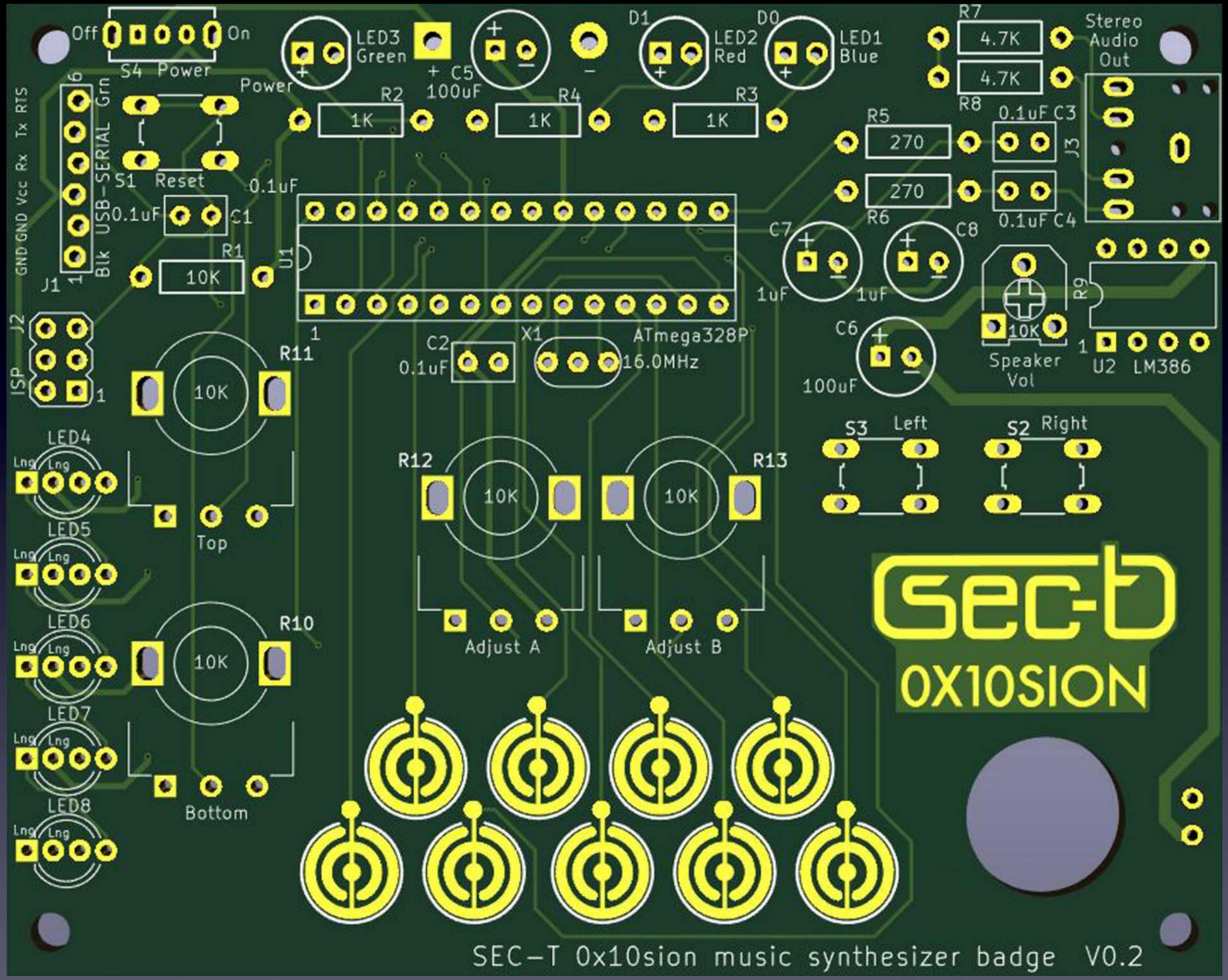
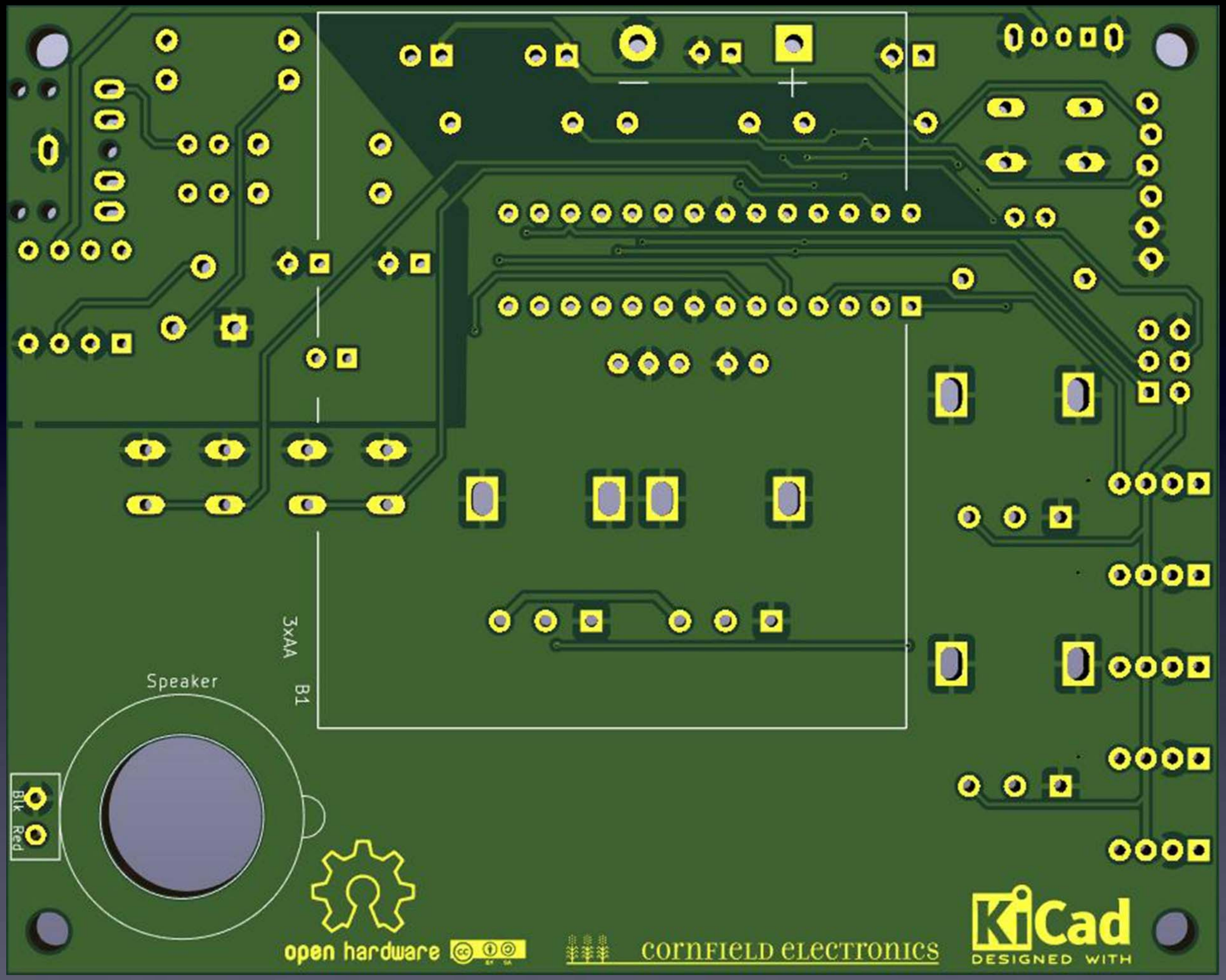


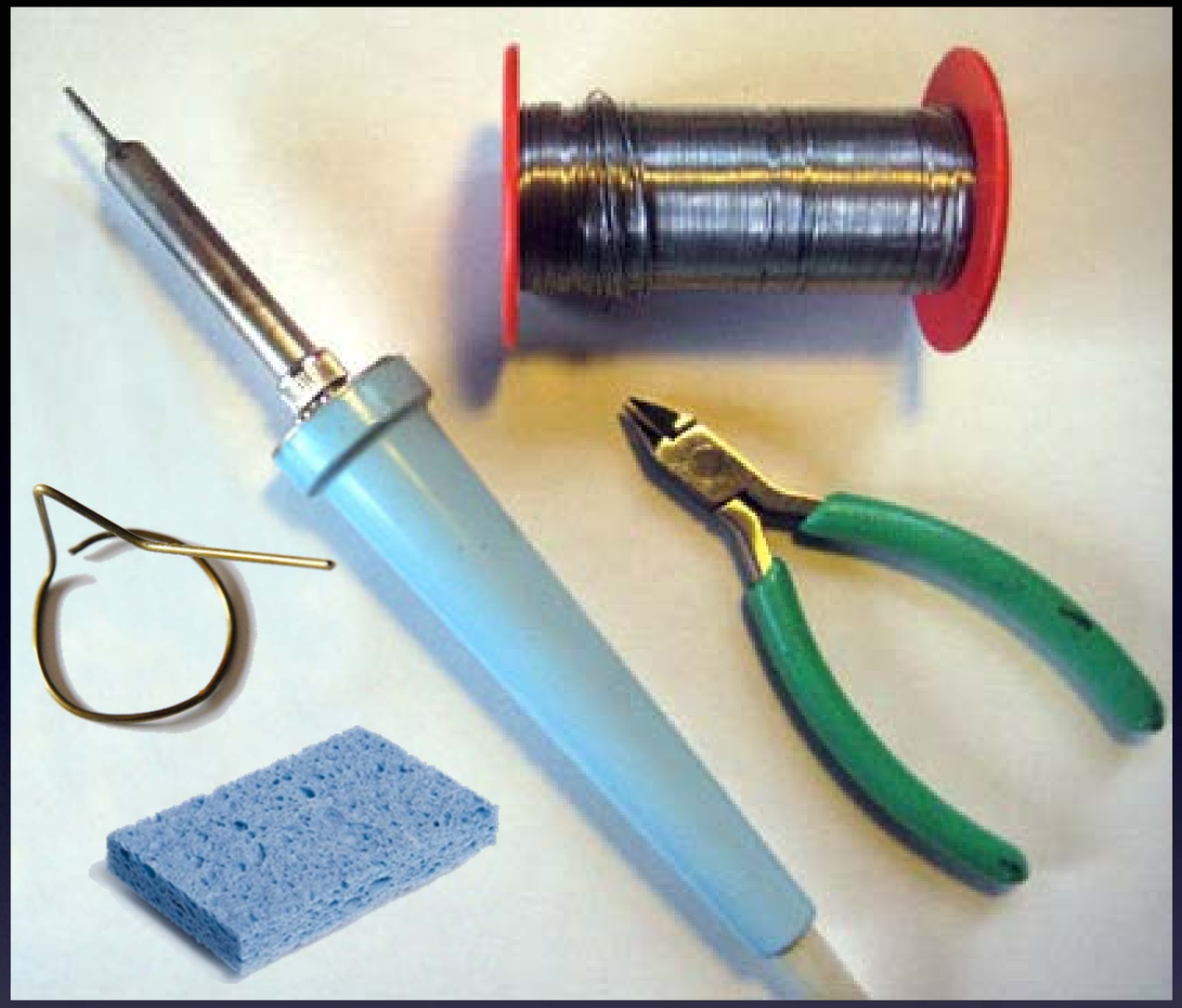
All of the parts



The board we'll solder the parts to (front)



The board we'll solder the parts to (back)



Note:

If you use Lead-Free solder
it is very helpful
to also have
flux paste in a syringe
And Isopropyl Alcohol

The tools you'll need:

- soldering Iron (35W or less)
- solder (*more details coming*)
- soldering iron stand
- cellulose kitchen sponge *(not plastic!)*
- *small* wire cutter

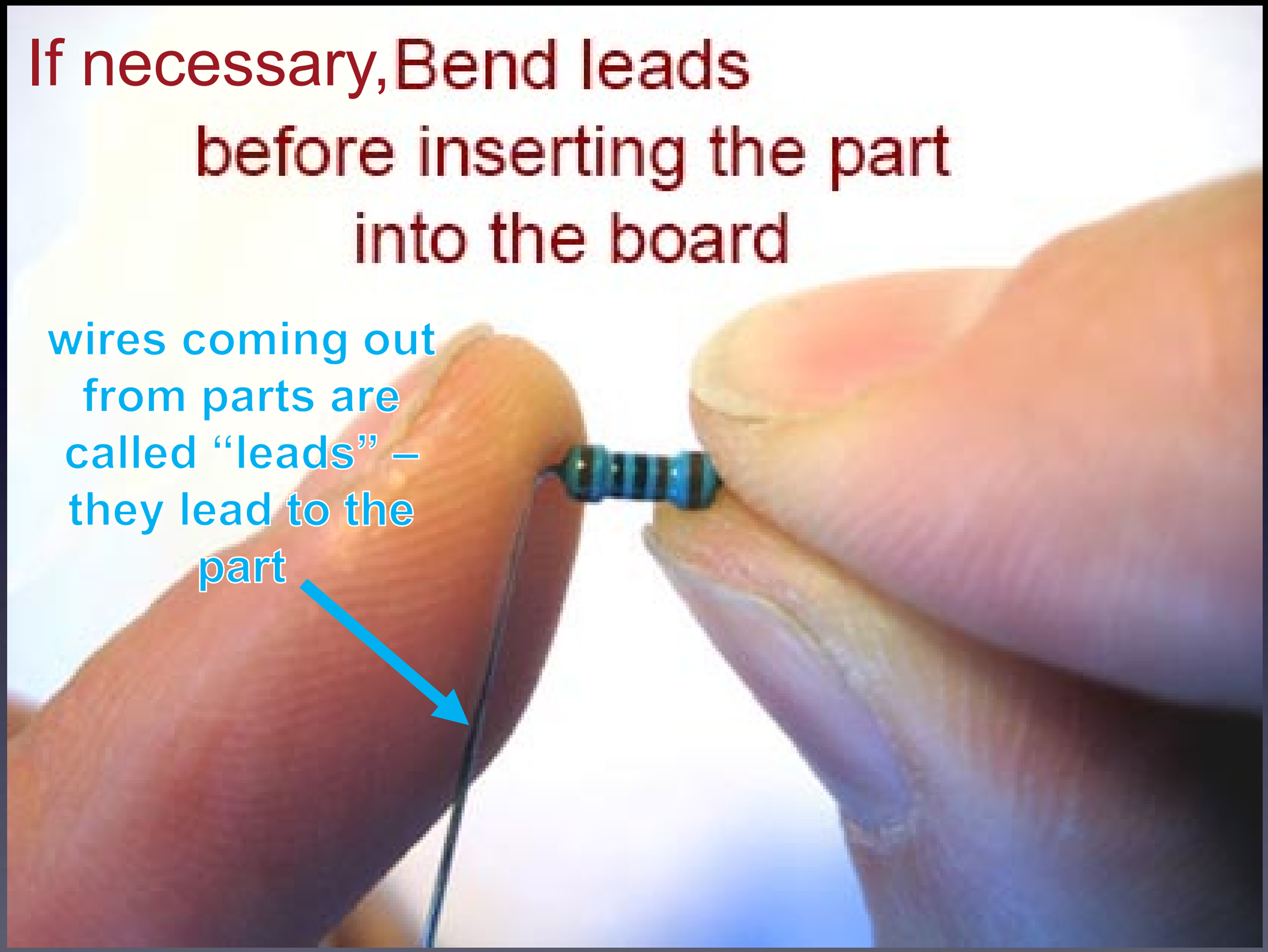
Our first part

R1: Brown, Black, Orange

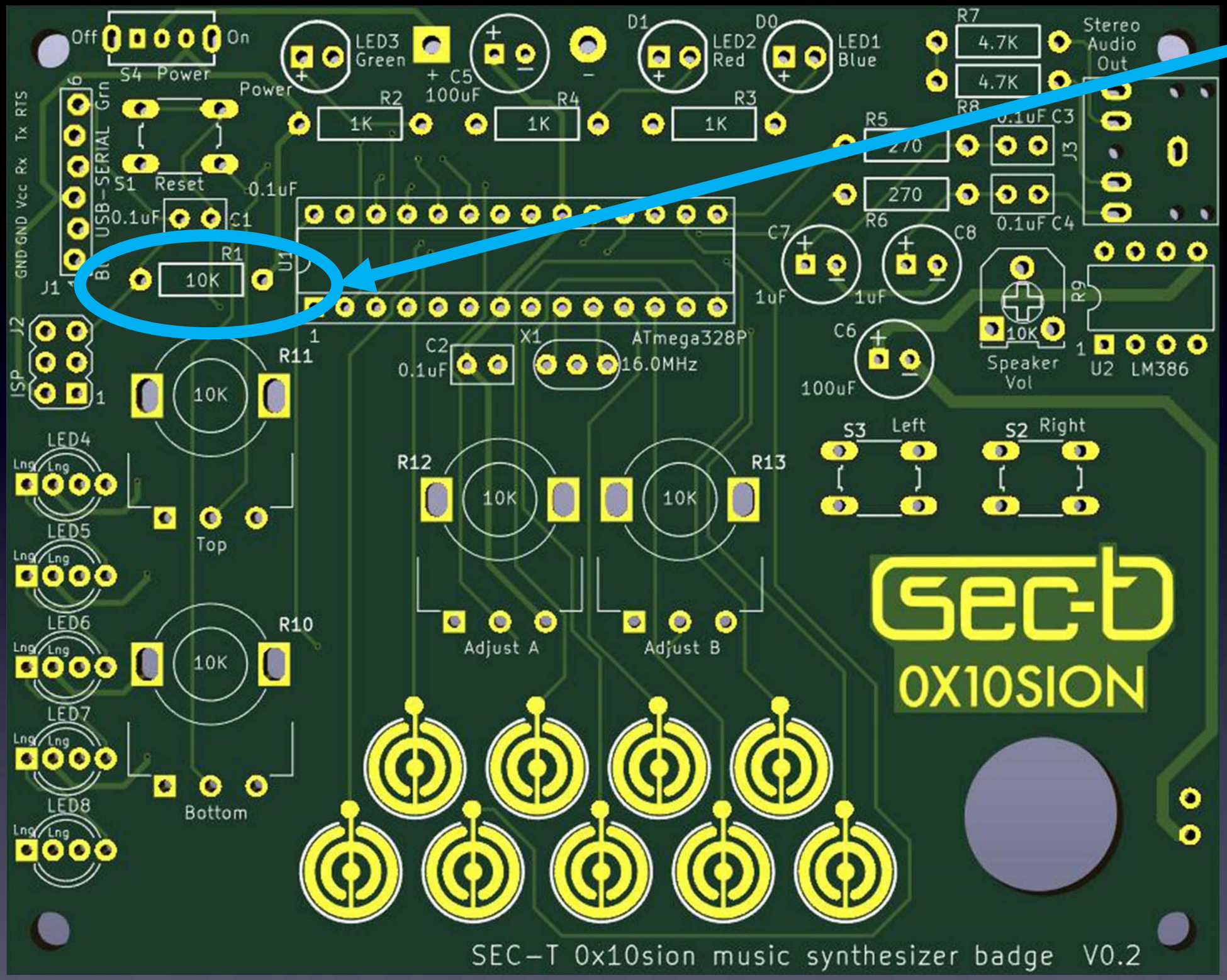
(the small one)

(*not* Brown, Black, ~~Red~~)

Some parts, such as resistors, need their leads bent first



R1 – this is how it will look *before* inserting it into the board

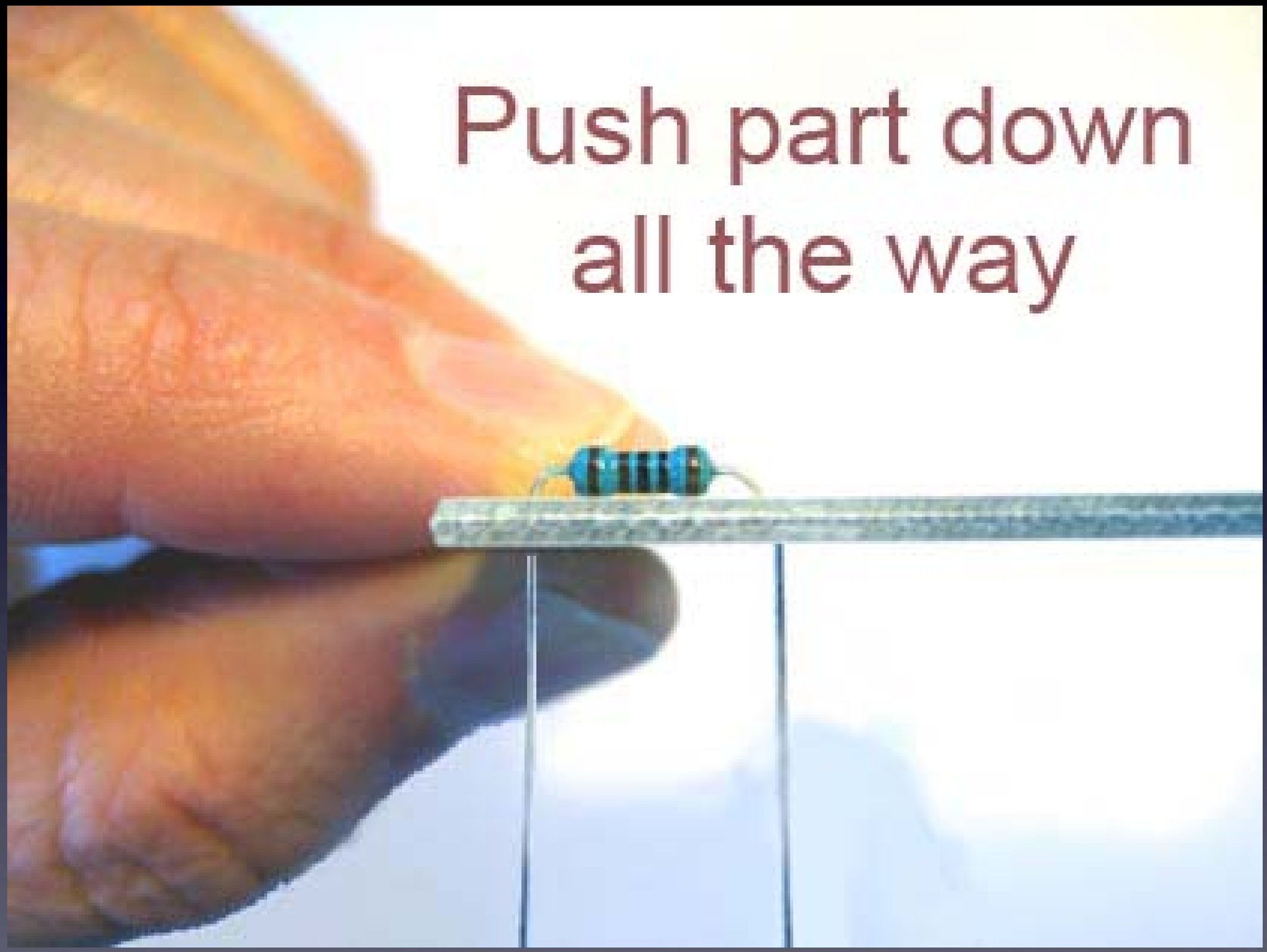


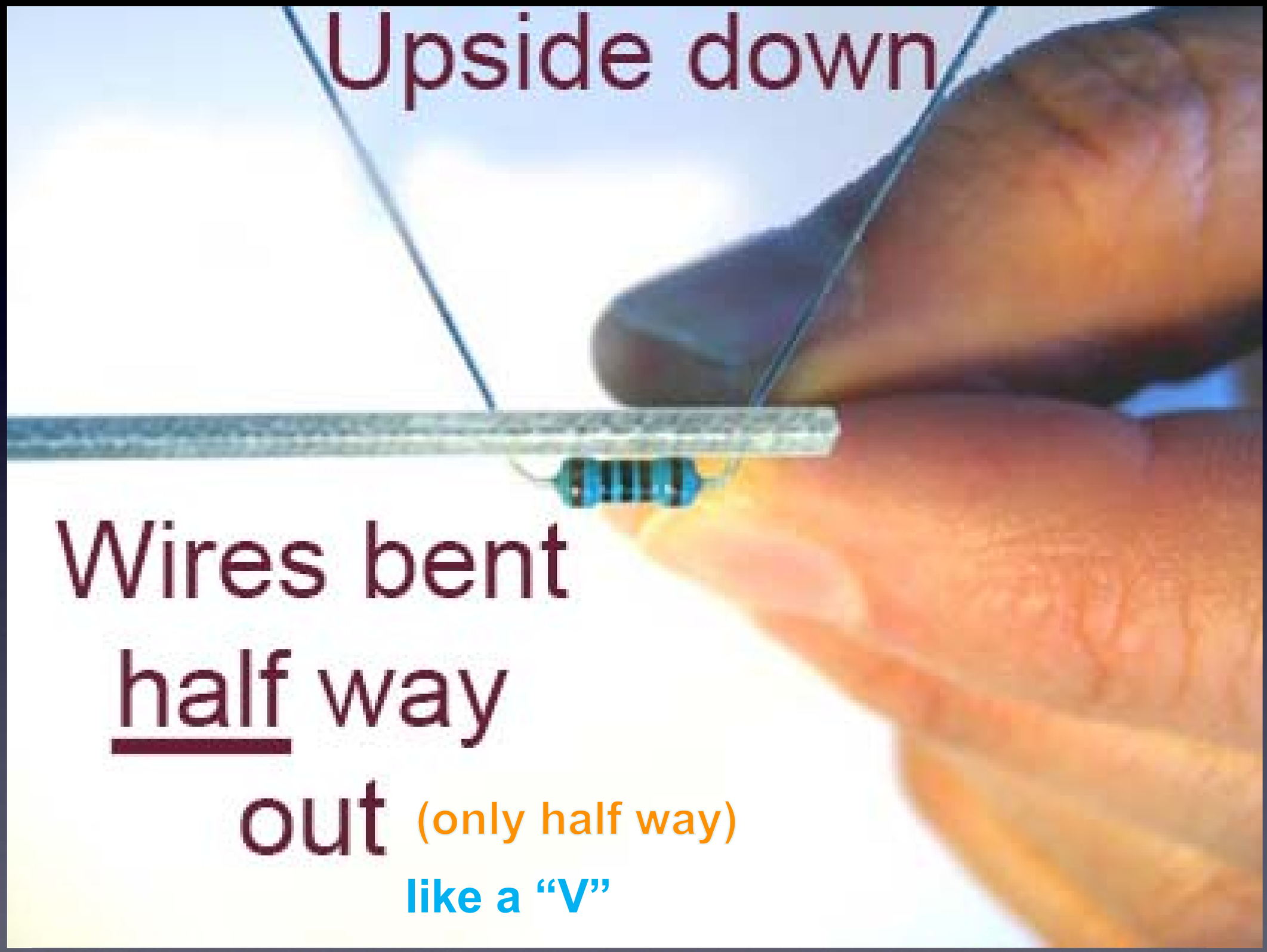
R1 – this is where it goes

Insert leads into pads

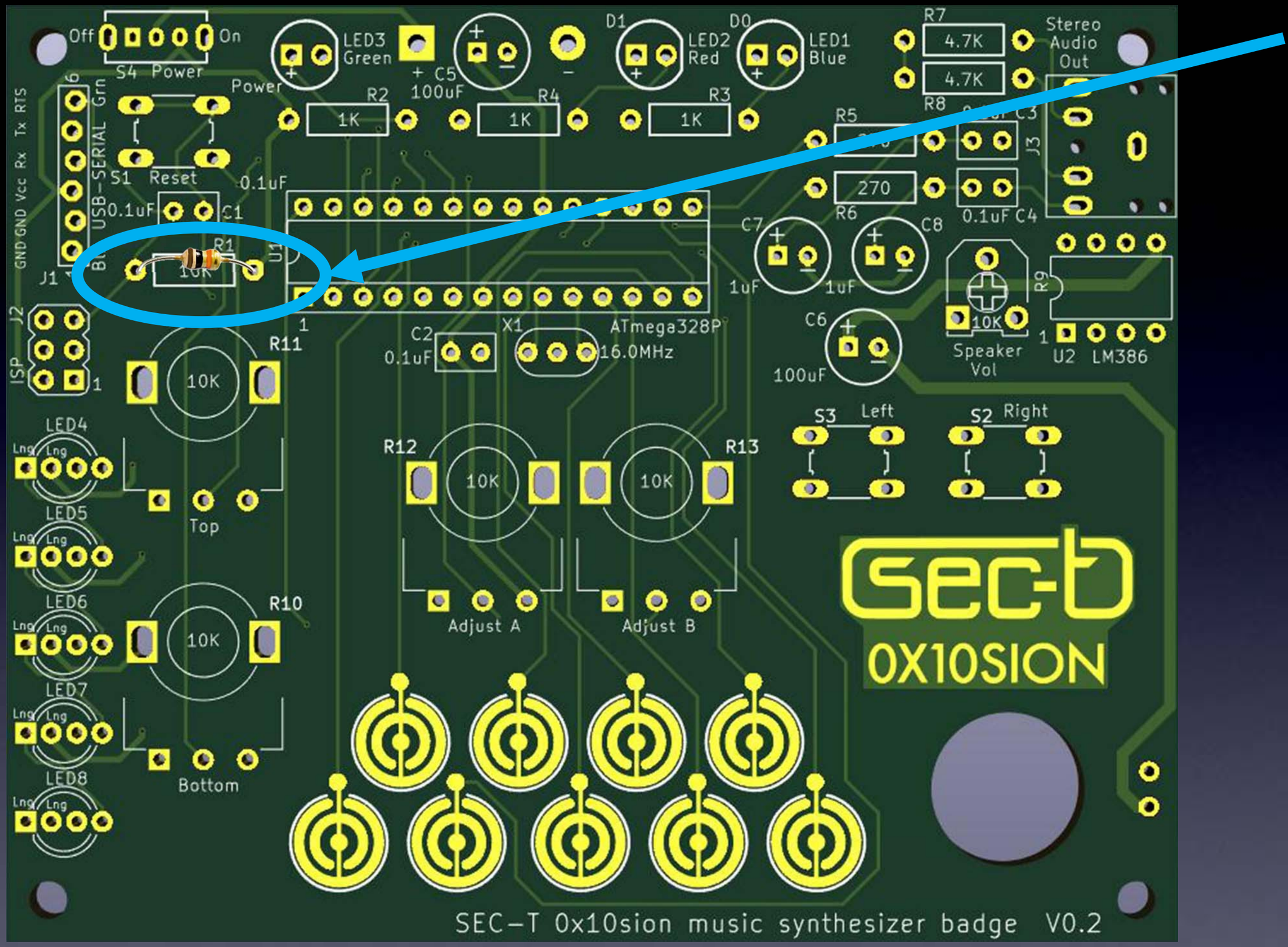
the circles with holes in them are called “pads”

there is one “pad” per lead for each part


Push part down
all the way



so that the part won’t fall out while soldering it



R1 – inserted into the board

How to hold a soldering iron

(Like a pencil – held from underneath)

Important

The best kind of solder for DIY electronics:

(Sn – Tin / Pb – Lead)

(60/40 is also good)

63/37 rosin core,

0.031” (0.8mm) diameter (or smaller)

Note:

Most
Lead-Free **solder**
has poisonous fumes!

A good kind of solder for DIY electronics:

This is the only good Lead-Free ***solder I have found!***

(after years of searching)

Kester
K100LD Rosin
(<u>not</u> "No Clean")
0.031" diameter (0.8mm)

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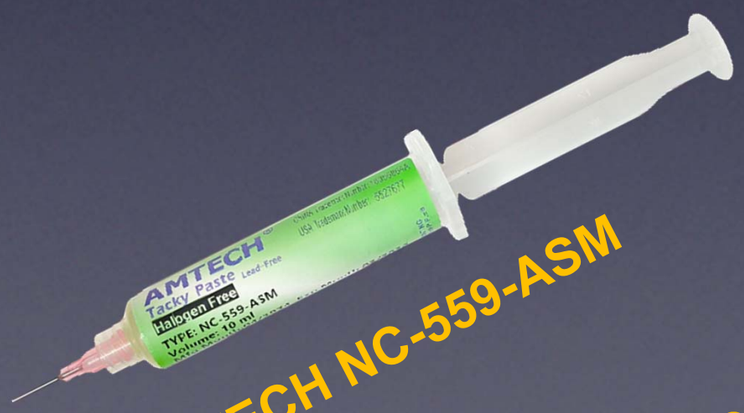


AMTECH NC-559-ASM

99%

Note:

If you use *Lead-Free* solder
it is *helpful*
to also have
***flux paste* in a syringe**
And Isopropyl Alcohol

Another good kind of solder for DIY electronics:

This is another good Lead-Free solder I have found!

Duratool
D01685 Rosin

0.7mm diameter

(as good as the Kester K100LD Rosin)

3 Safety Tips…

Safety Tip #1:

Hot !!

(When you touch the tip,
you *will* let go quickly every time!)

Safety Tip #2:

Soldering chemicals are toxic

But they easily wash off your hands with soap and water

Safety Tip #3:

(coming soon)

2 secrets to good soldering…

Secret #1:

Clean the tip!

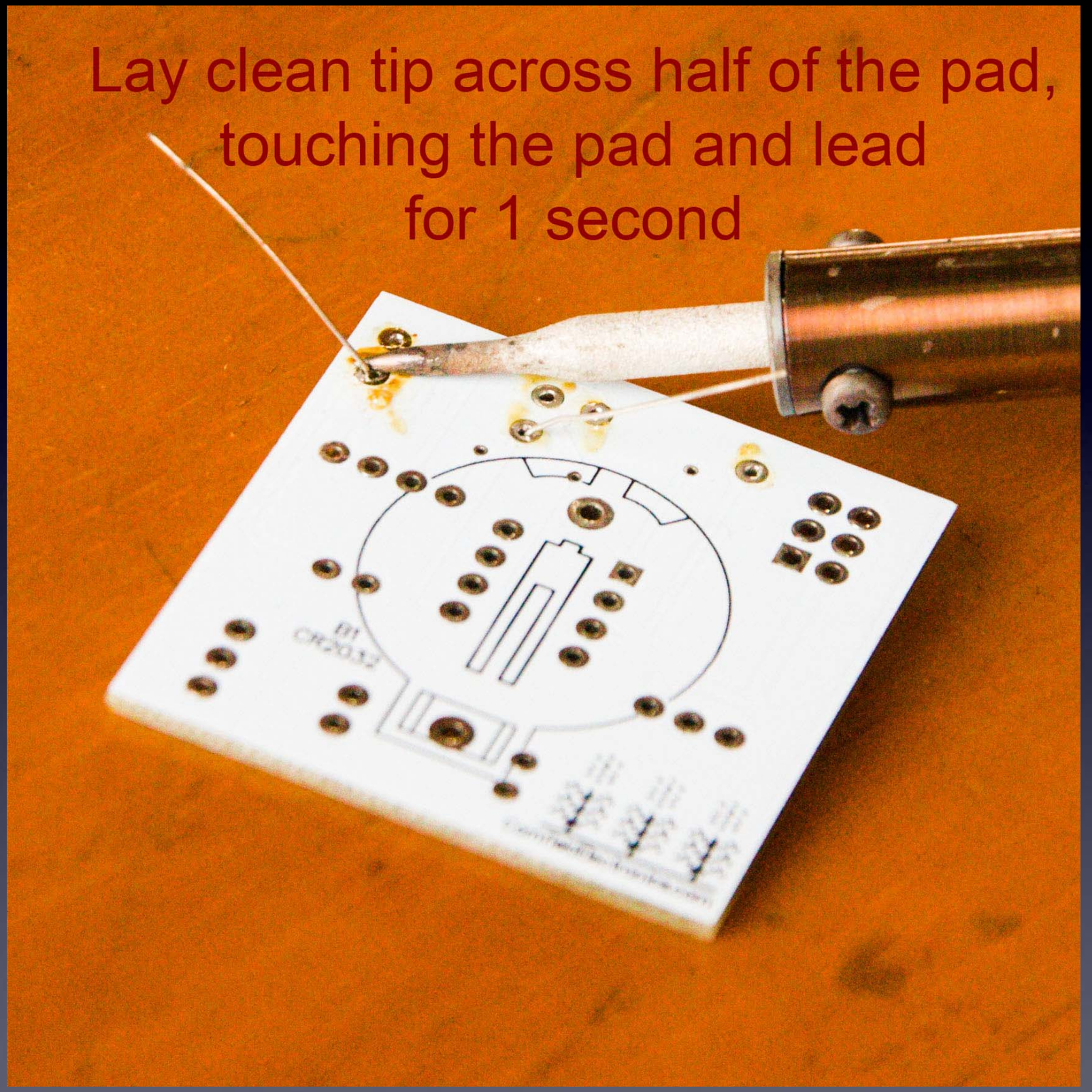
(before every solder connection)

Bang (lightly) 3 times,

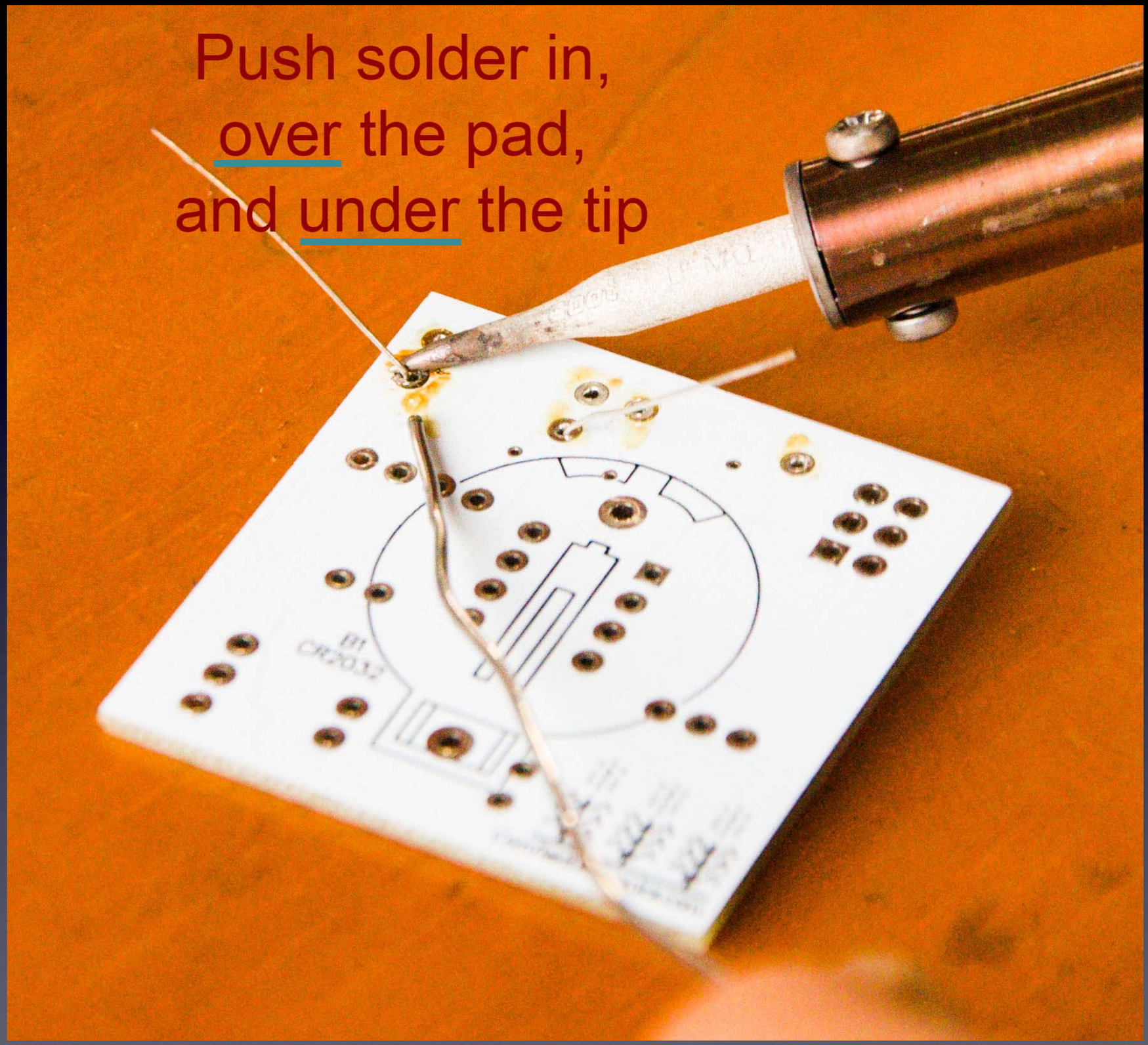
Swipe, Rotate, Swipe (on the sponge):

Keep the tip shiny silver!

knock solder off the tip

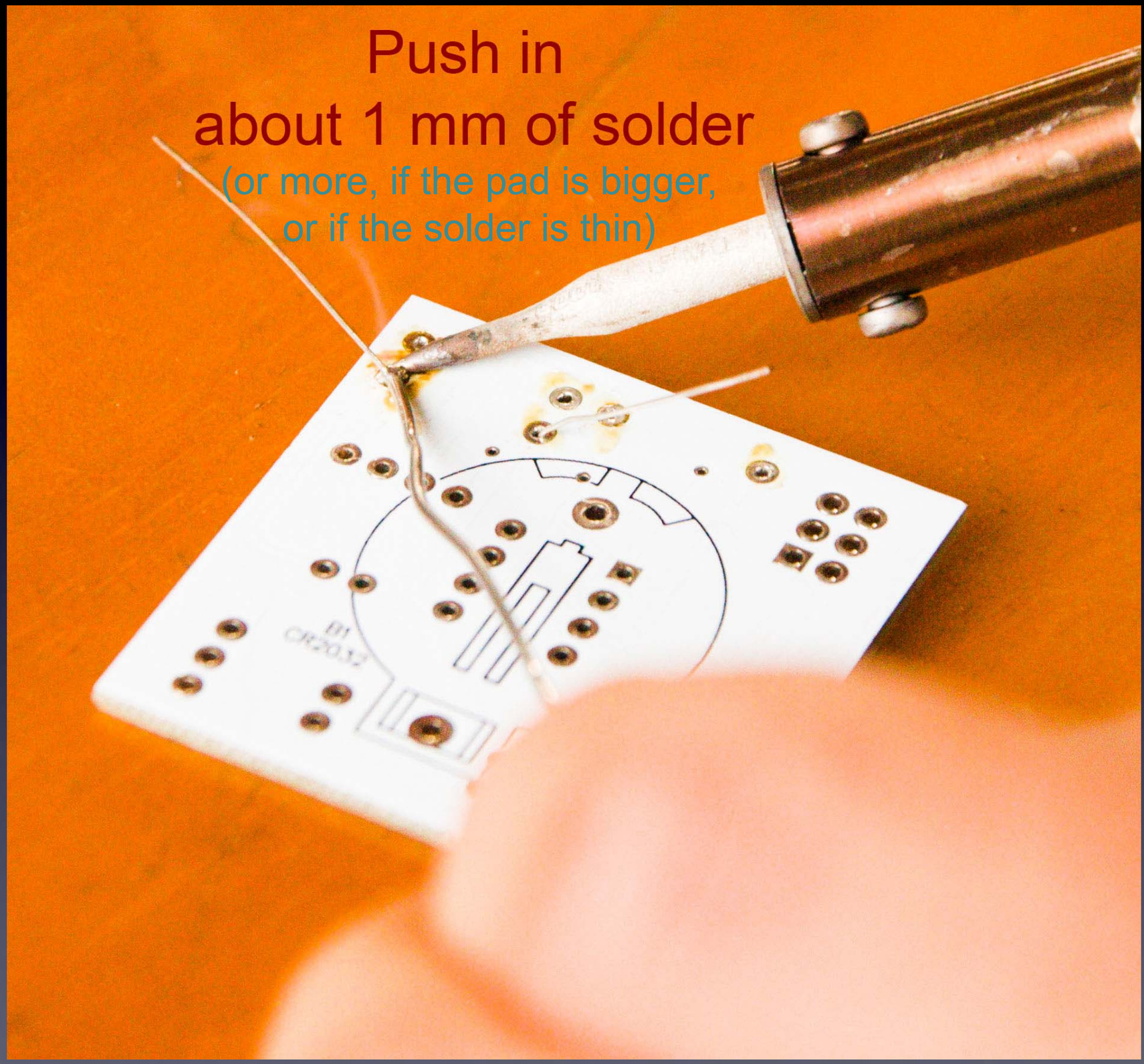

Lay clean tip across half of the pad,
touching the pad and lead
for 1 second

Do this quickly (slowly doesn't work well) – solder in & out in about 1 second

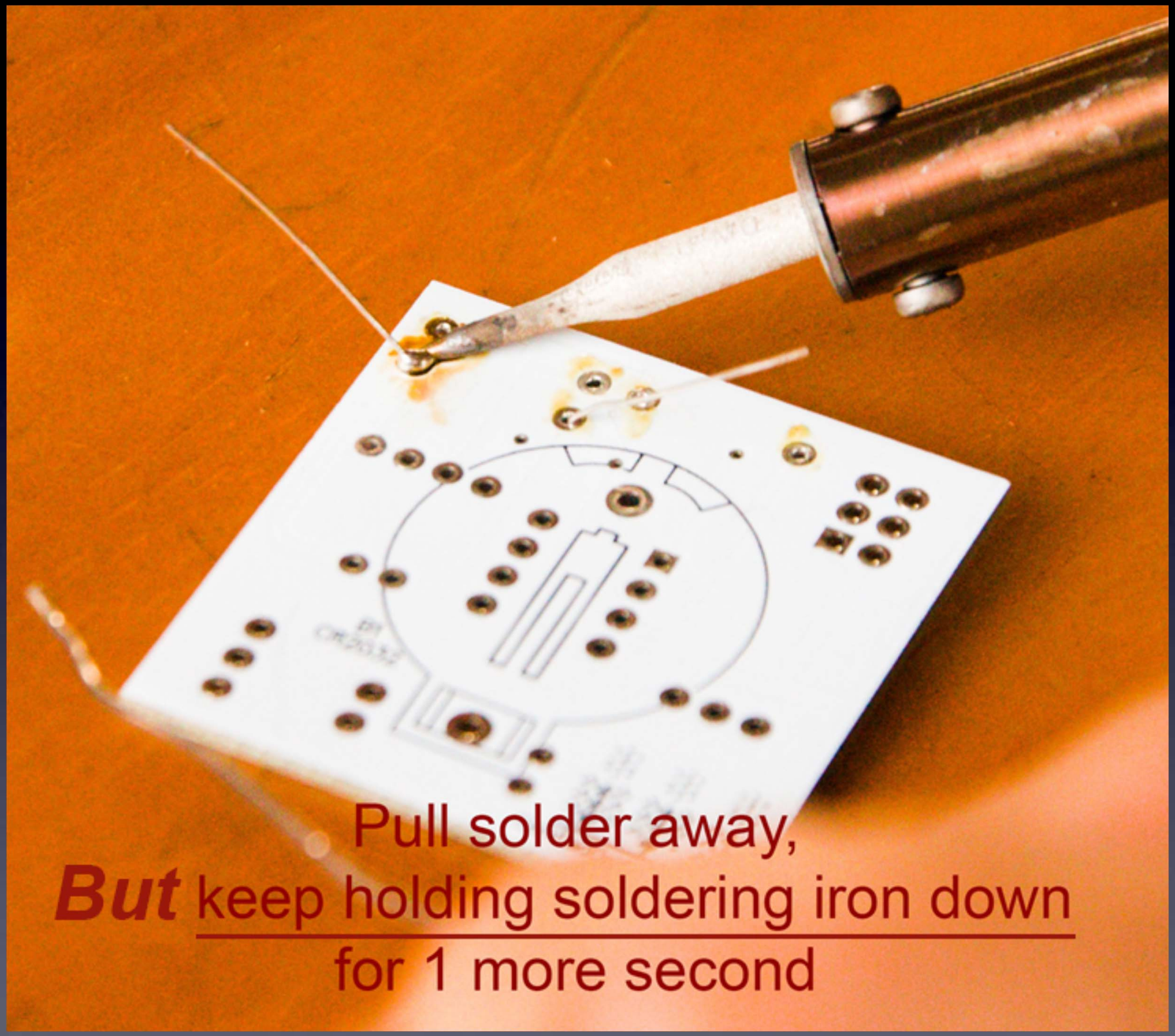


Make sure solder melts on the underside of the soldering iron tip
(not the side or top of the soldering iron tip)!

Do this quickly (slowly doesn't work well) – solder in & out in about 1 second



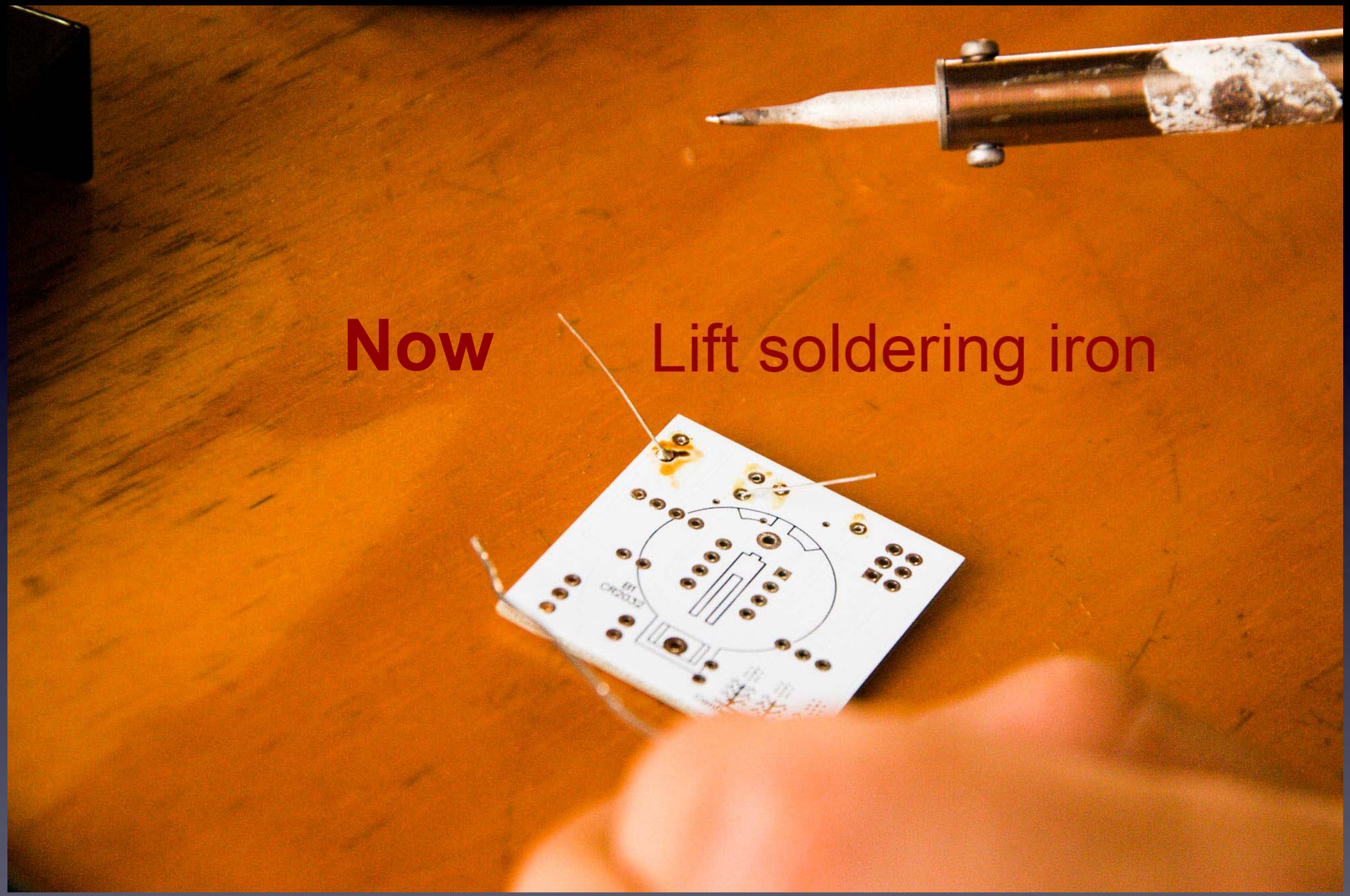
Make sure solder melts on the <u>underside</u> of the soldering iron tip
(not the side or top of the soldering iron tip)!

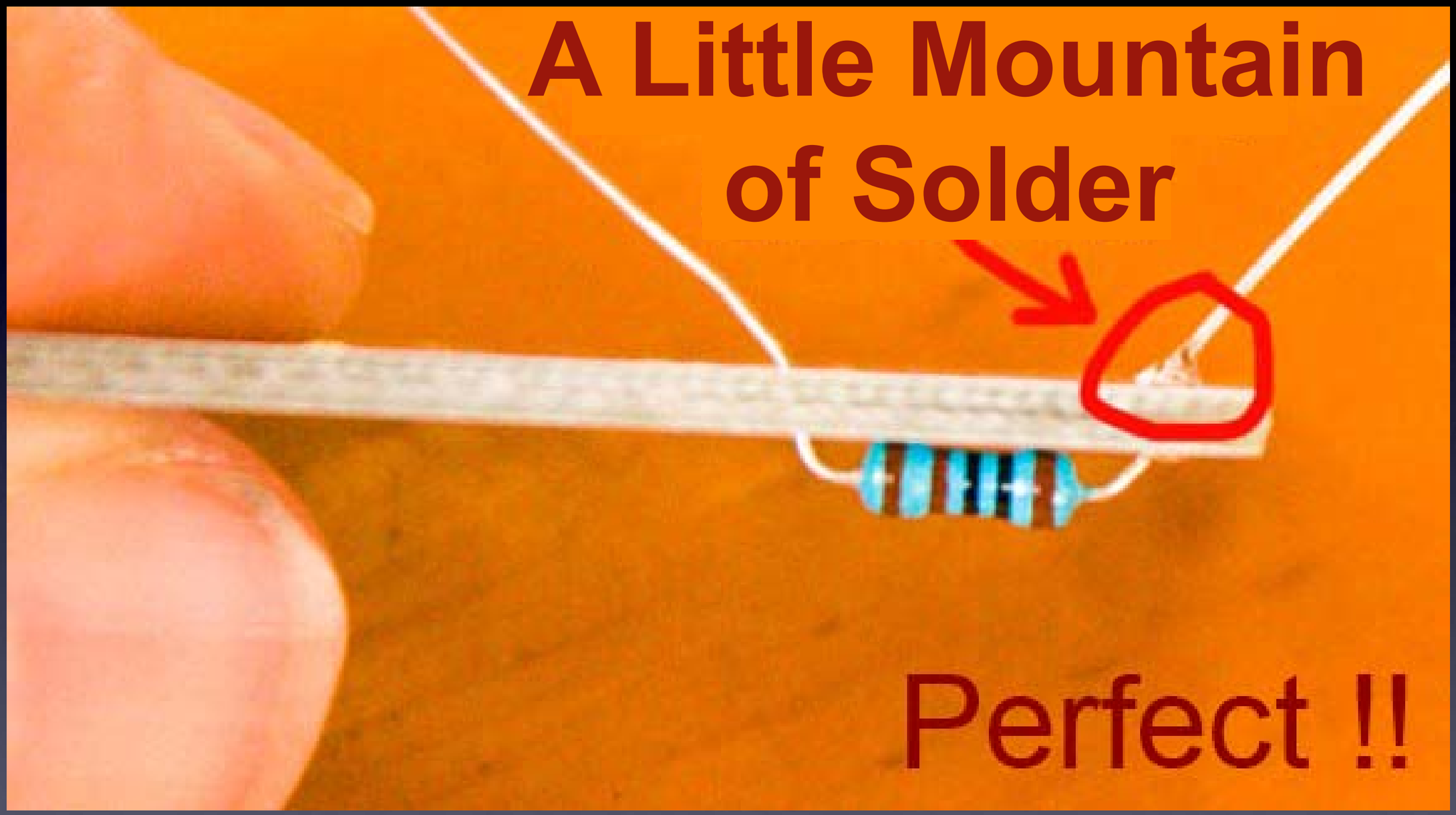

Pull solder away,
But keep holding soldering iron down
for 1 more second

Secret #2:

Keep hot tip down

1 second
for solder to flow !!


Now
Lift soldering iron



If you can see any of the pad, or the hole, you need more solder – so, just do all the steps again to make it perfect.

The Rhythm !

is just as important as the preceding steps!

The Rhythm !

and speed (about 1 second per step)

The Rhythm !

and speed (about 1 second per step)

Clean the tip

The Rhythm !

and speed (about 1 second per step)

Tip Down

The Rhythm !

and speed (about 1 second per step)

Solder In

The Rhythm !

and speed (about 1 second per step)

Solder Out

The Rhythm !

and speed (about 1 second per step)

WAIT !

The Rhythm !

and speed (about 1 second per step)

Lift Tip

The Rhythm !

and speed (about 1 second per step)

Let’s go through that rhythm

one more time…

The Rhythm !

and speed (about 1 second per step)

The Rhythm !

and speed (about 1 second per step)

Clean the tip

The Rhythm !

and speed (about 1 second per step)

Tip Down

The Rhythm !

and speed (about 1 second per step)

Solder In

The Rhythm !

and speed (about 1 second per step)

Solder Out

The Rhythm !

and speed (about 1 second per step)

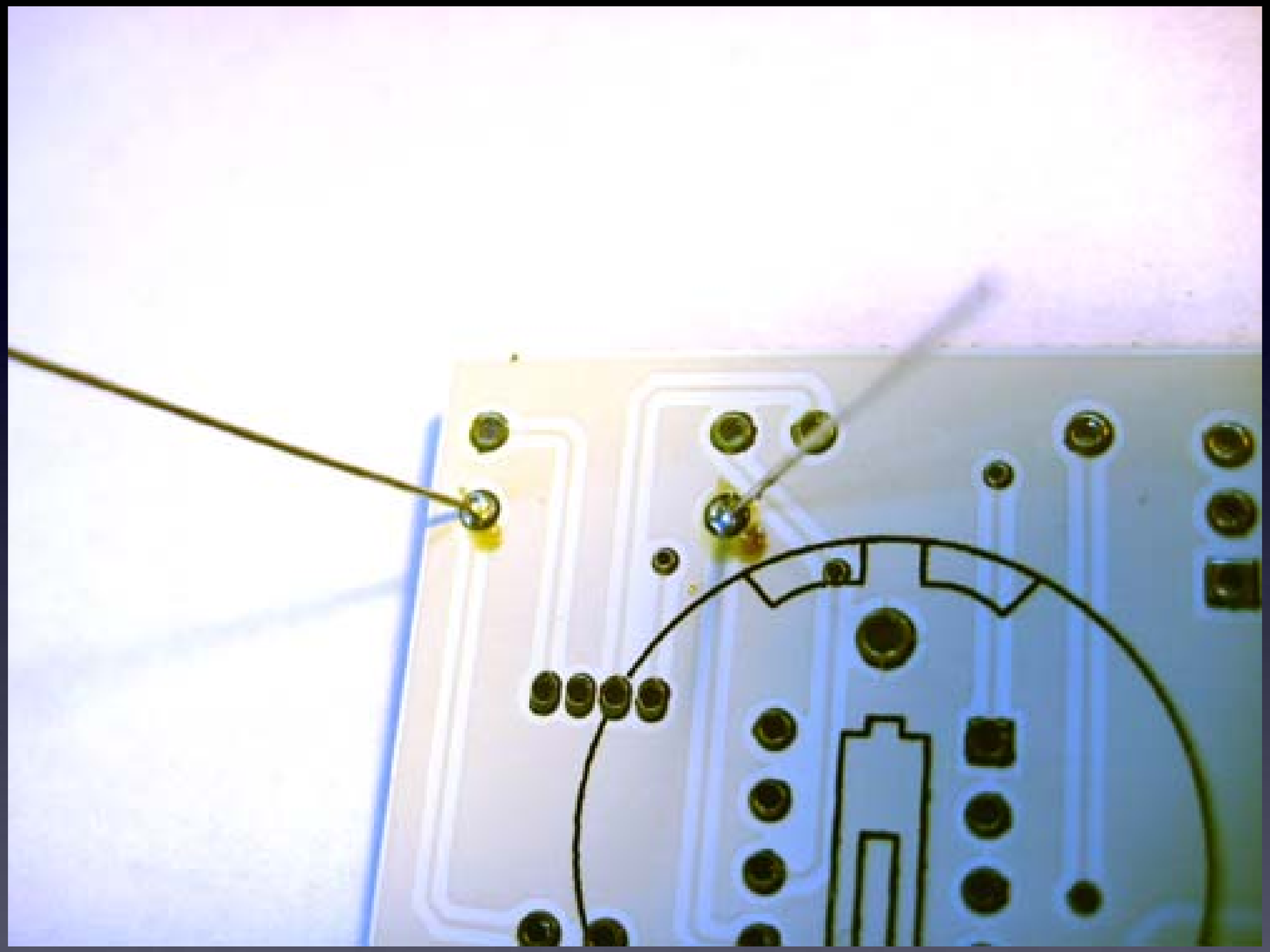
WAIT !

The Rhythm !

and speed (about 1 second per step)

Lift Tip

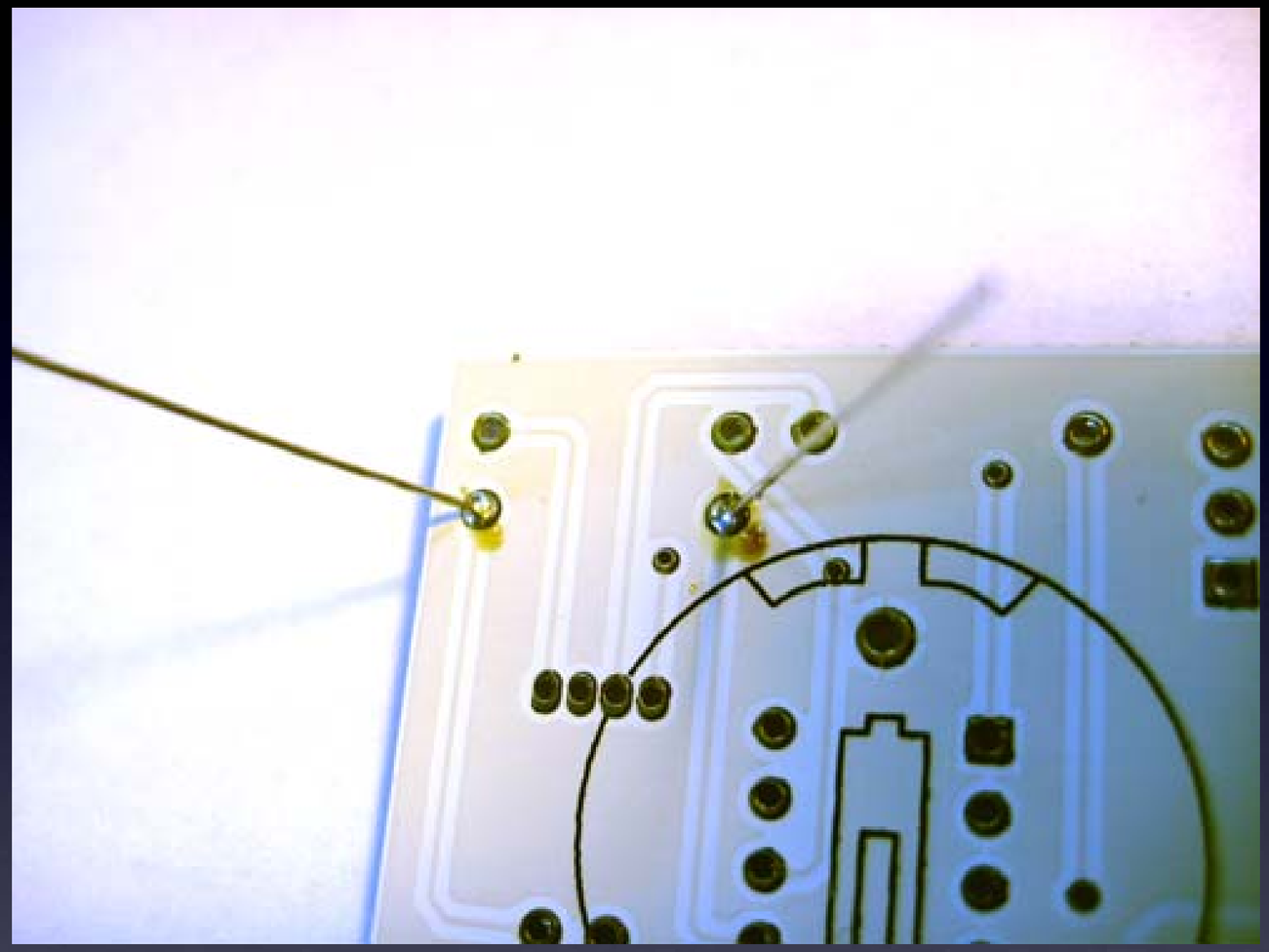
Solder all of the leads of the part to the board



For this part, there are two leads

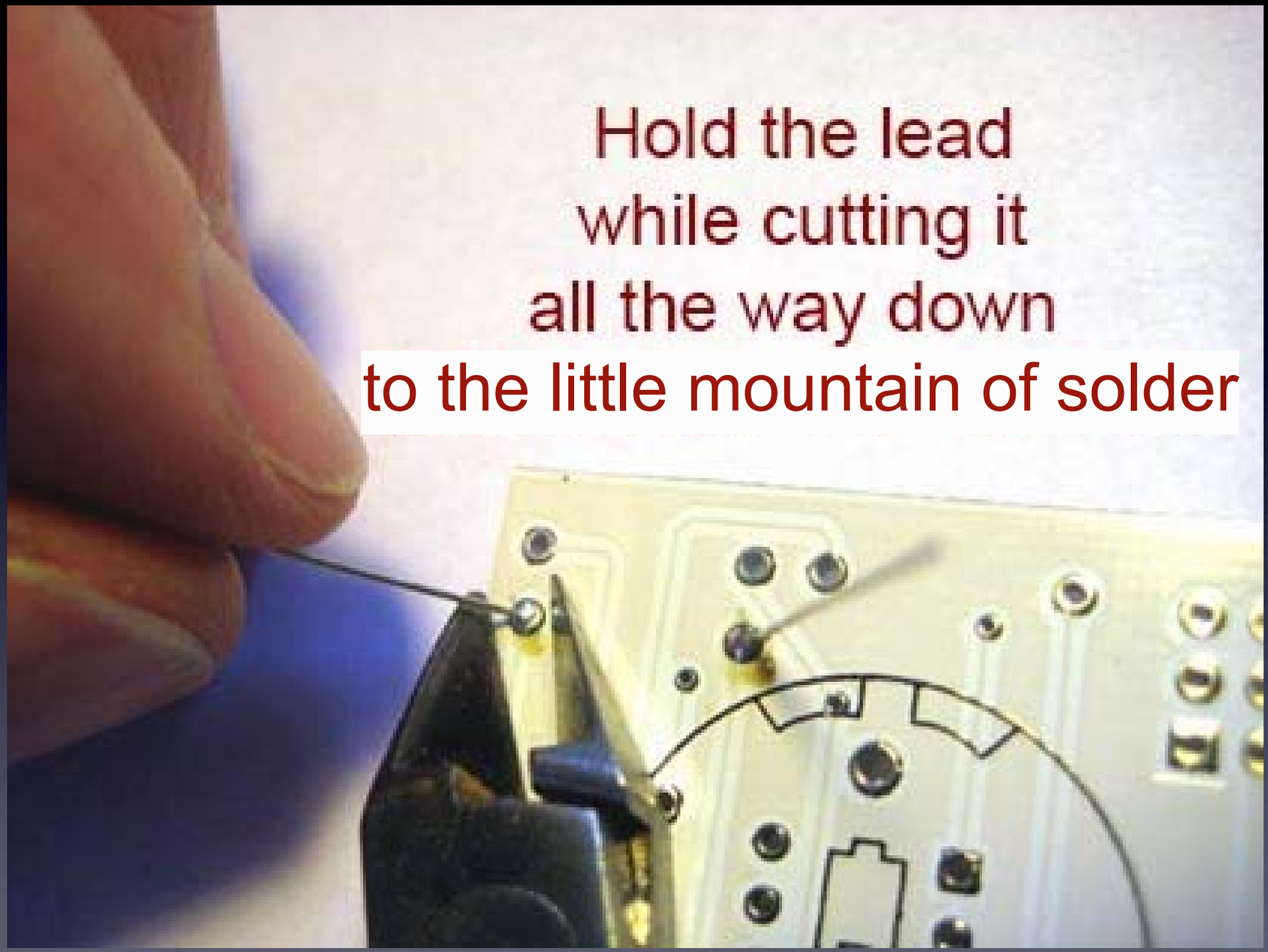
Here you can see two good solder connections

Two good solder connections



- Little mountains (not flat)
- Pads totally covered in solder
- Can’t see the hole
- No connections to other pads

Now cut the leads short



Cutting with the tip of the wire cutter gives you more control

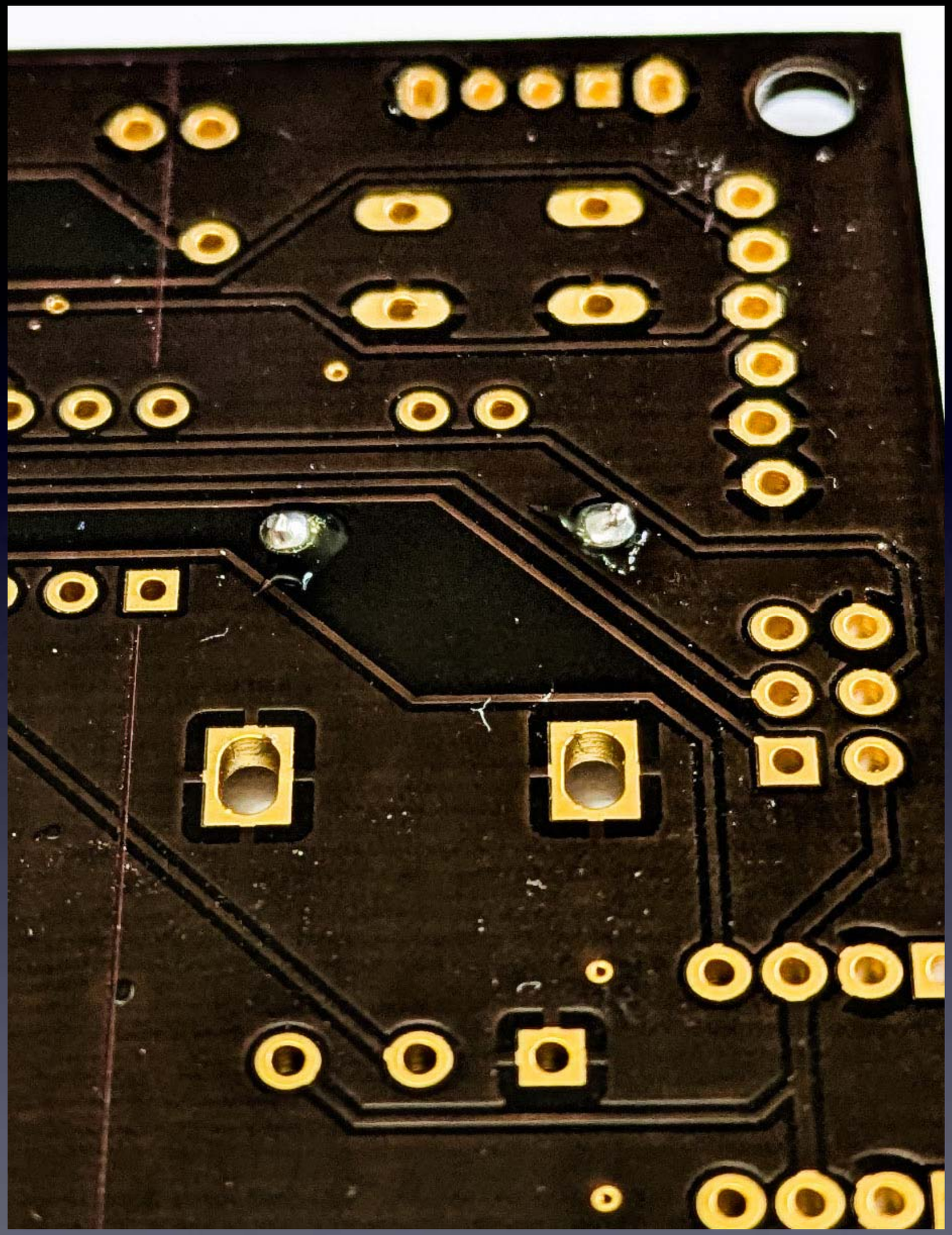
Safety Tip #3:

Hold or cover the lead !

(or it will fly into your eye!)

(They like doing that – so please hold or cover the lead when you cut.)

No wires sticking out



R1 soldered to the board

Notice that:

- each connection is a small bump (not flat)
- you cannot see any pad (it's totally covered with solder)
- you cannot see the hole (it's totally covered with solder)

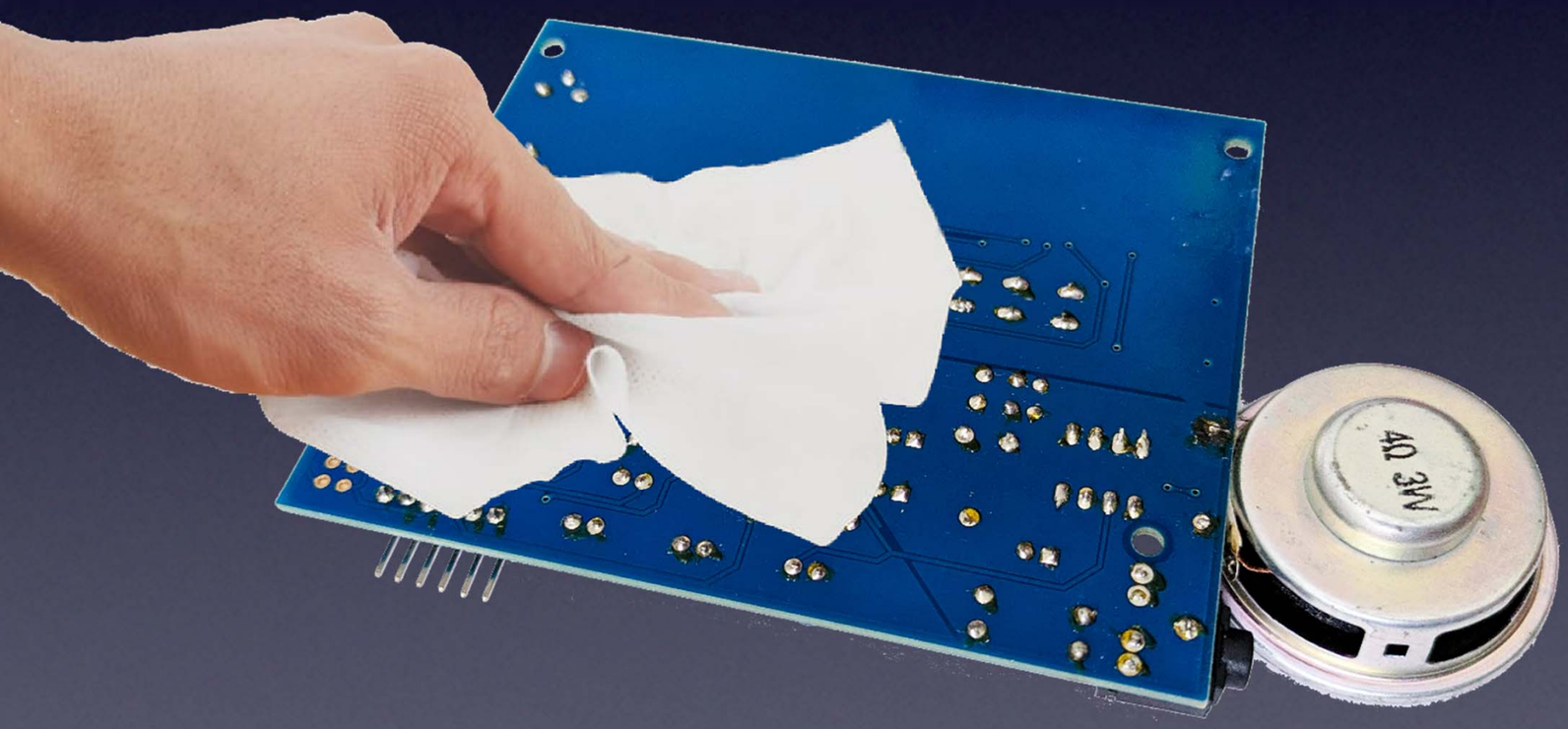
One part at a time

Till all the parts are soldered

And it will look like this when you’re done.

If you used any *flux paste for re-working problems*

The bottom of the PCB will be sticky from the flux



You can clean it with a cloth wet with Isopropyl Alcohol

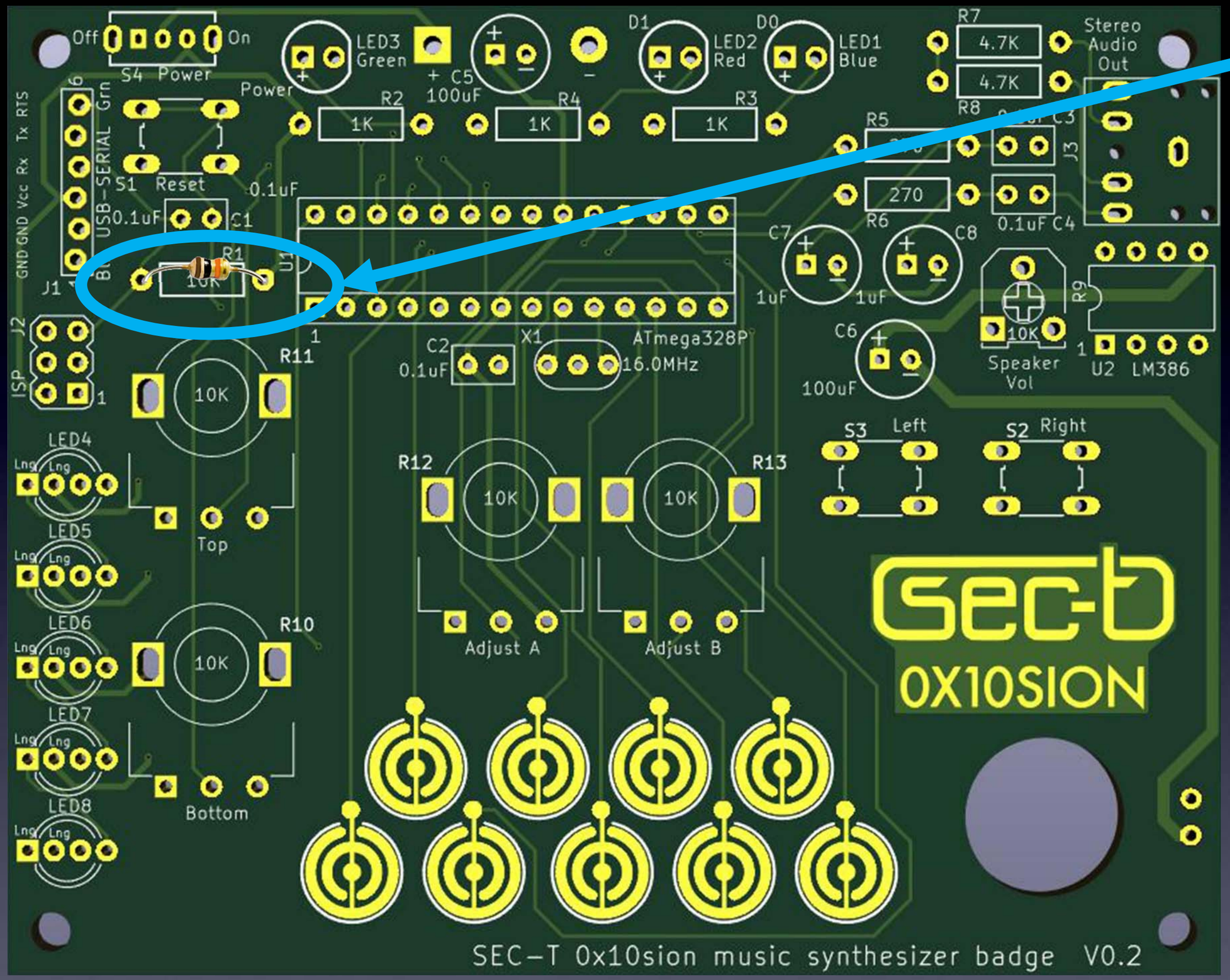
Then put in the batteries,

Turn it on,

And it works!

(Or you start debugging.)

Let’s start!



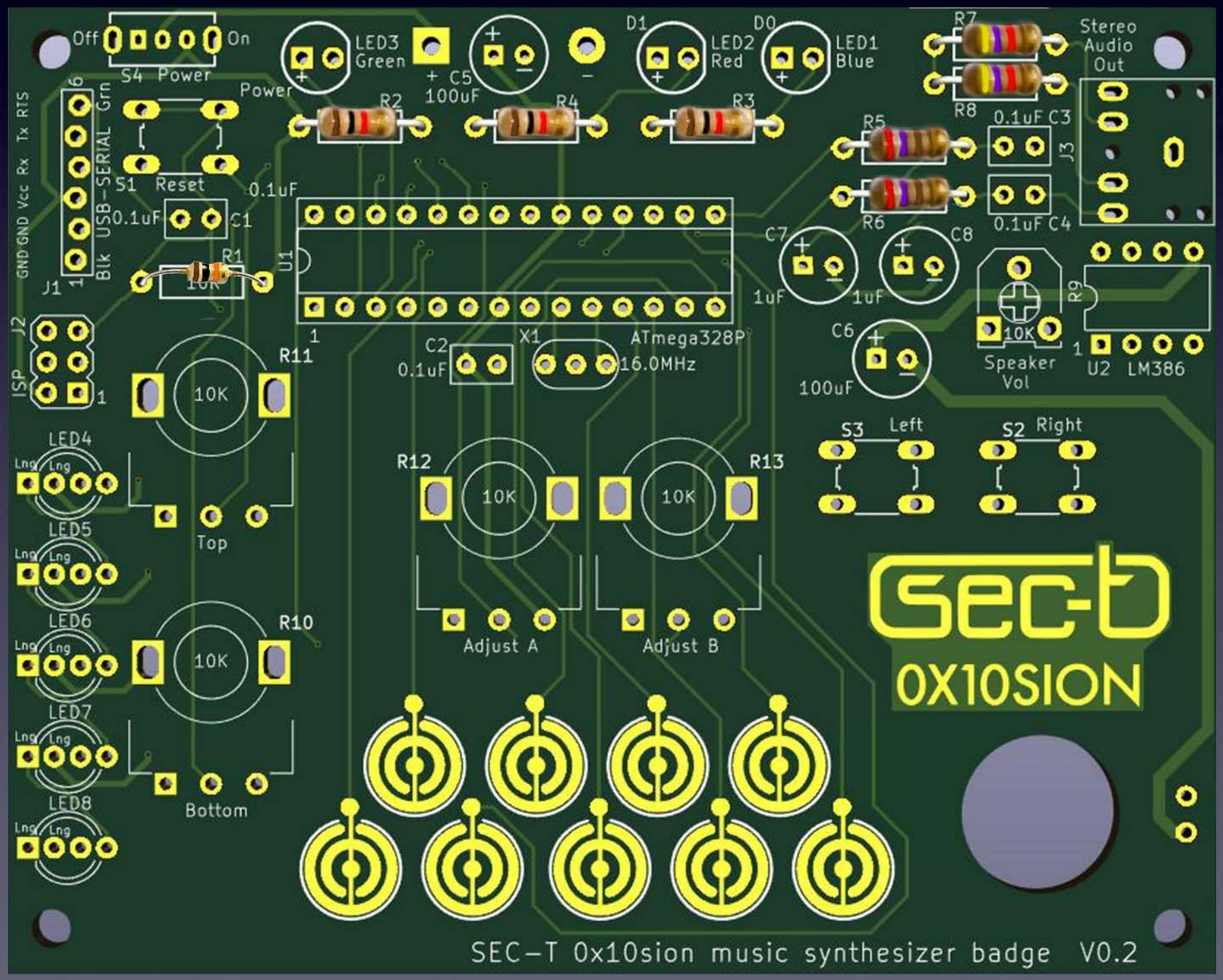
If you haven’t done so already, solder R1: brown, black, orange

R1: 10K: (the small one) Brown, Black, Orange

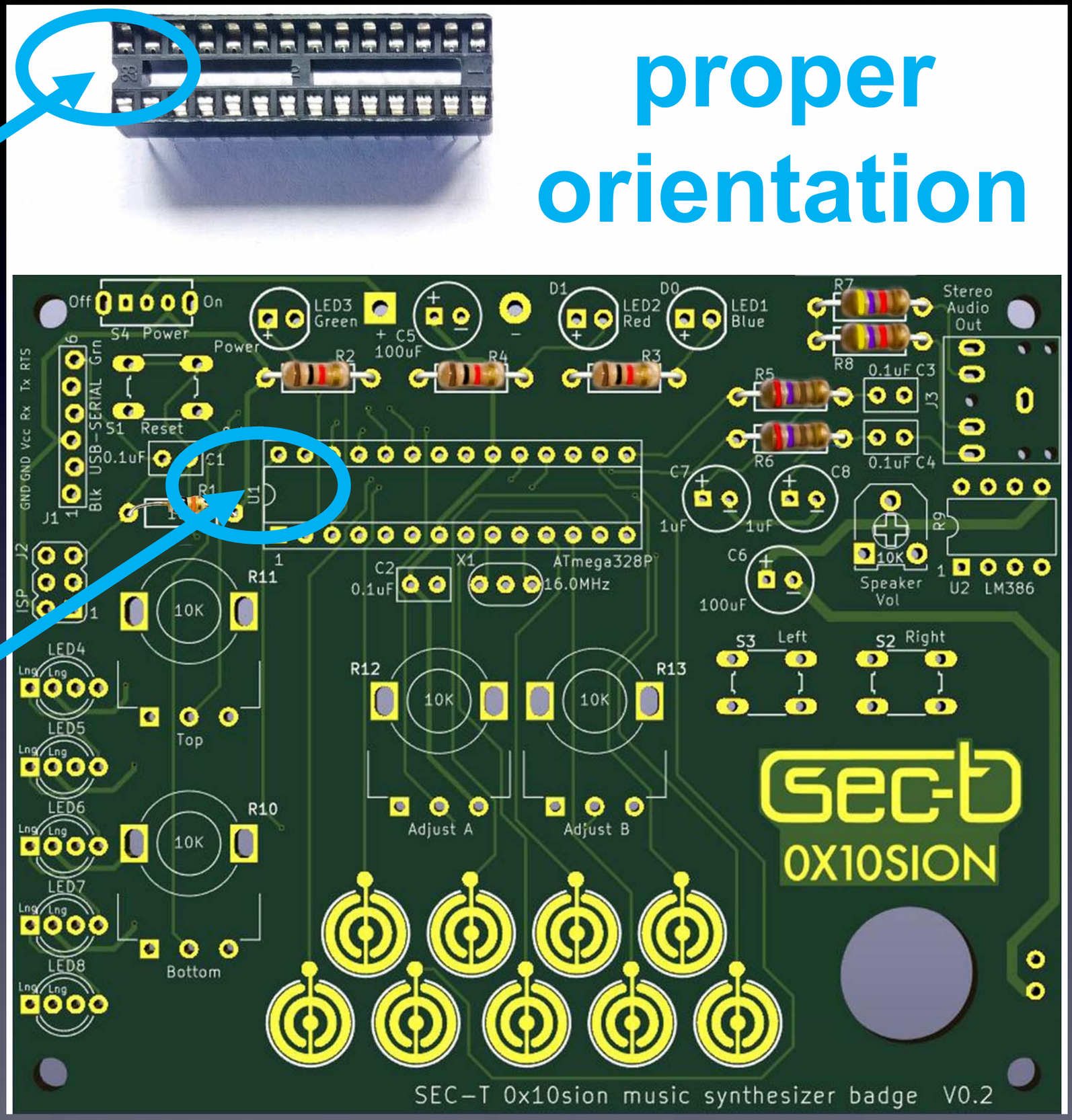
R2, R3, R4: 1K: Brown, Black, Red

R5, R6: 270: Red, Violet, Brown

R7, R8: 4.7K: Yellow, Violet, Red

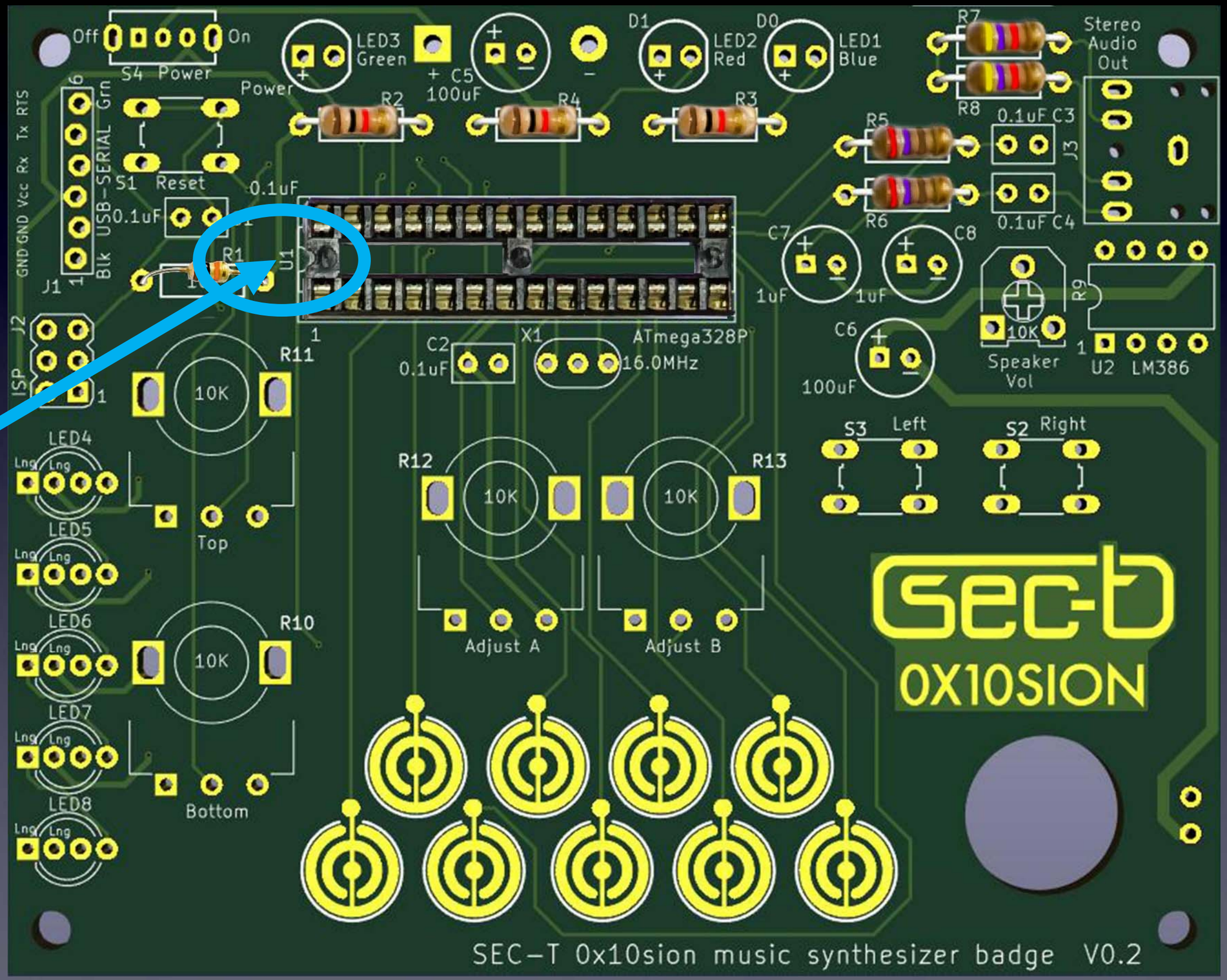


U1: microcontroller socket

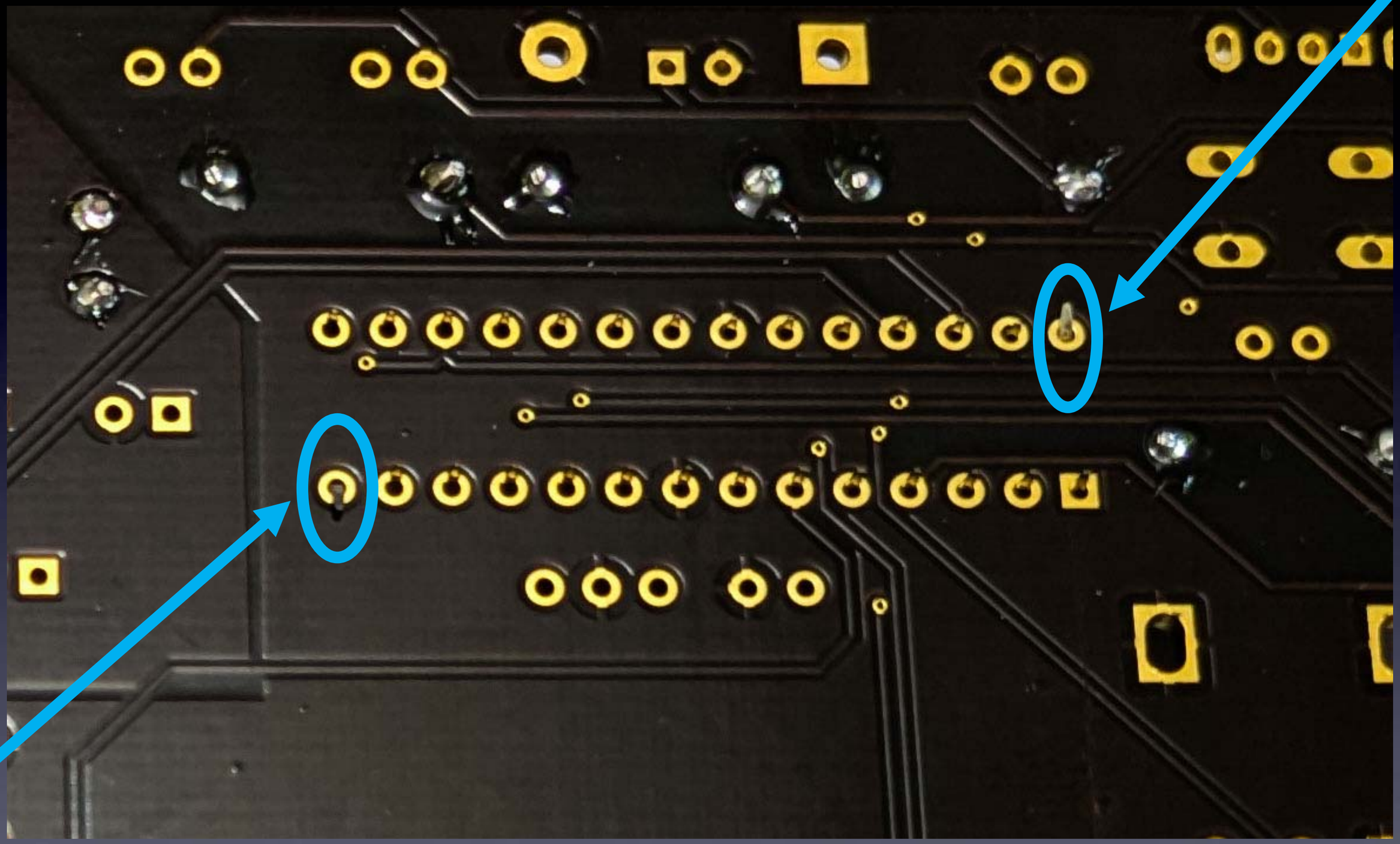


U1: microcontroller socket:

inserted correctly

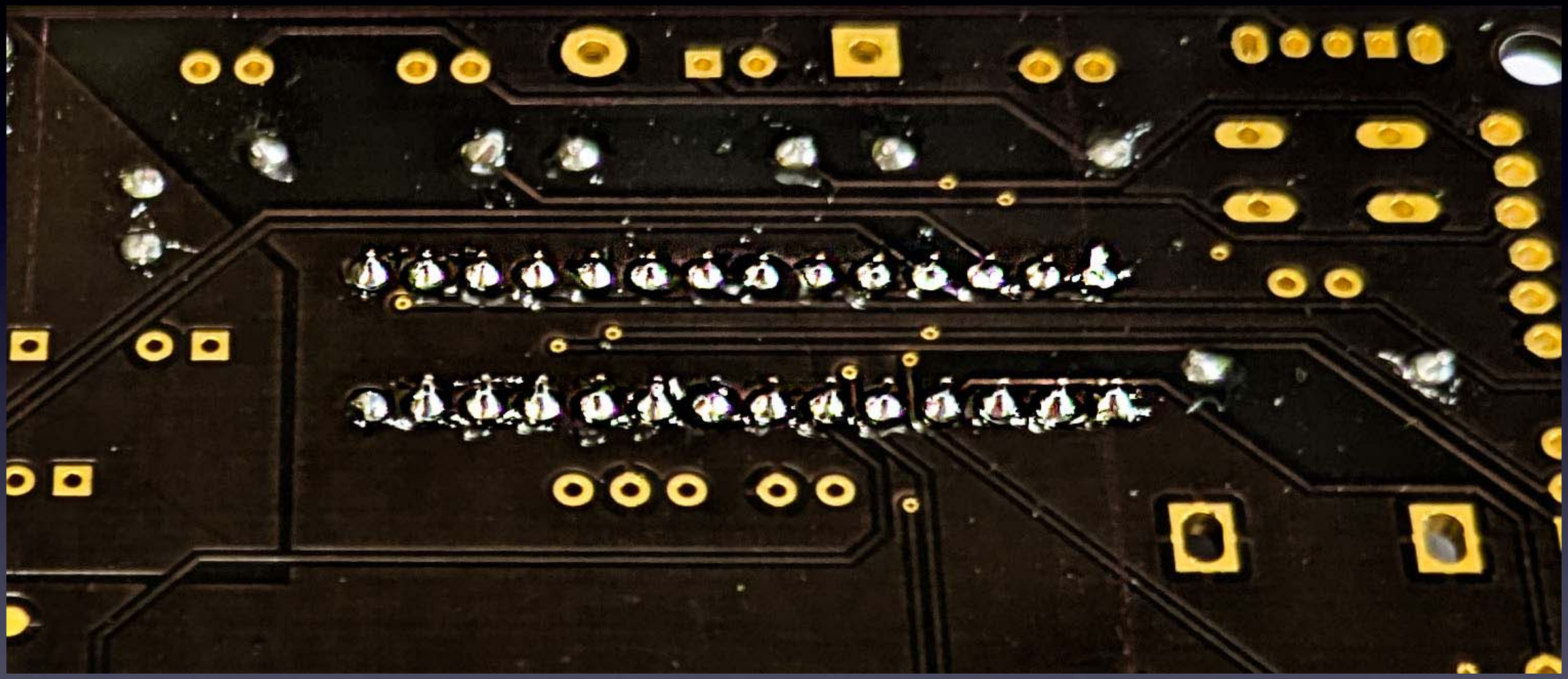


U1: microcontroller socket



bend pins down on two corners,
and solder all 28 leads to the board

U1: microcontroller socket

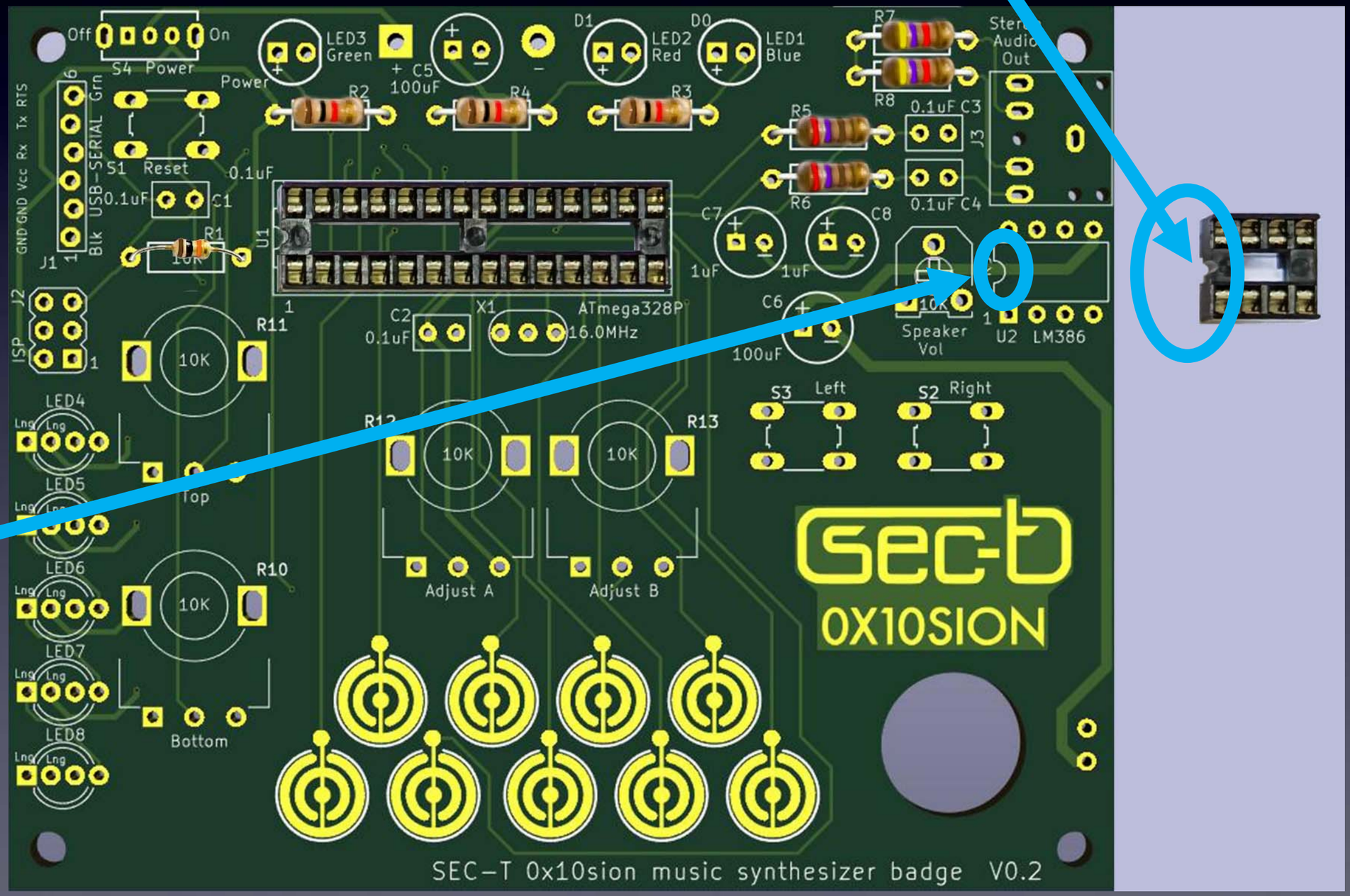


All 28 leads soldered to the board:

→ Notice that each pad is totally covered with solder. ←

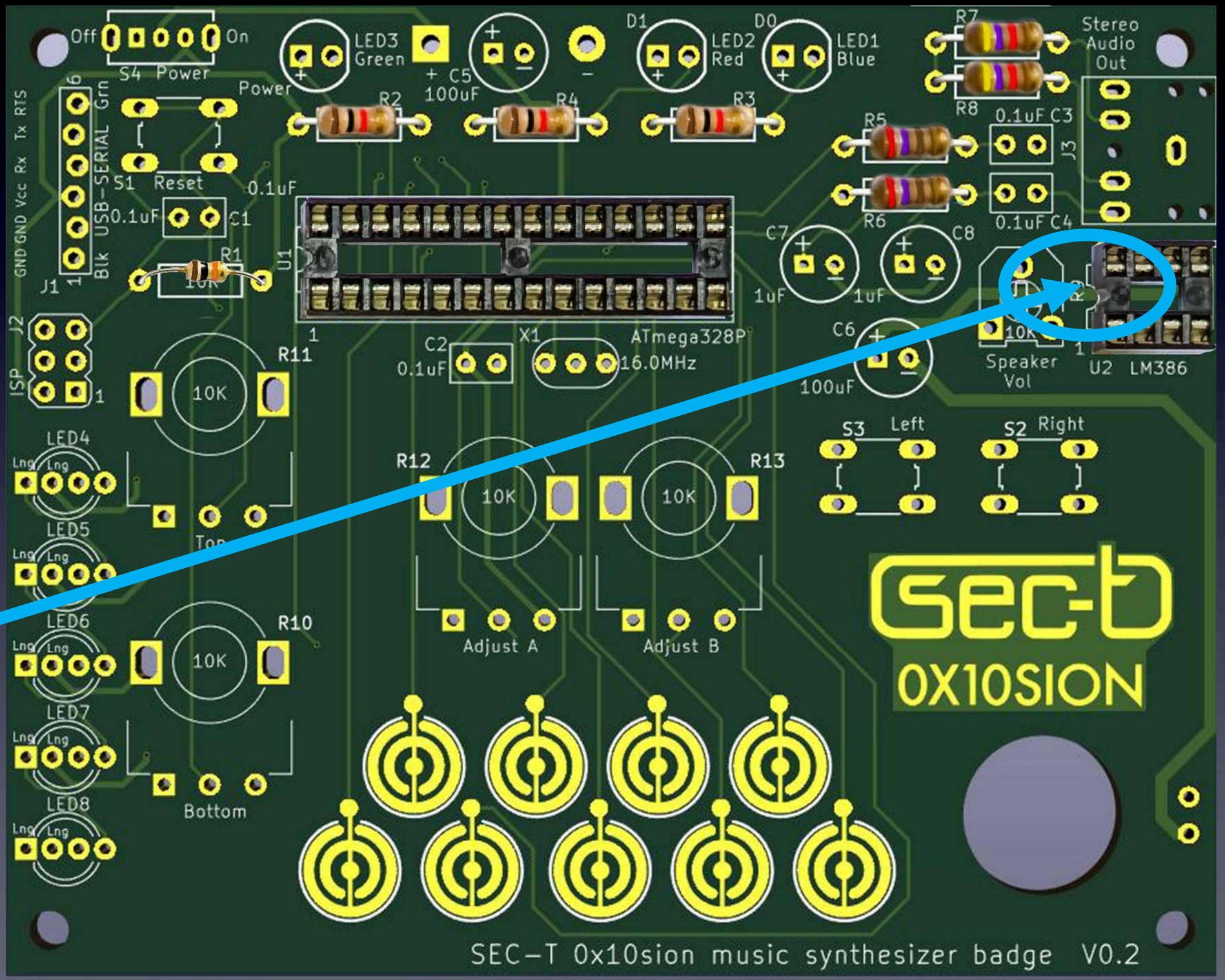
U2: amplifier chip socket

proper orientation

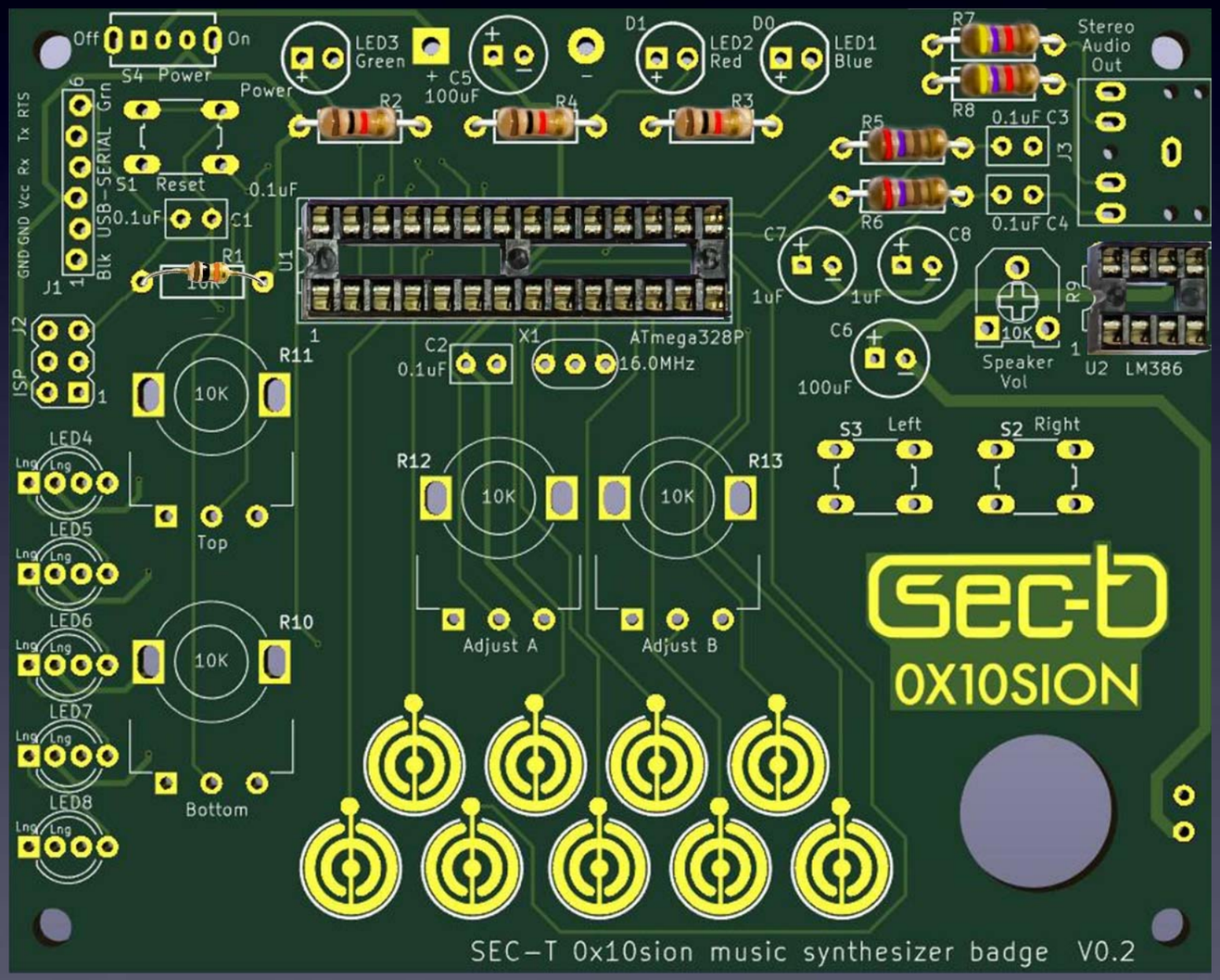


U2: amplifier chip socket

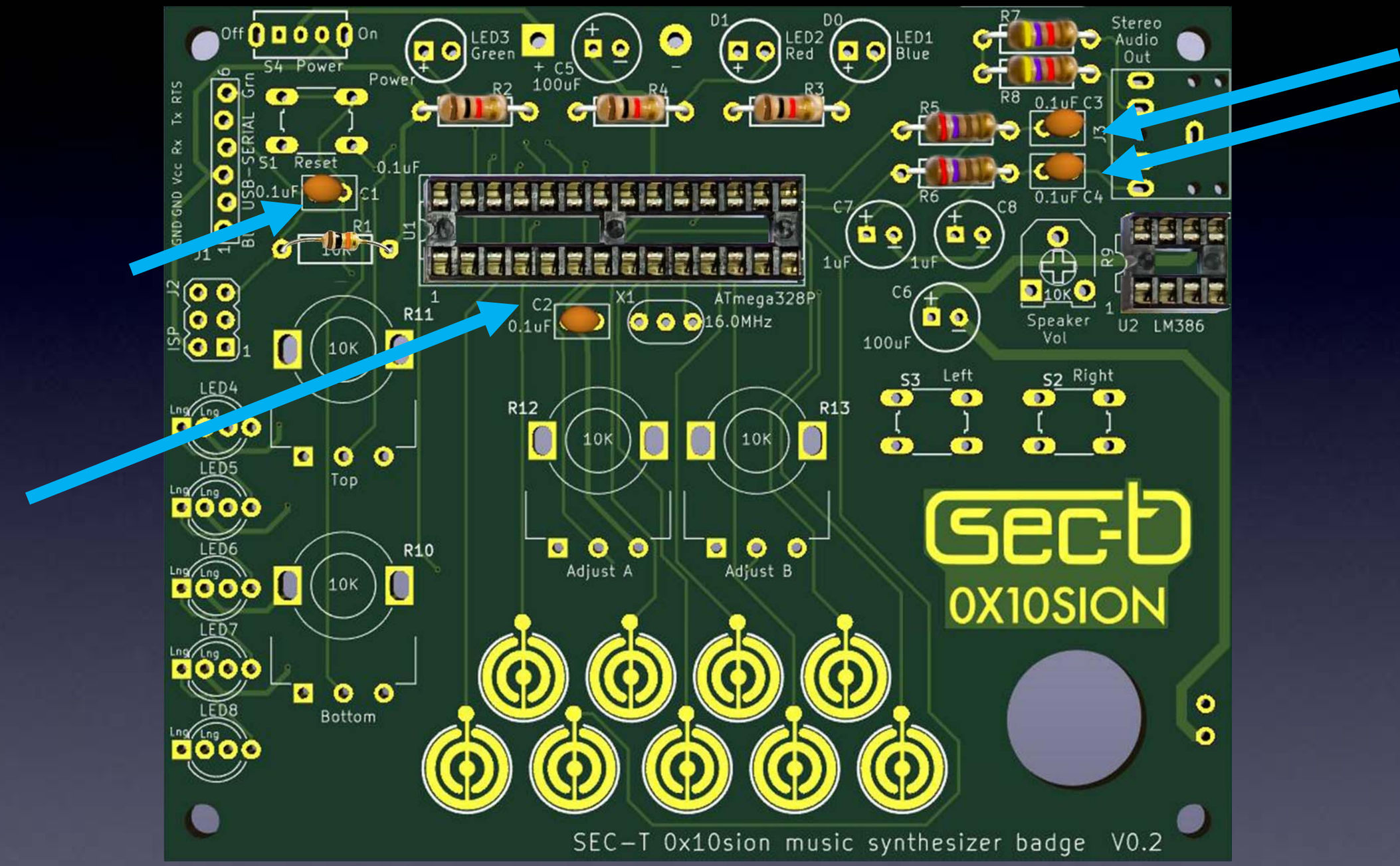
inserted correctly



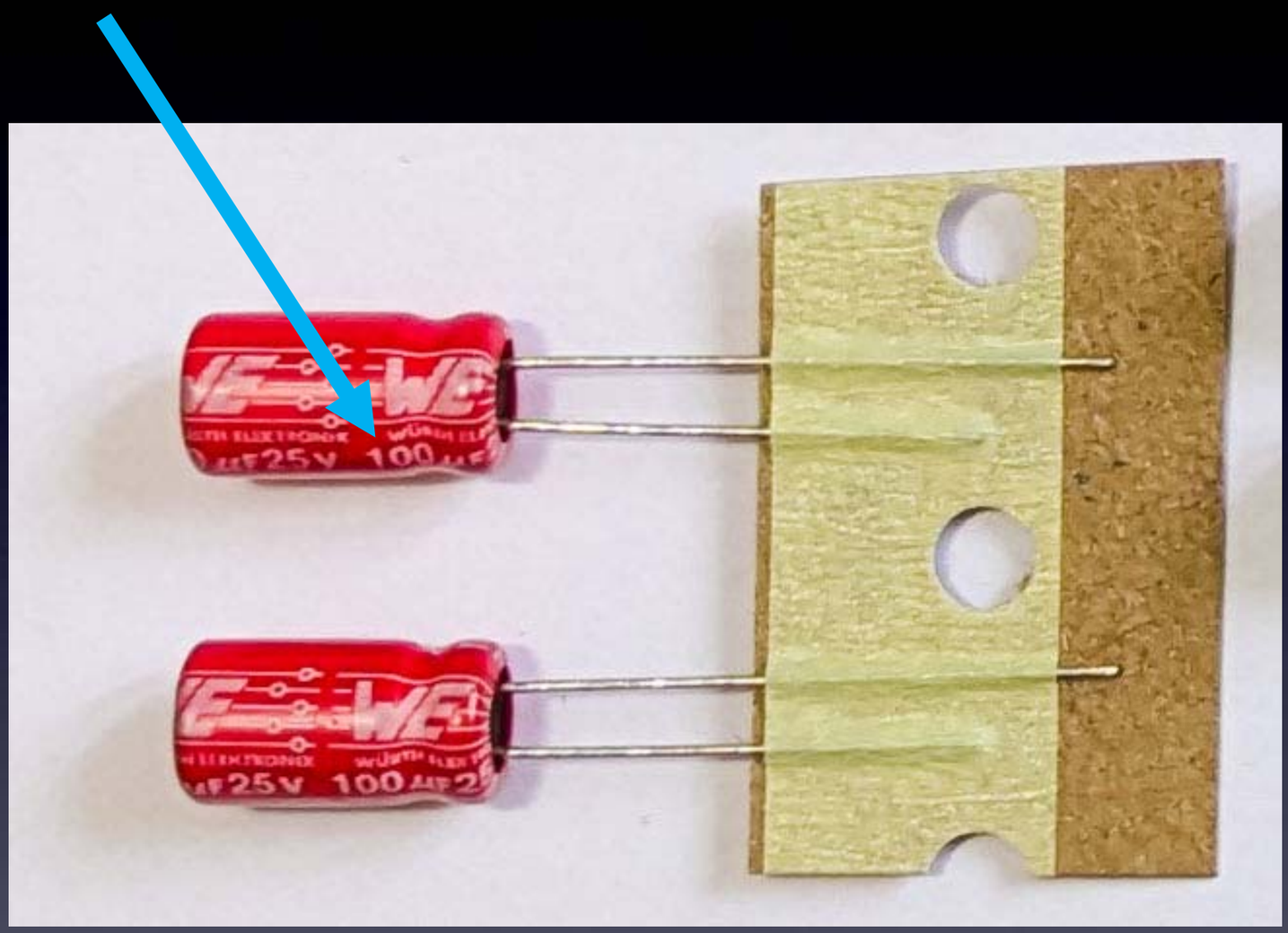
U2: amplifier chip socket



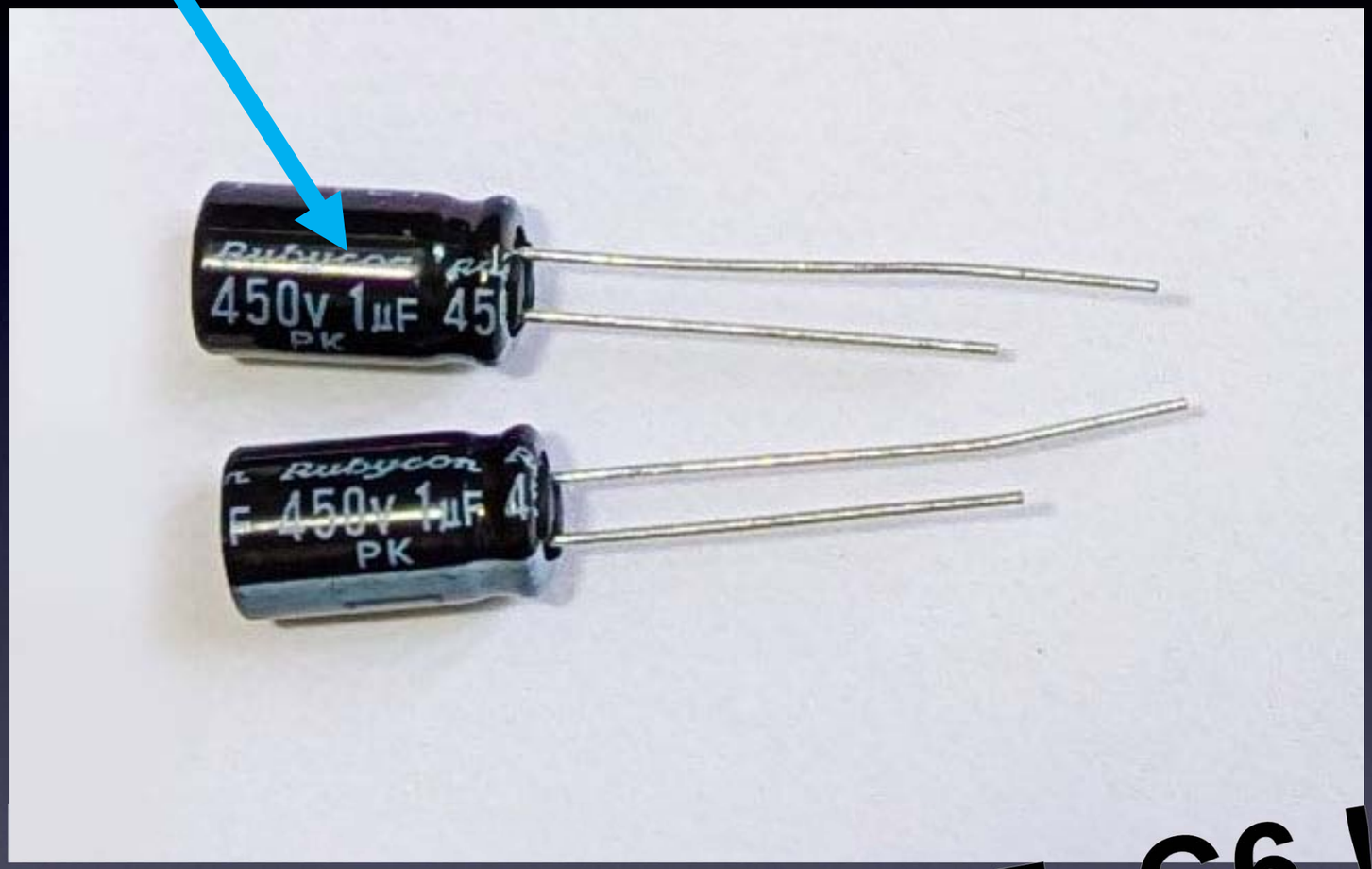
bend pins down on two corners,
and solder all 8 leads to the board


Off
On
S4 Power
LED3 Green
+ C5 100uF
D1
LED2 Red
D0
LED1 Blue
R7
R8
Stereo Audio Out
Power
R2
R4
R3
R5
R6
0.1uF C3
0.1uF C4
J3
S1 Reset
0.1uF
0.1uF C1
R1
10K
U1
C7
C8
1uF
1uF
C6
100uF
R9
10K
Speaker Vol
U2 LM386
J1
GND GND Vcc Rx Tx RTS
USB-SERIAL Grn
J2
ISP
C2
0.1uF
X1
16.0MHz
ATmega328P
R11
10K
S3 Left
S2 Right
LED4
LED5
LED6
LED7
LED8
Top
Bottom
R10
R12
R13
Adjust A
Adjust B
Sec-t
0X10SION
SEC-T 0x10sion music synthesizer badge V0.2

C1, C2, C3, C4

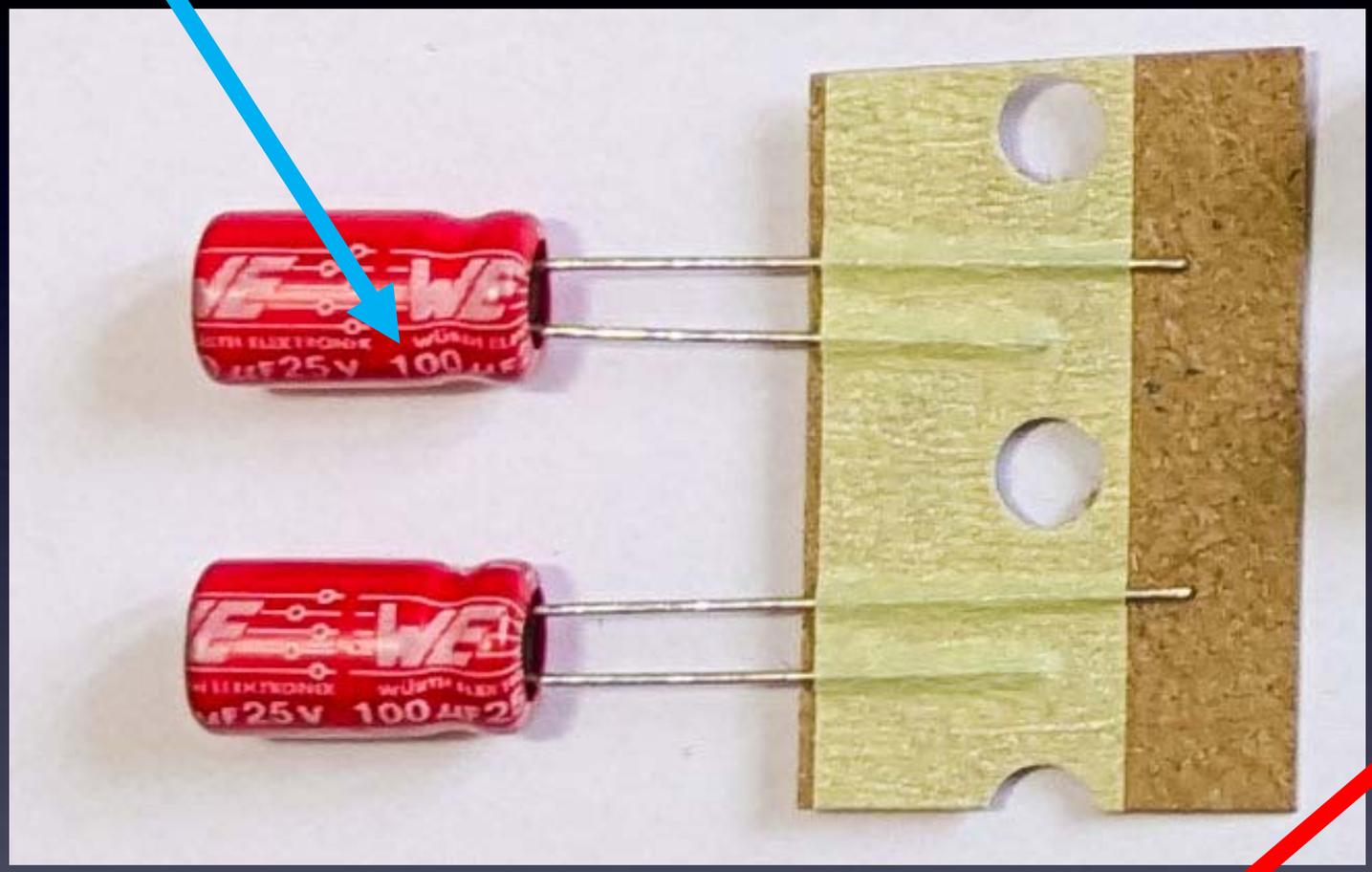


C5, C6: 100uF

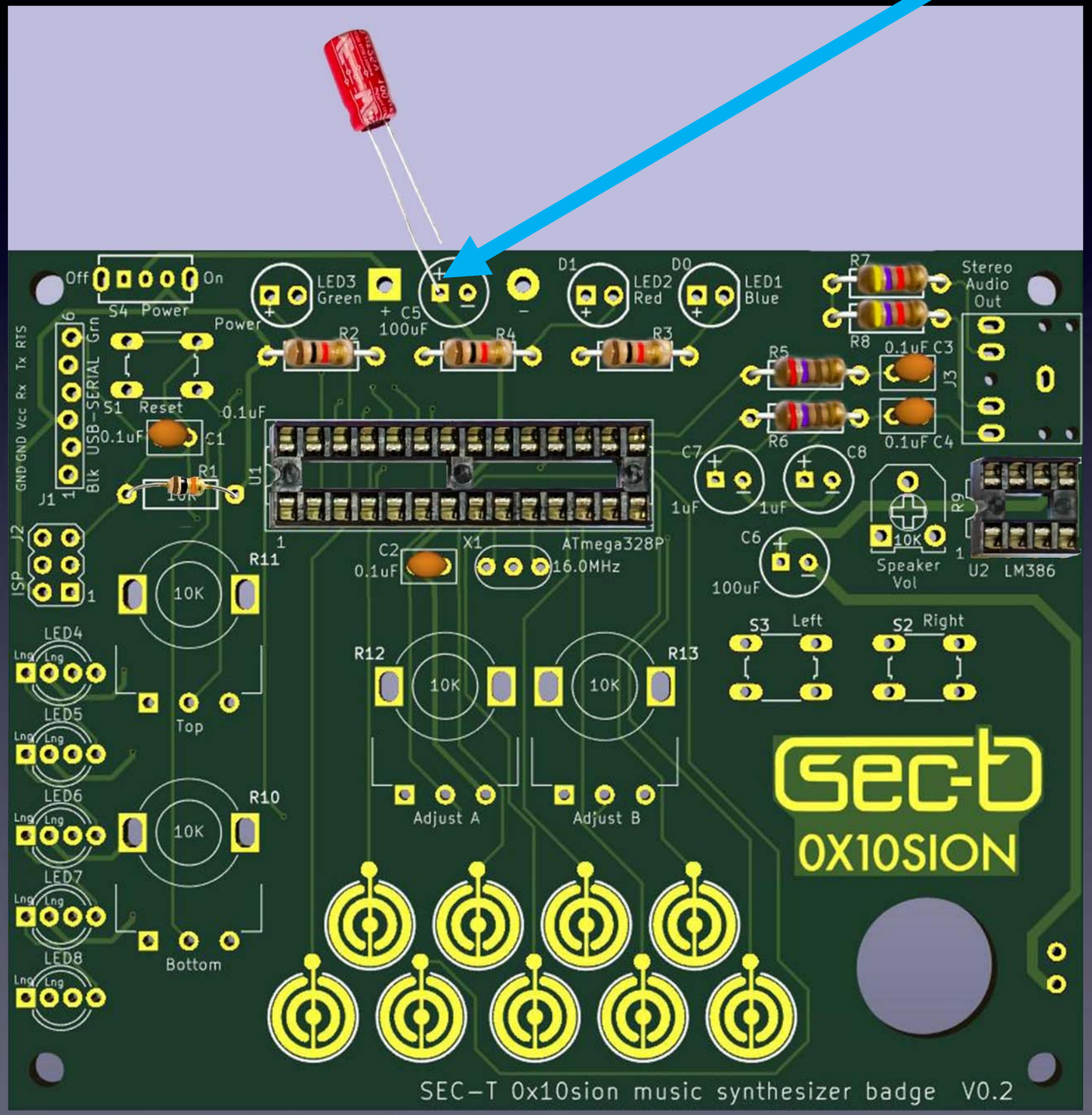


Different than C5, C6 !

C7, C8: 1uF

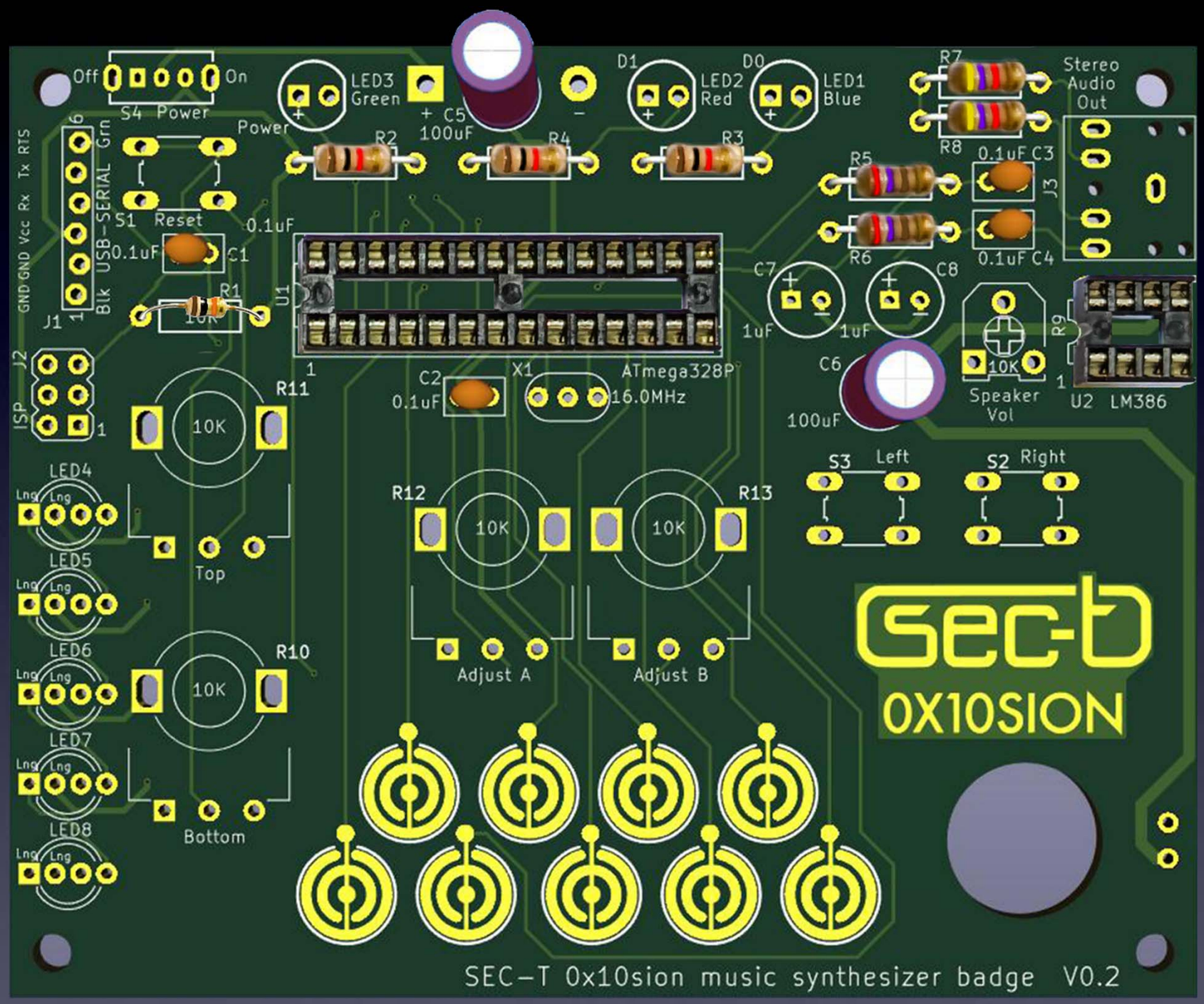


C5, C6: 100uF

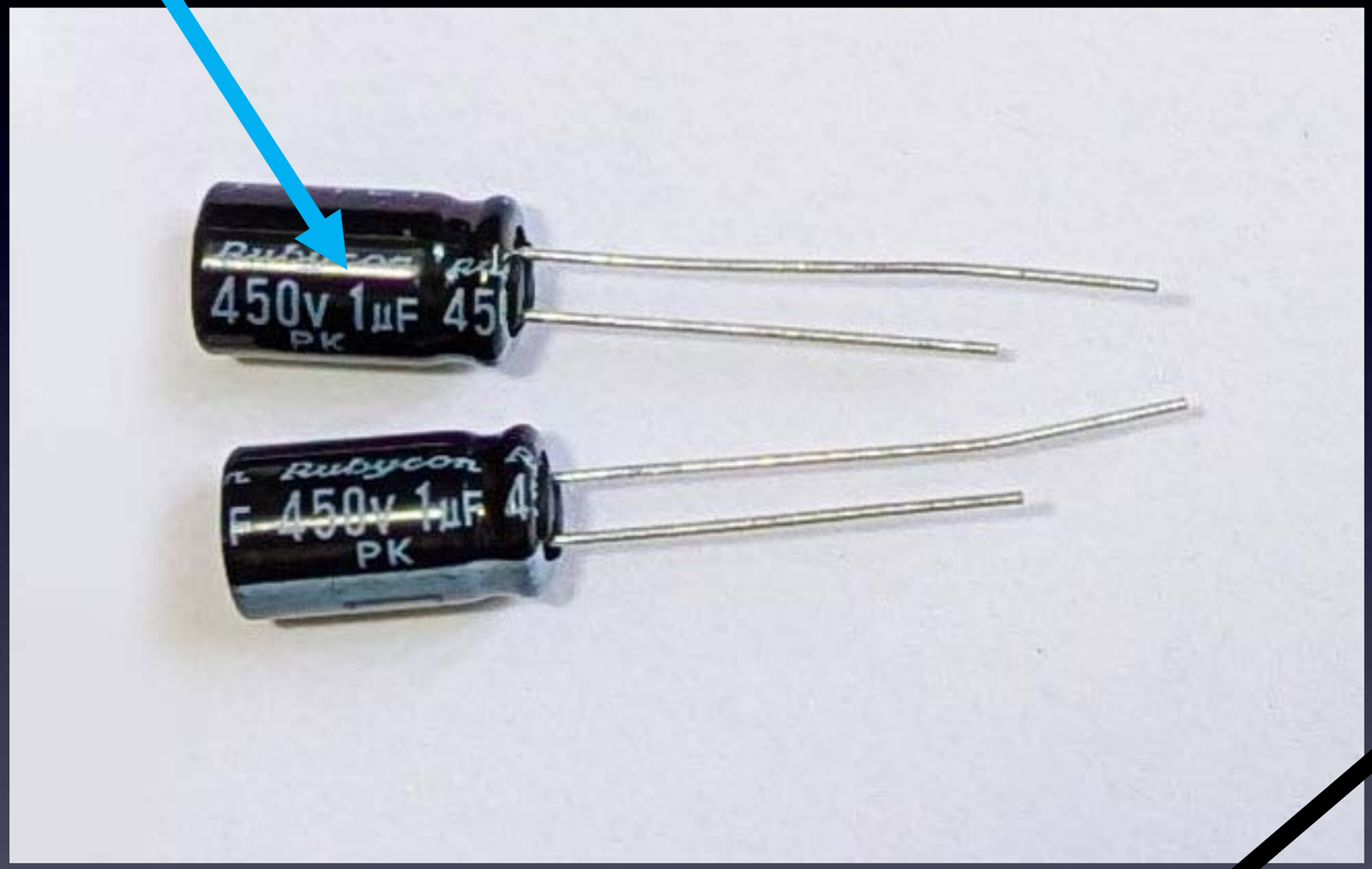


C5, C6: Long Lead “+”

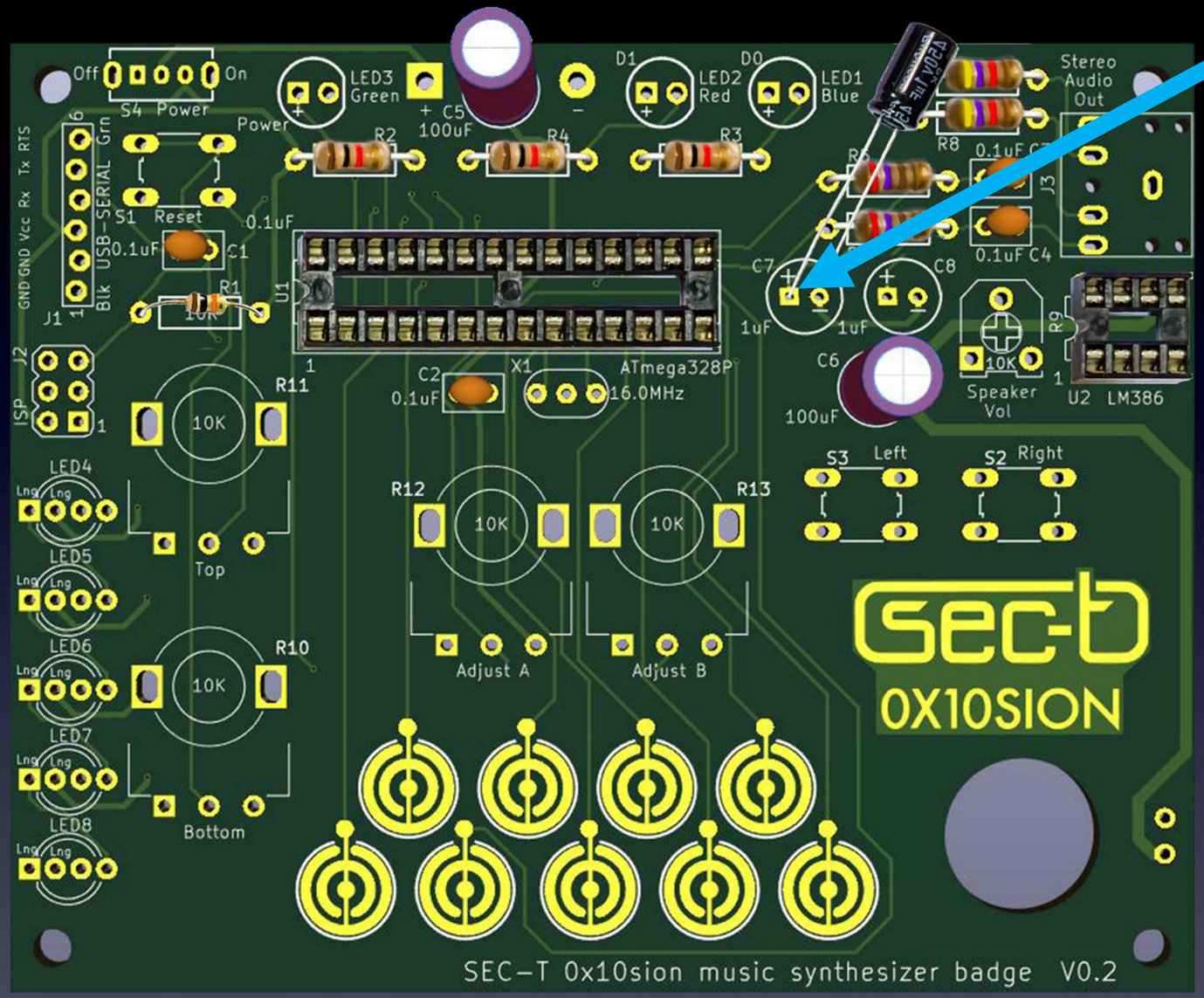
Use 100uF !!



C5, C6: 100uF – soldered to board

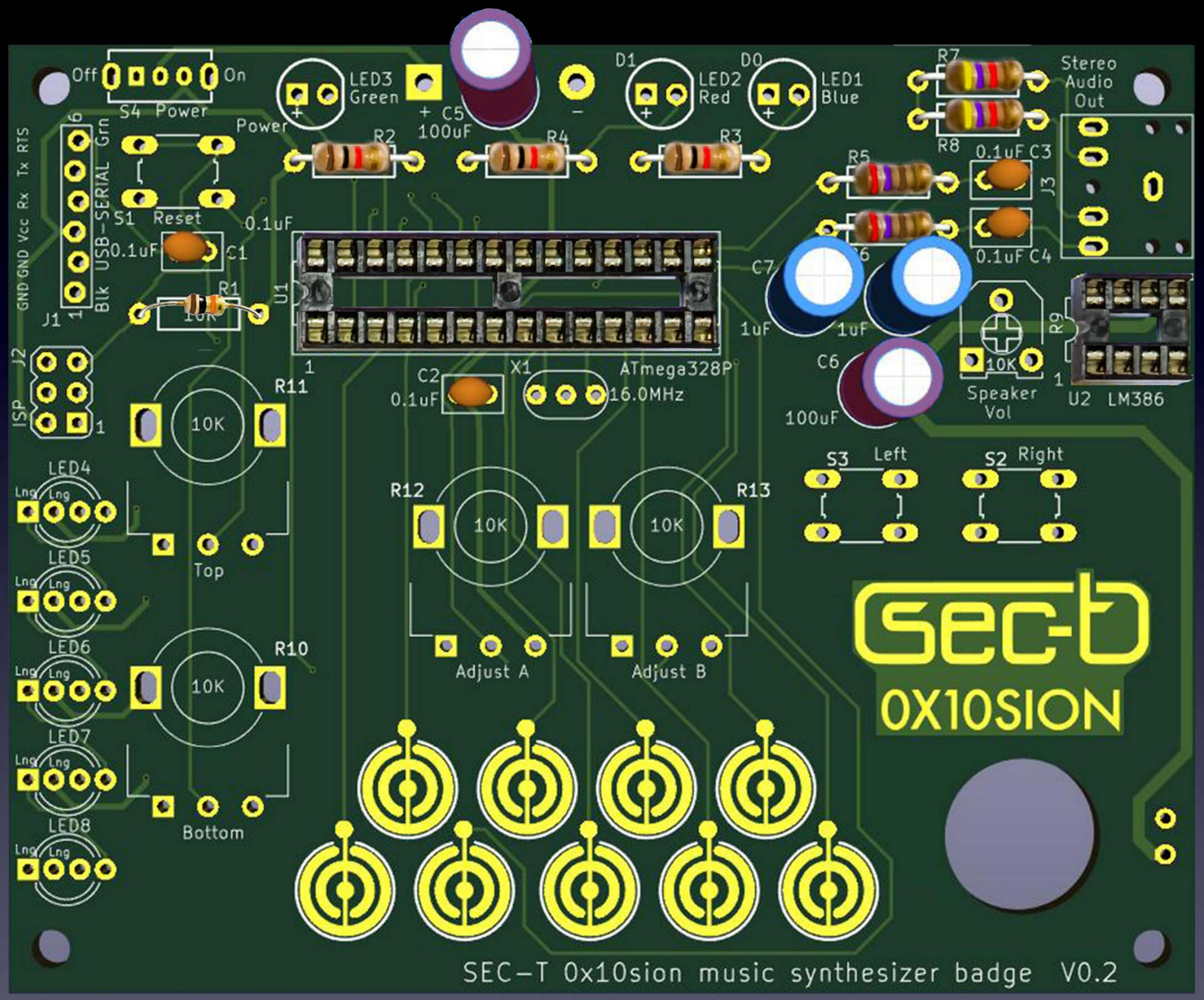


C7, C8: 1uF



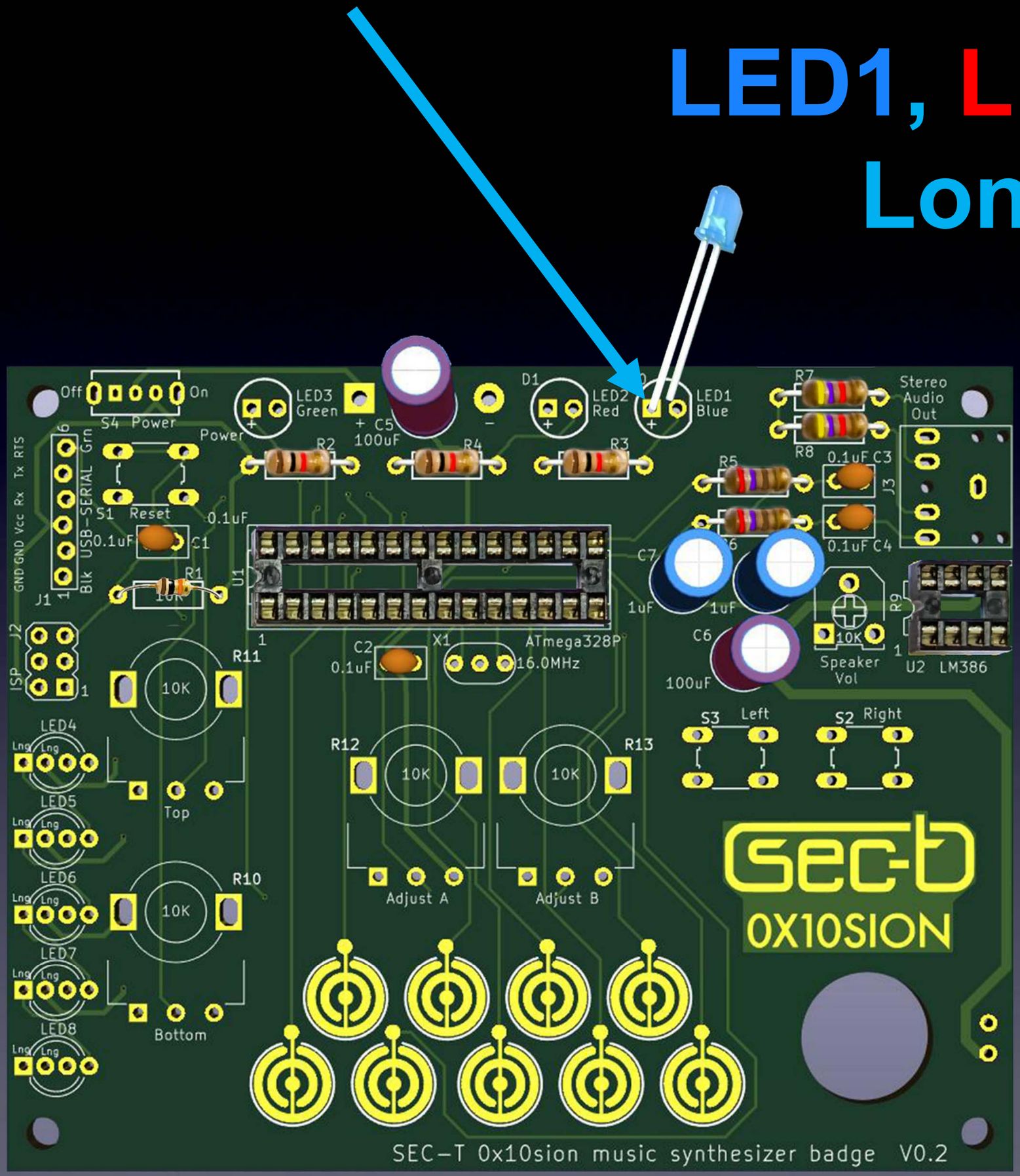
C7, C8: Long Lead “+”

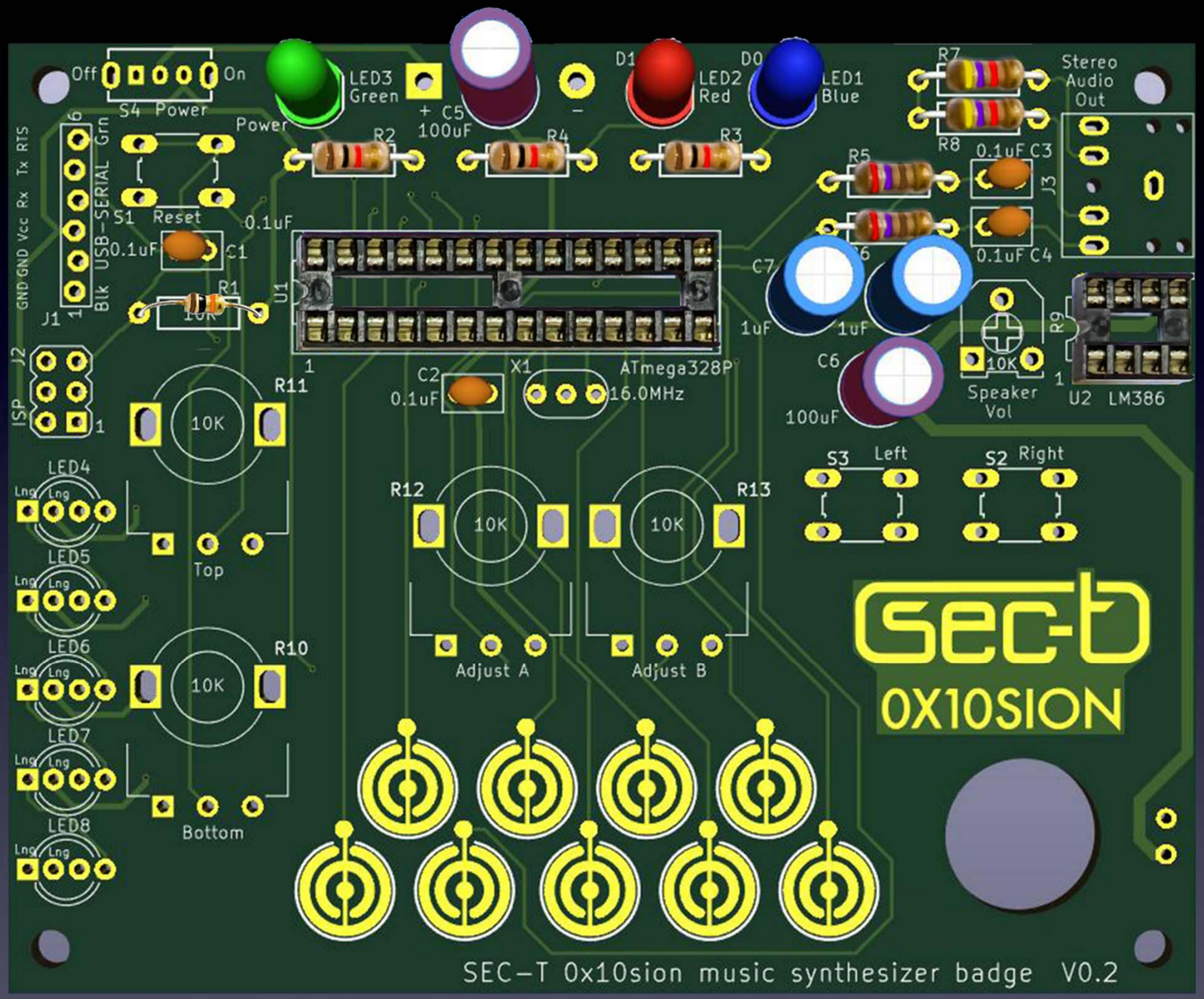
Use 1uF !!



C7, C8: 1uF – soldered to board

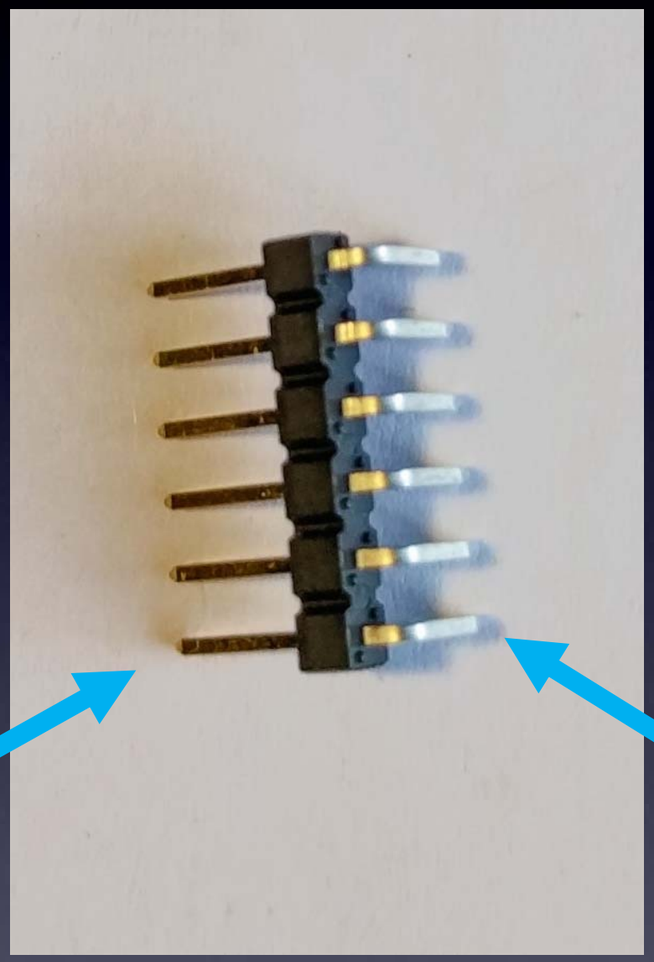
LED1, LED2, LED3:
Long Lead “+”





LED3, LED2, LED1

Green, Red, Blue – soldered to board



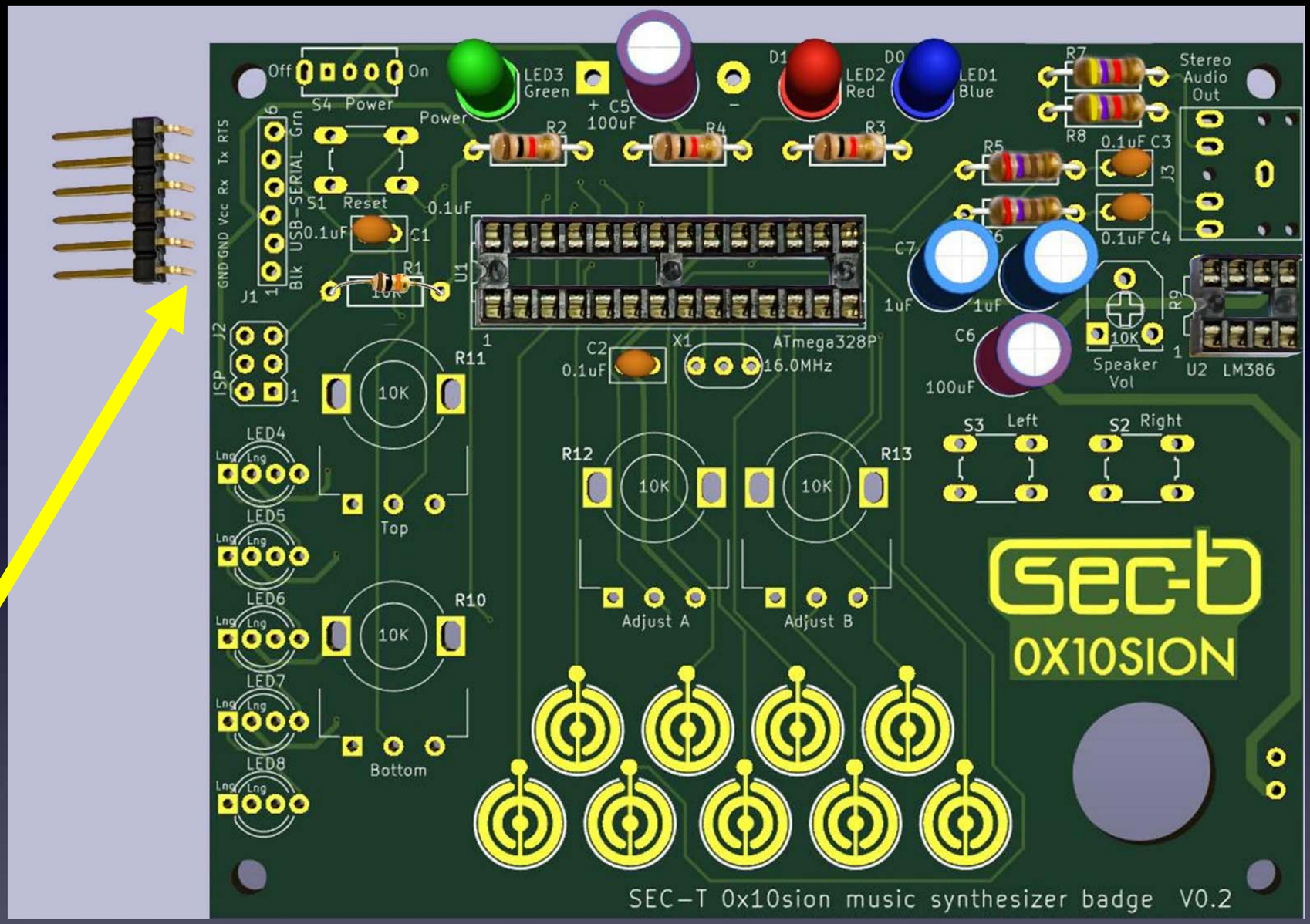
long leads

short leads

J1

Short leads into board

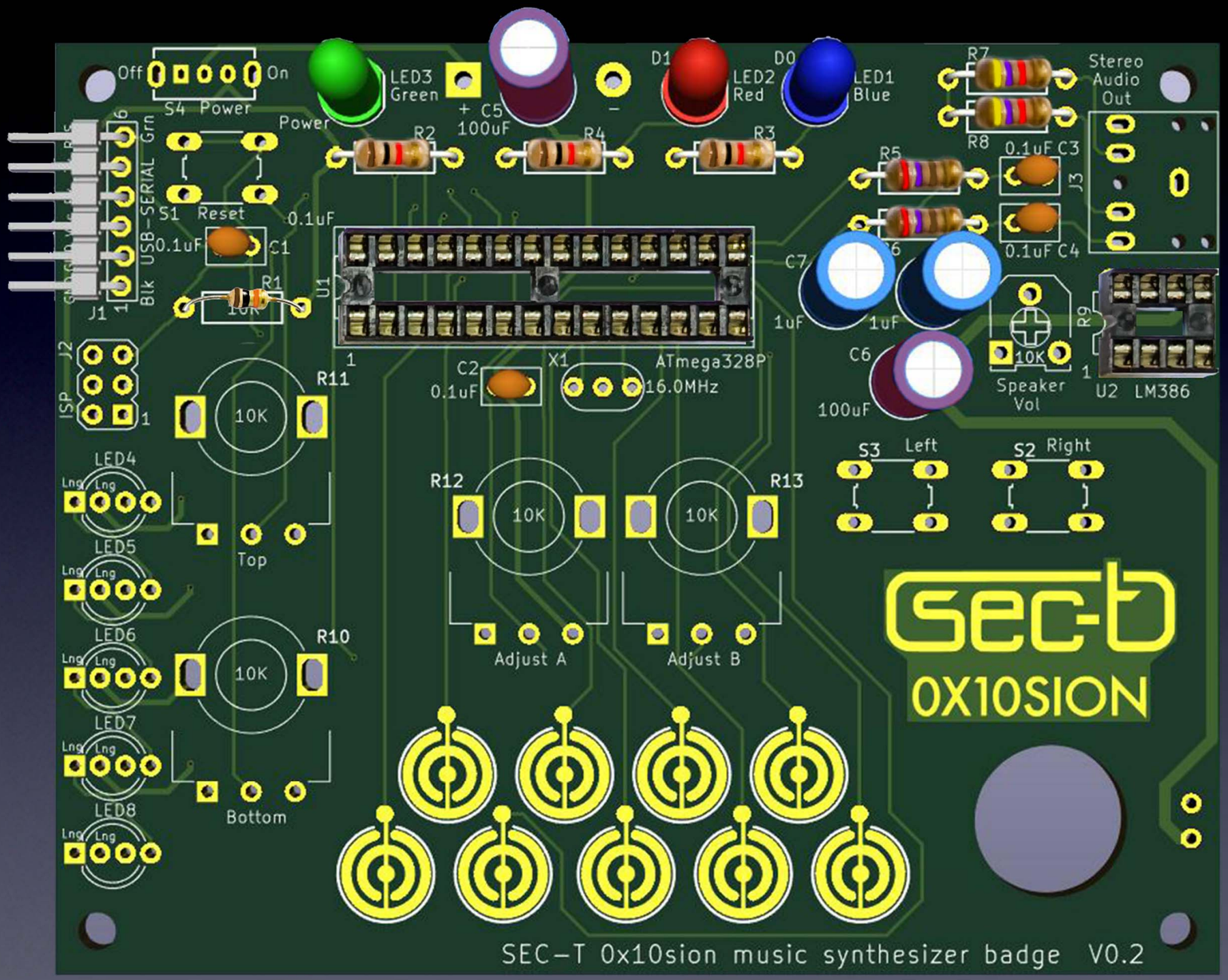
J1

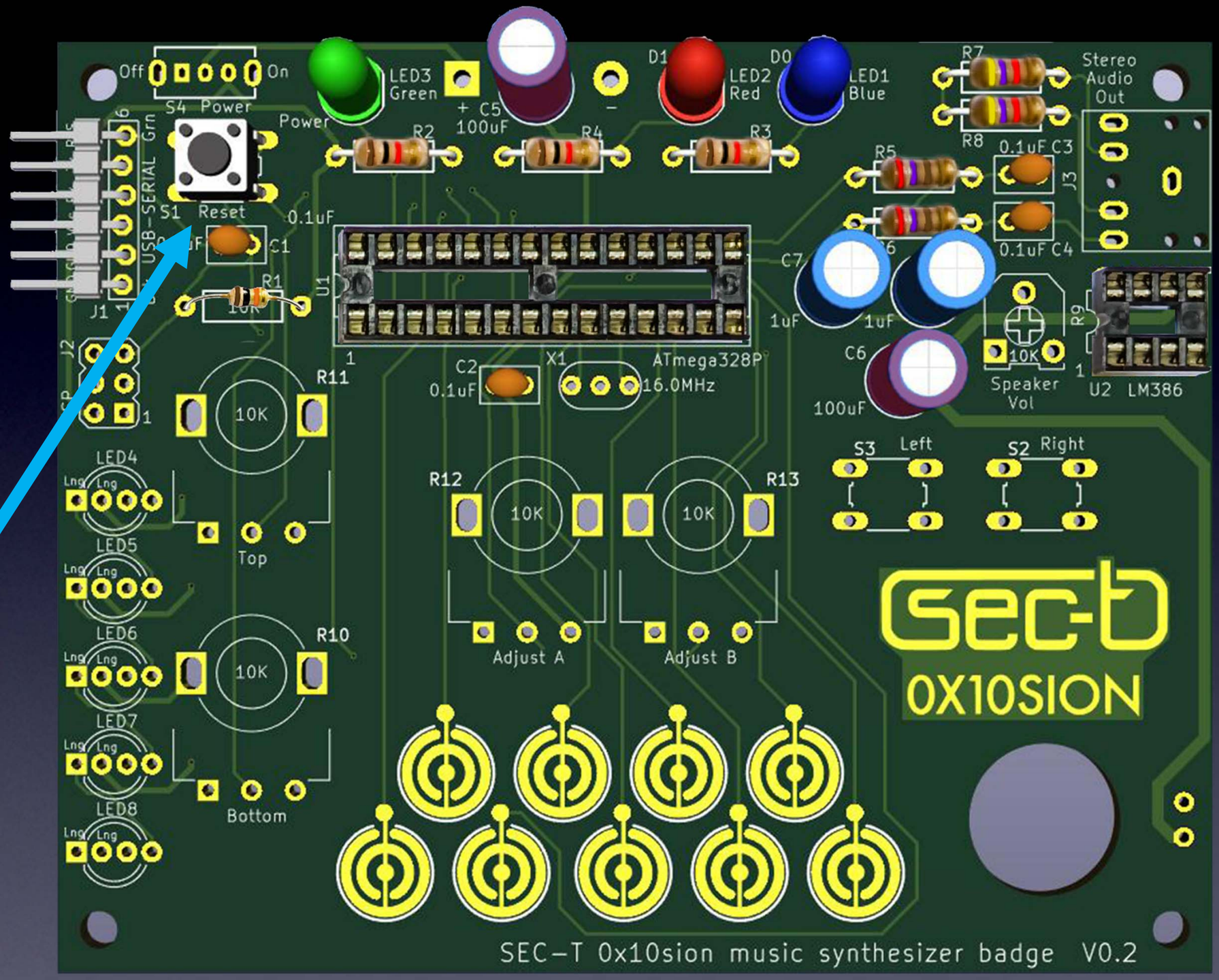


IMPORTANT:
short leads
go into the board

→ long leads sticking out from board

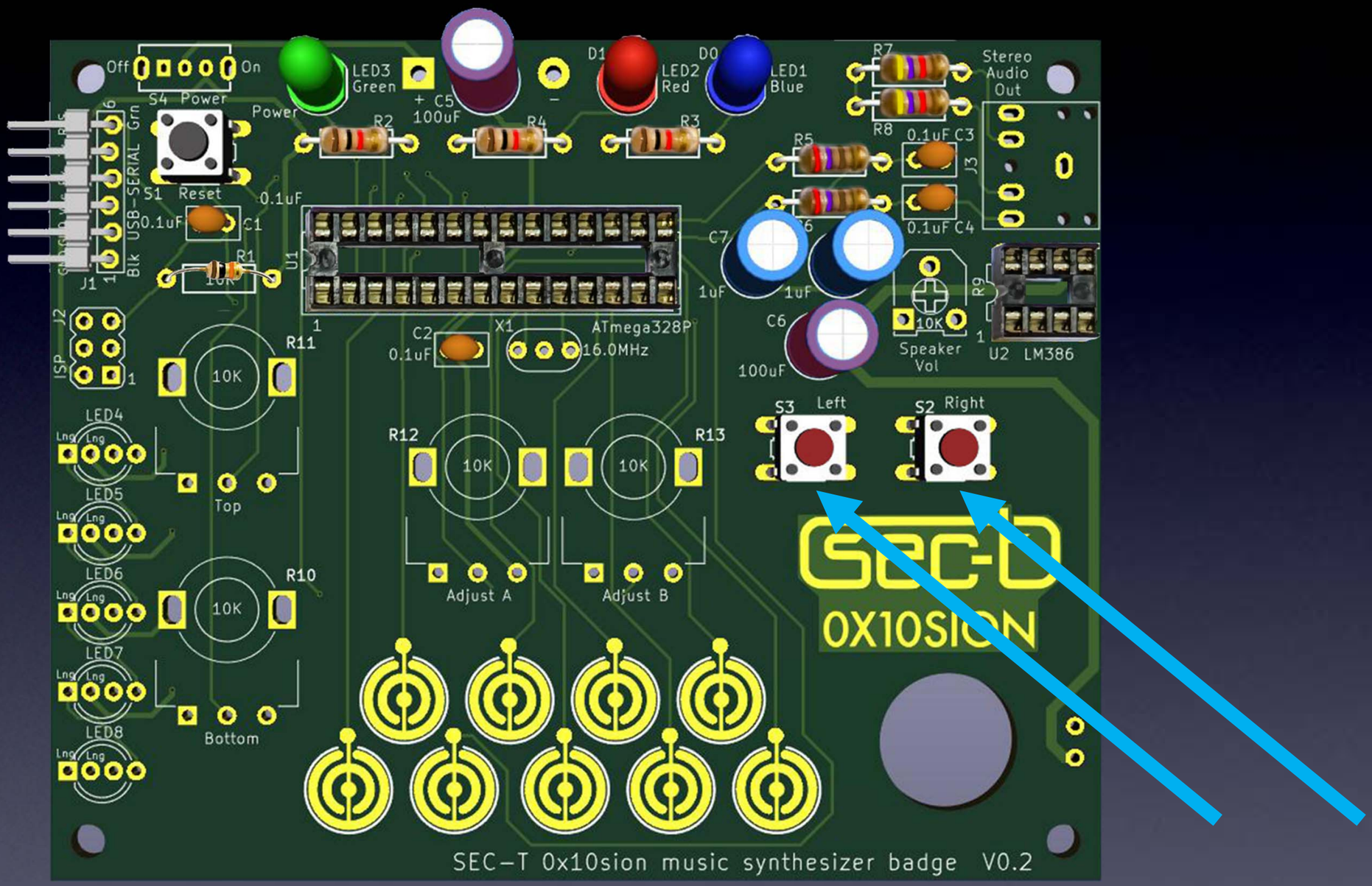
J1

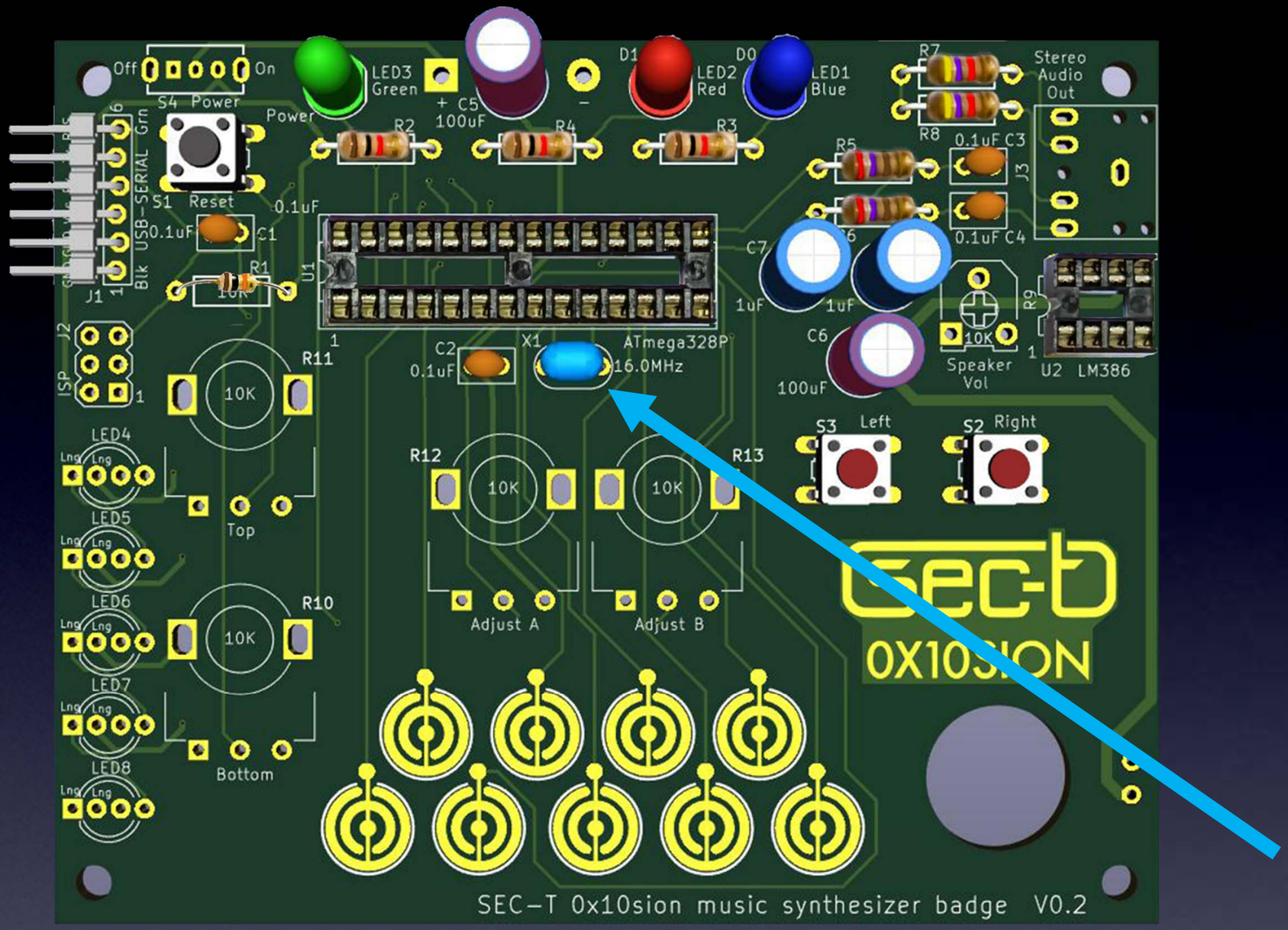

Off
On
S4 Power
Power
LED3
Green
+ C5
100uF
D1
LED2
Red
D0
LED1
Blue
R7
R8
Stereo
Audio
Out
R2
R4
R3
R5
R6
0.1uF C3
0.1uF C4
J3
S1 Reset
0.1uF
0.1uF C1
R1
10K
J1
Blk USB-SERIAL Grn
U1
C7
1uF
1uF
C6
100uF
10K
Speaker
Vol
R9
U2 LM386
C2
0.1uF
X1
ATmega328P
16.0MHz
J2
ISP
R11
10K
S3 Left
S2 Right
LED4
LED5
LED6
LED7
LED8
Top
Bottom
R10
10K
R12
10K
R13
10K
Adjust A
Adjust B
sec-t
0X10SION
SEC-T 0x10sion music synthesizer badge V0.2



S1: black Reset button

S2, S3: Red buttons

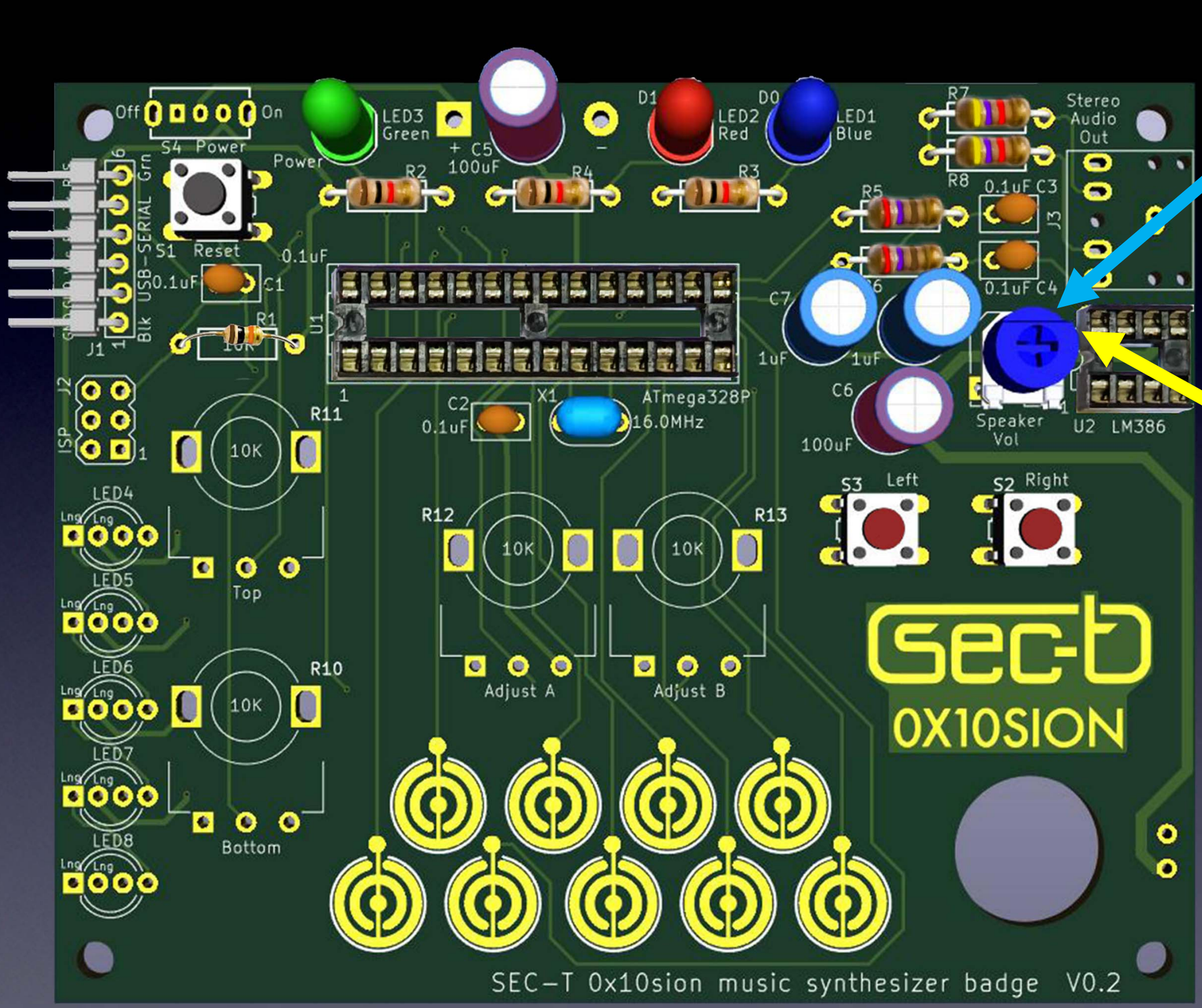




X1

The orientation of X1 does not matter.

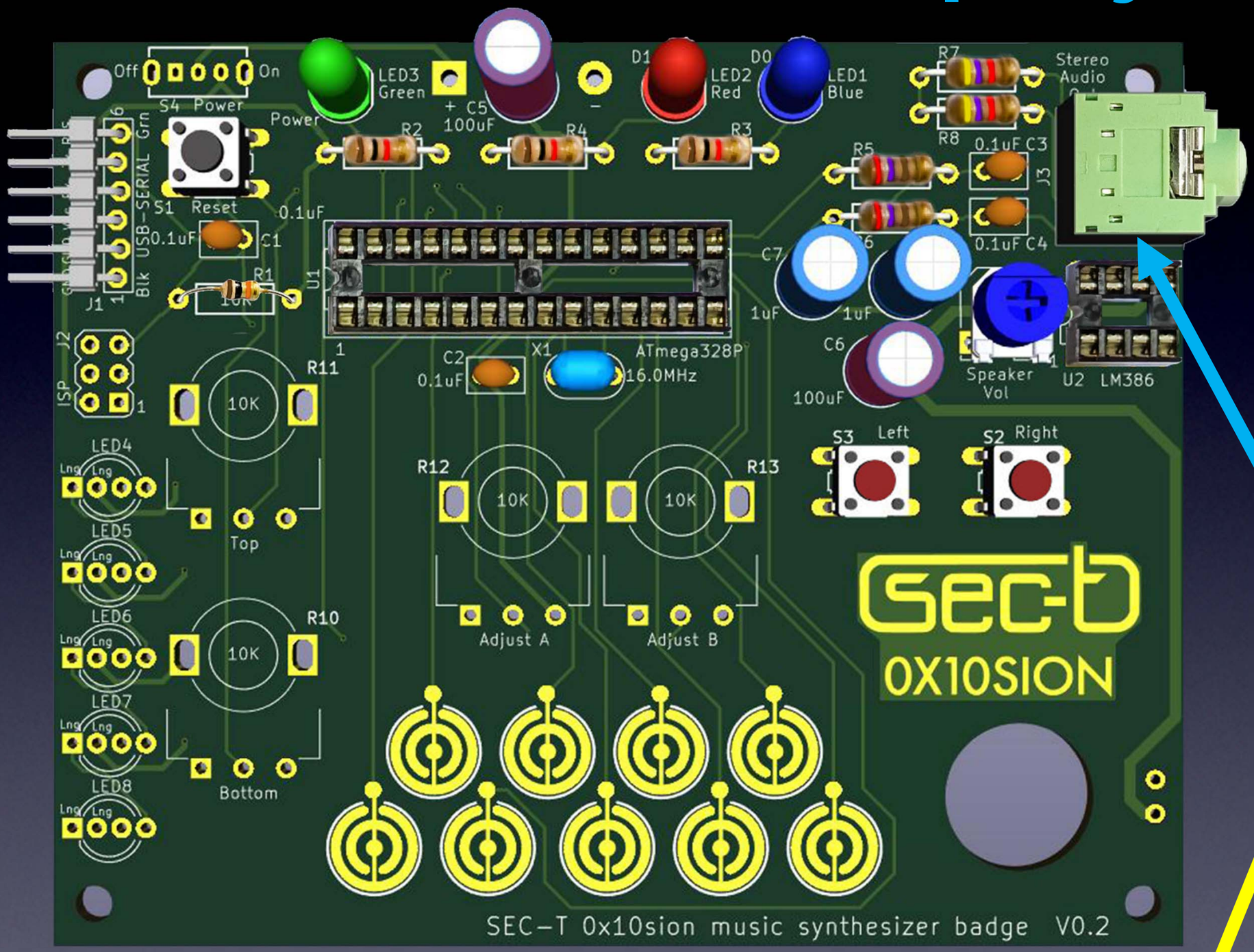
R9: volume control



If necessary, rotate the blue top so that it looks like this photo *(rotated half-way)*

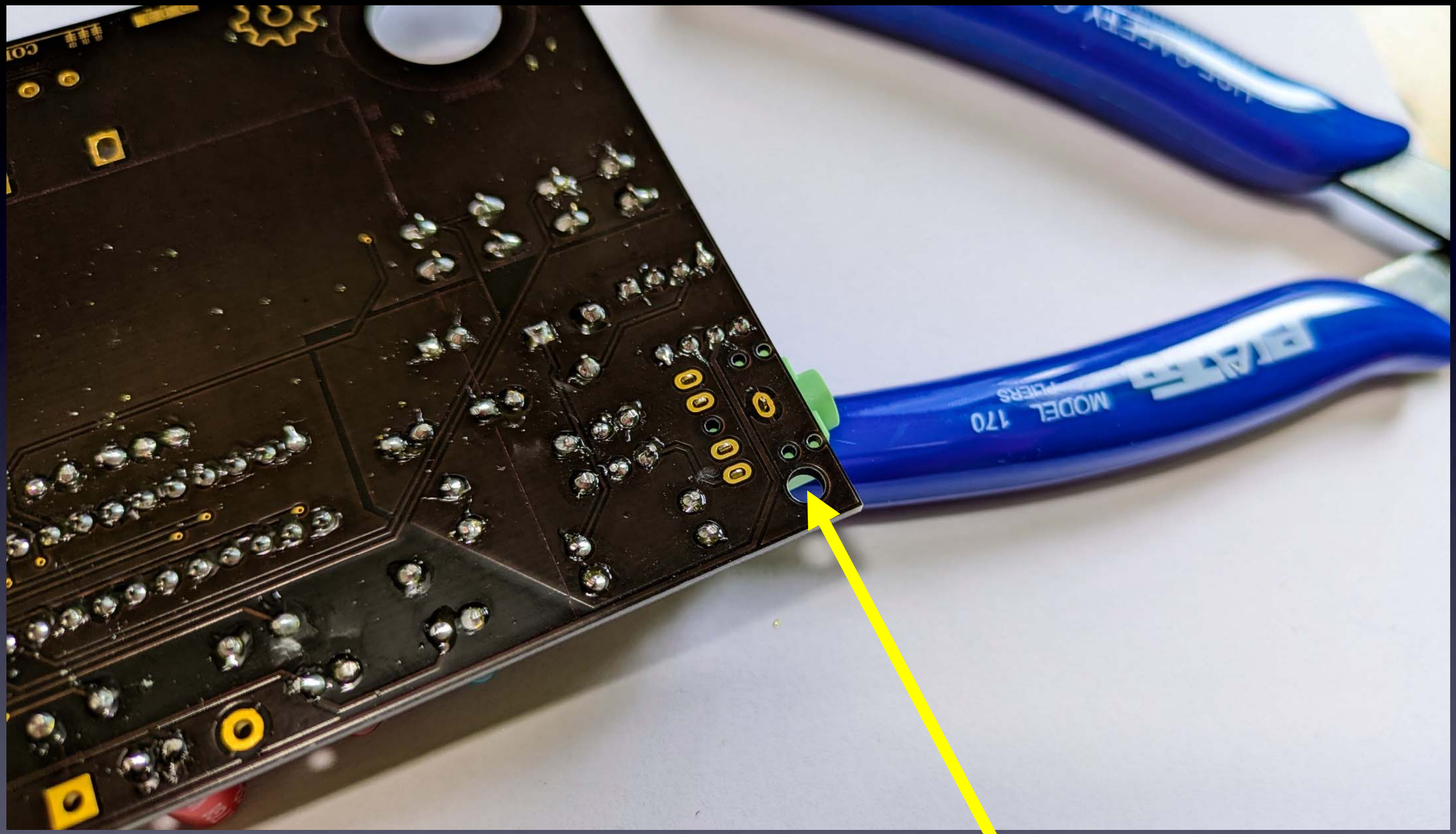
NOTE: R9 only controls the volume to the Speaker, not to Stereo Audio Out (J3)

J3: Stereo Audio Output jack



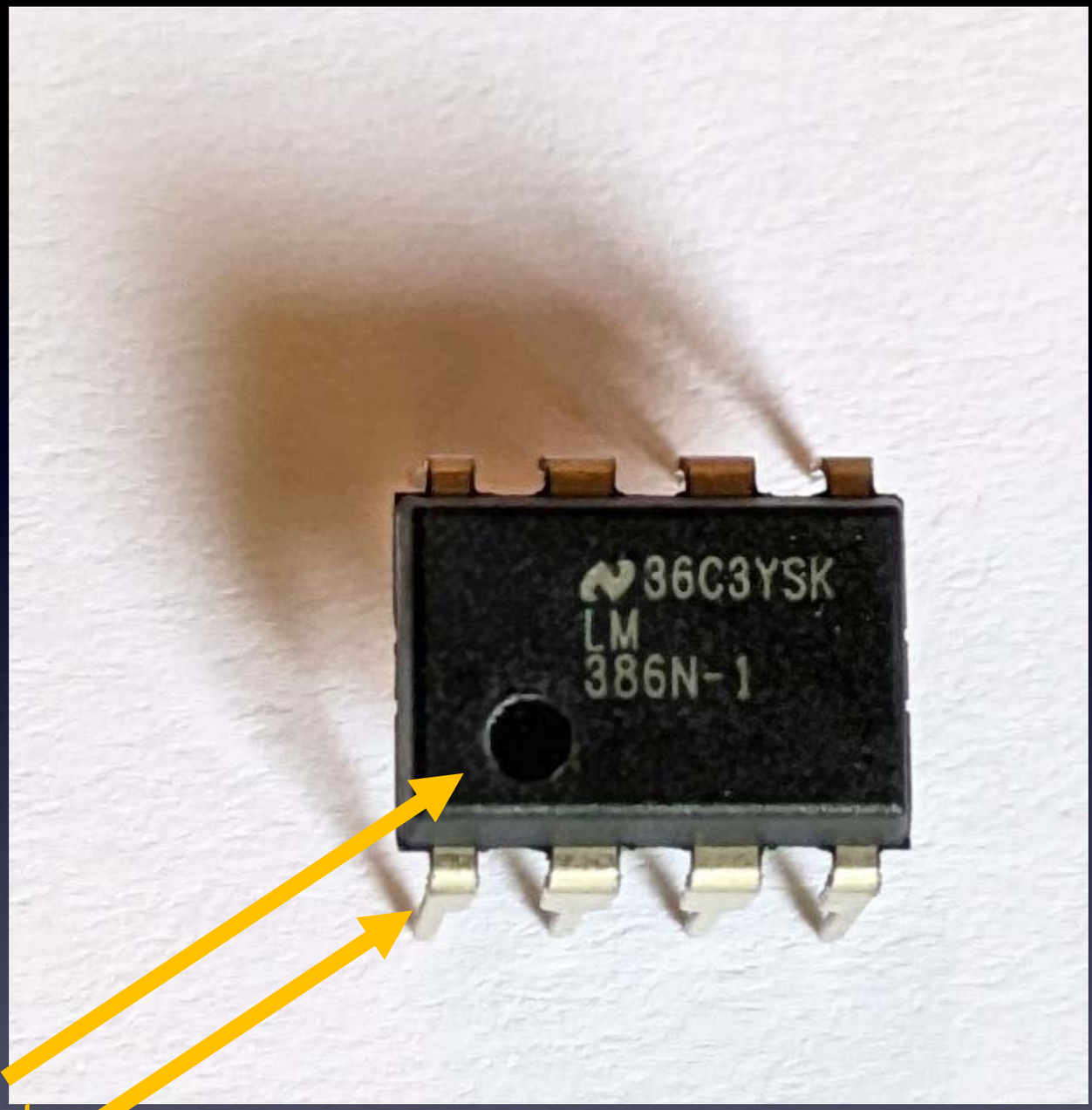
→ See next photo for how to solder this part…

J3: Stereo Audio Output jack



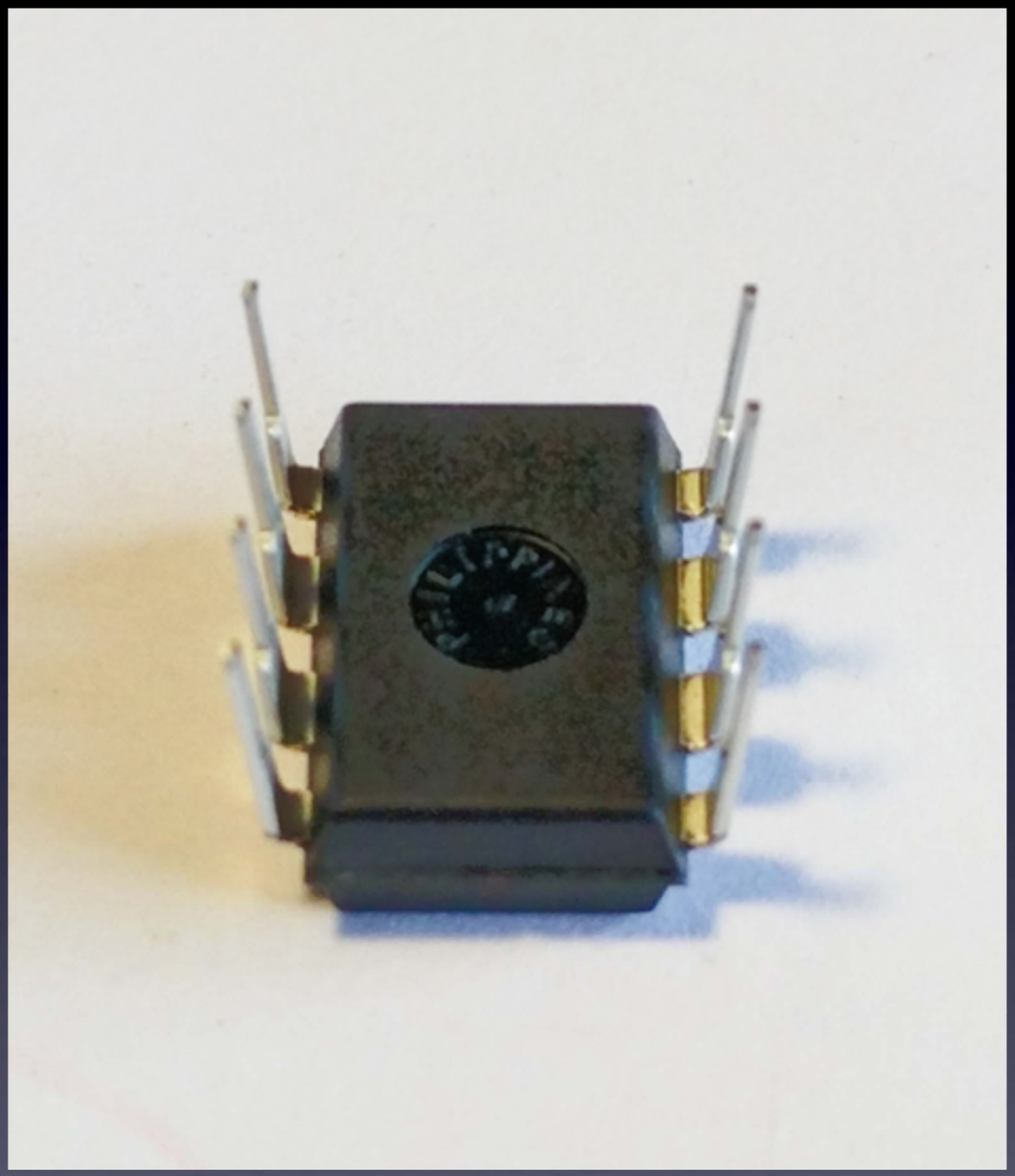
Resting on the wire cutters, J3 is held in place here, so you can solder all 5 leads.

U2: amplifier chip



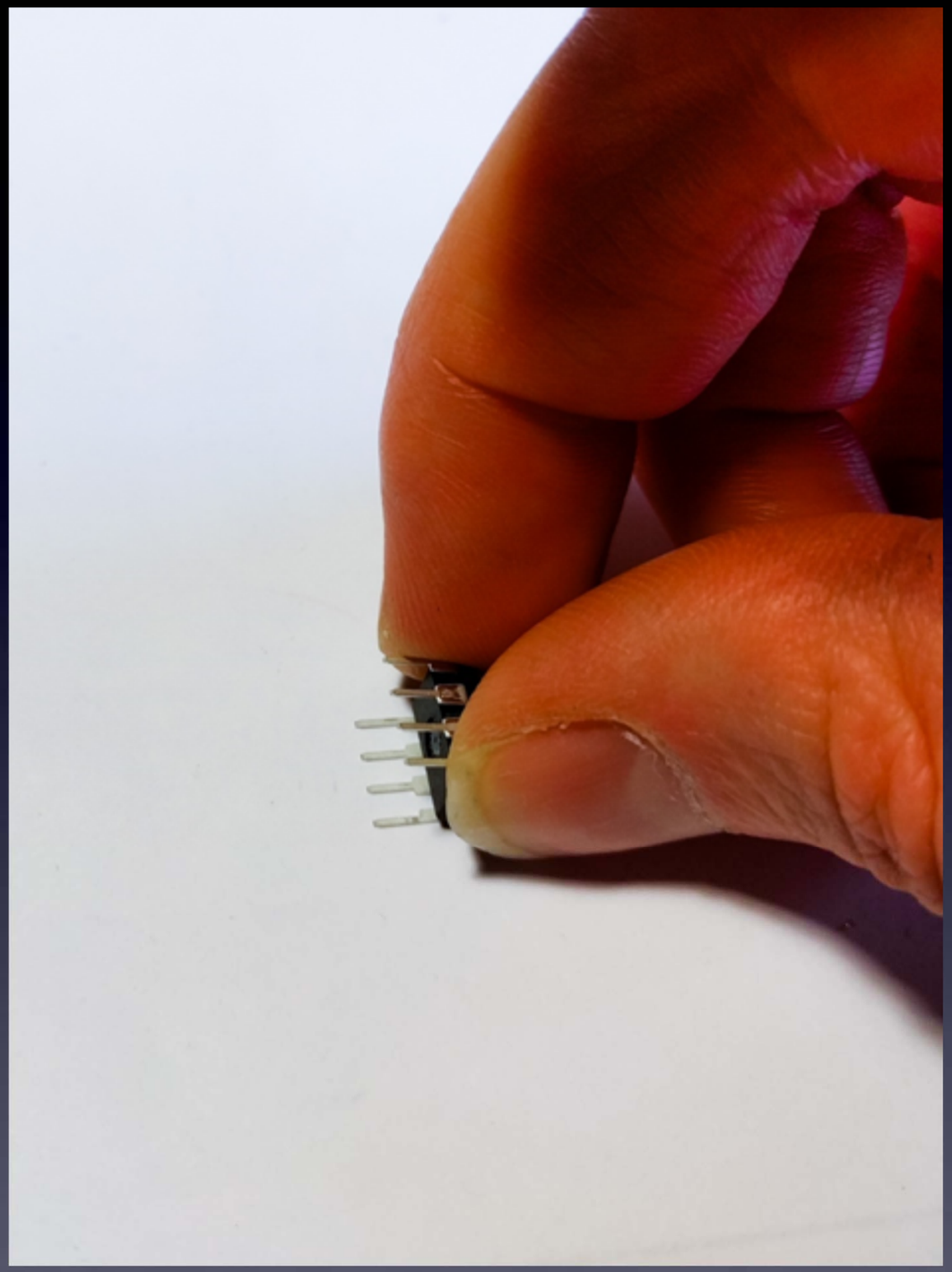
Indented black dot
Pin 1

U2: amplifier chip



When chips are new,
their pins are bent out.

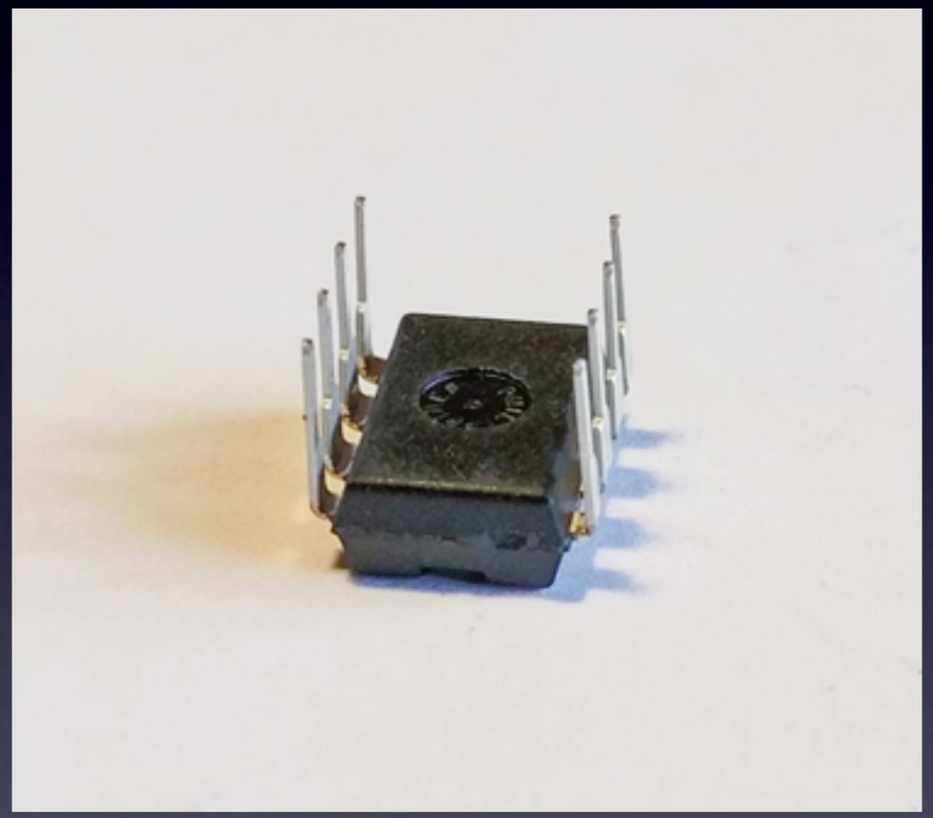
U2: amplifier chip



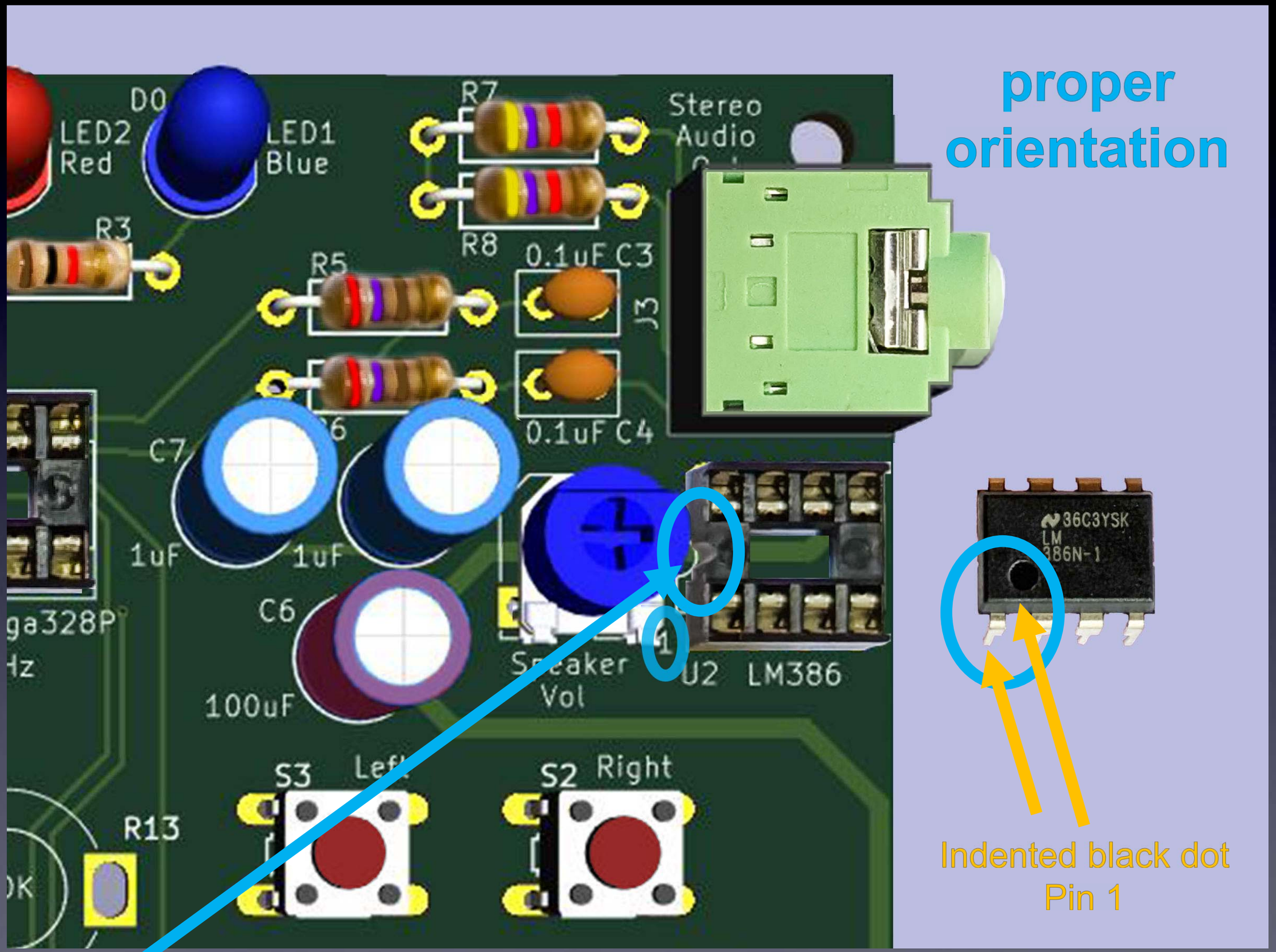
We need the pins bent *straight and parallel*.
Use your work table to (gently) bend the leads.

U2: amplifier chip

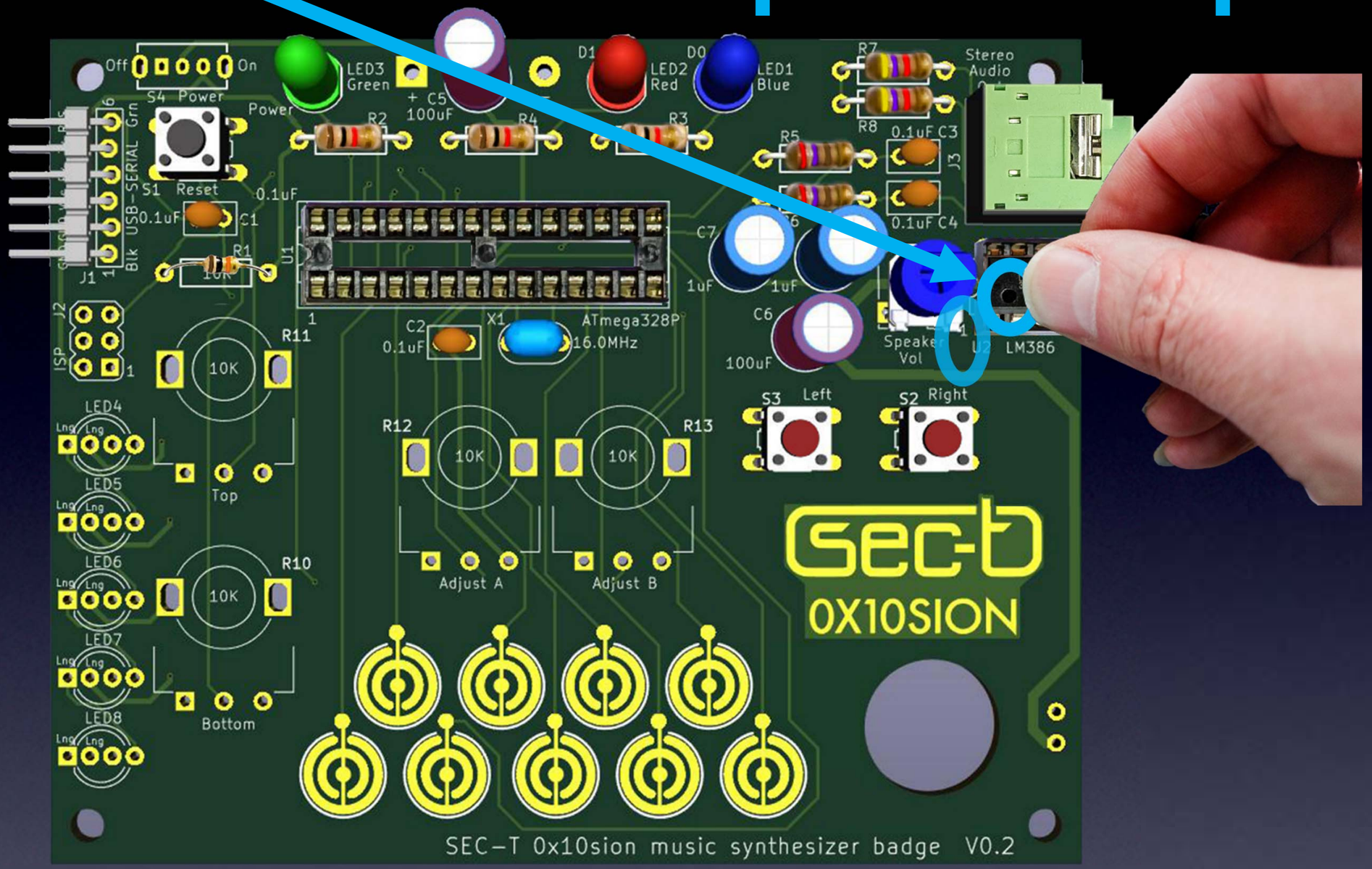
Gently bend leads so they're straight and parallel



U2: amplifier chip

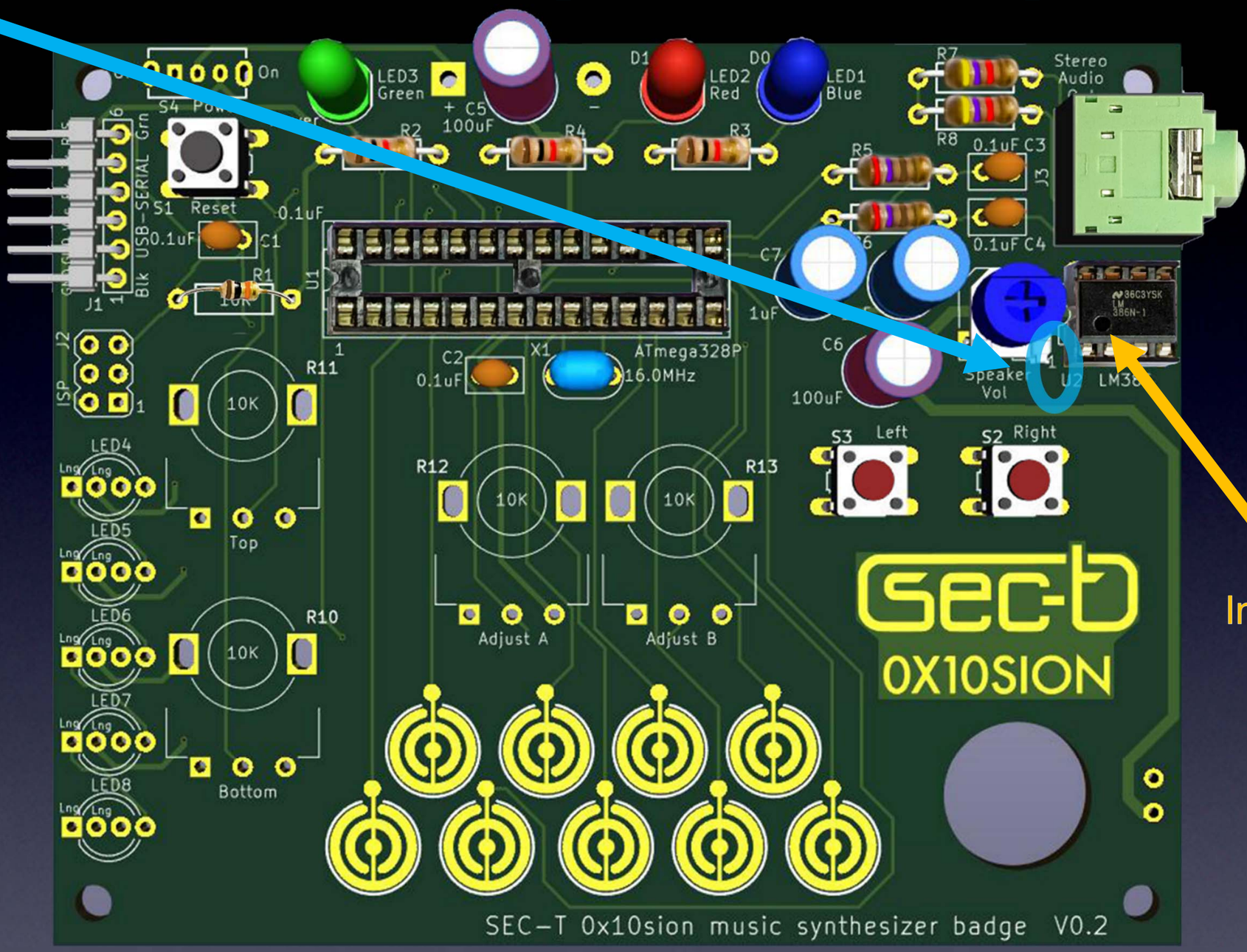


U2: amplifier chip



Make sure all 8 pins rest in their holes in the socket
→ with the proper orientation

U2: amplifier chip

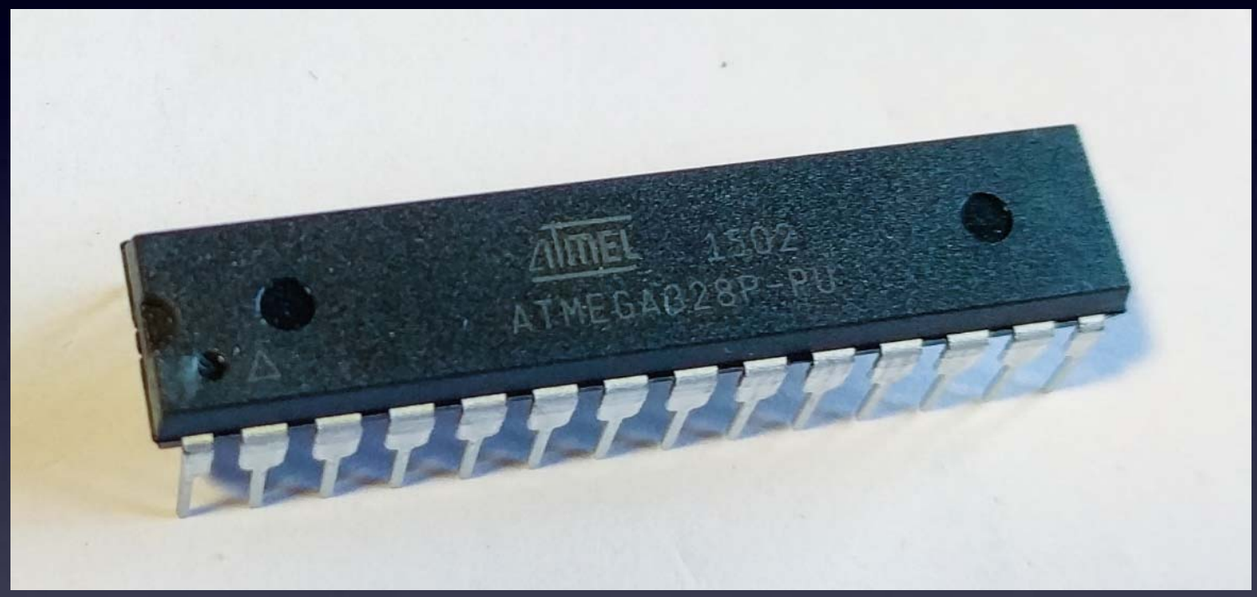


Indented black dot
Pin 1

Inspect all pins, and be sure each went into its hole in the socket – not bent.

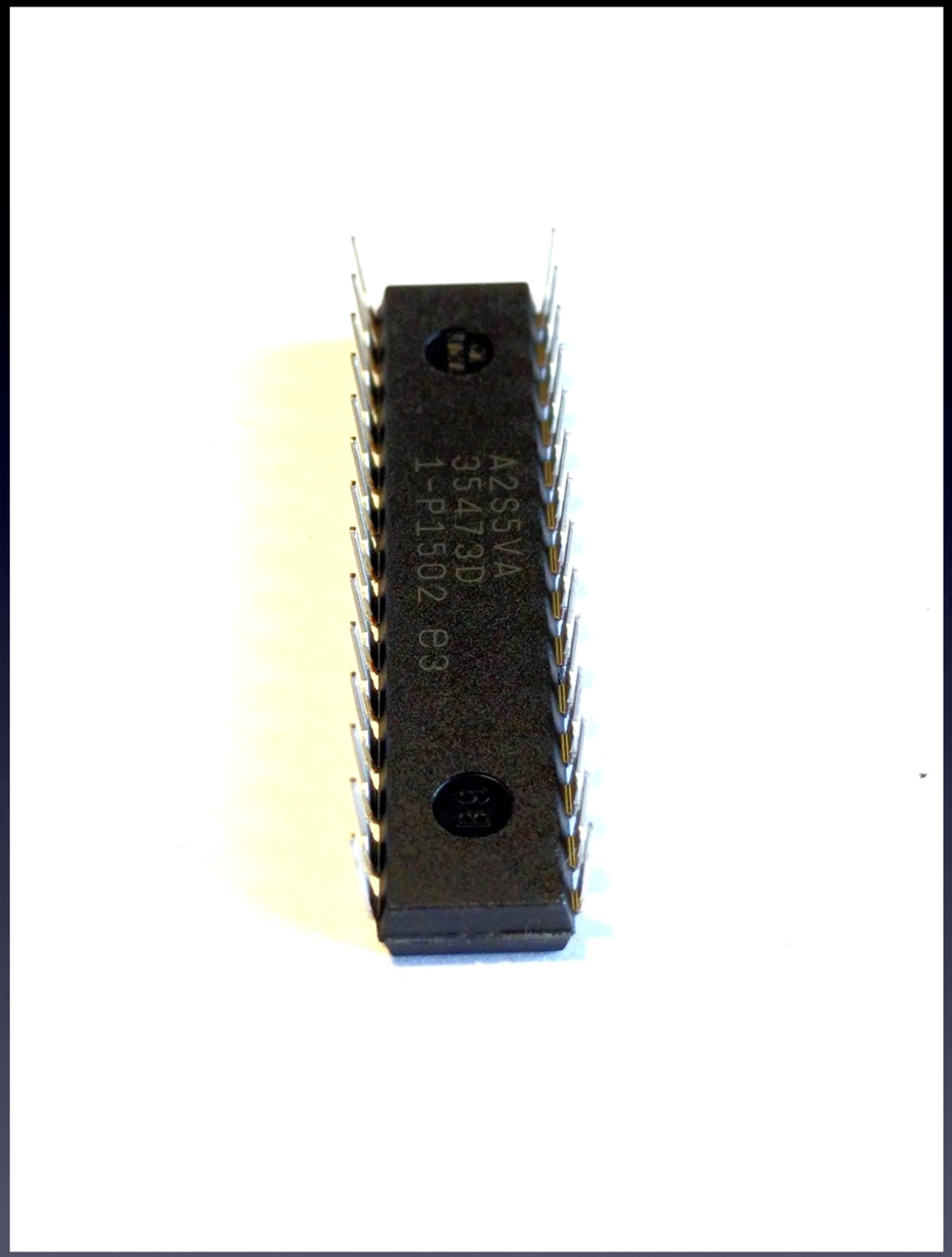
If any pins are bent, (gently) pry out chip, straighten pins, and insert again.

U1: microcontroller



U1: microcontroller

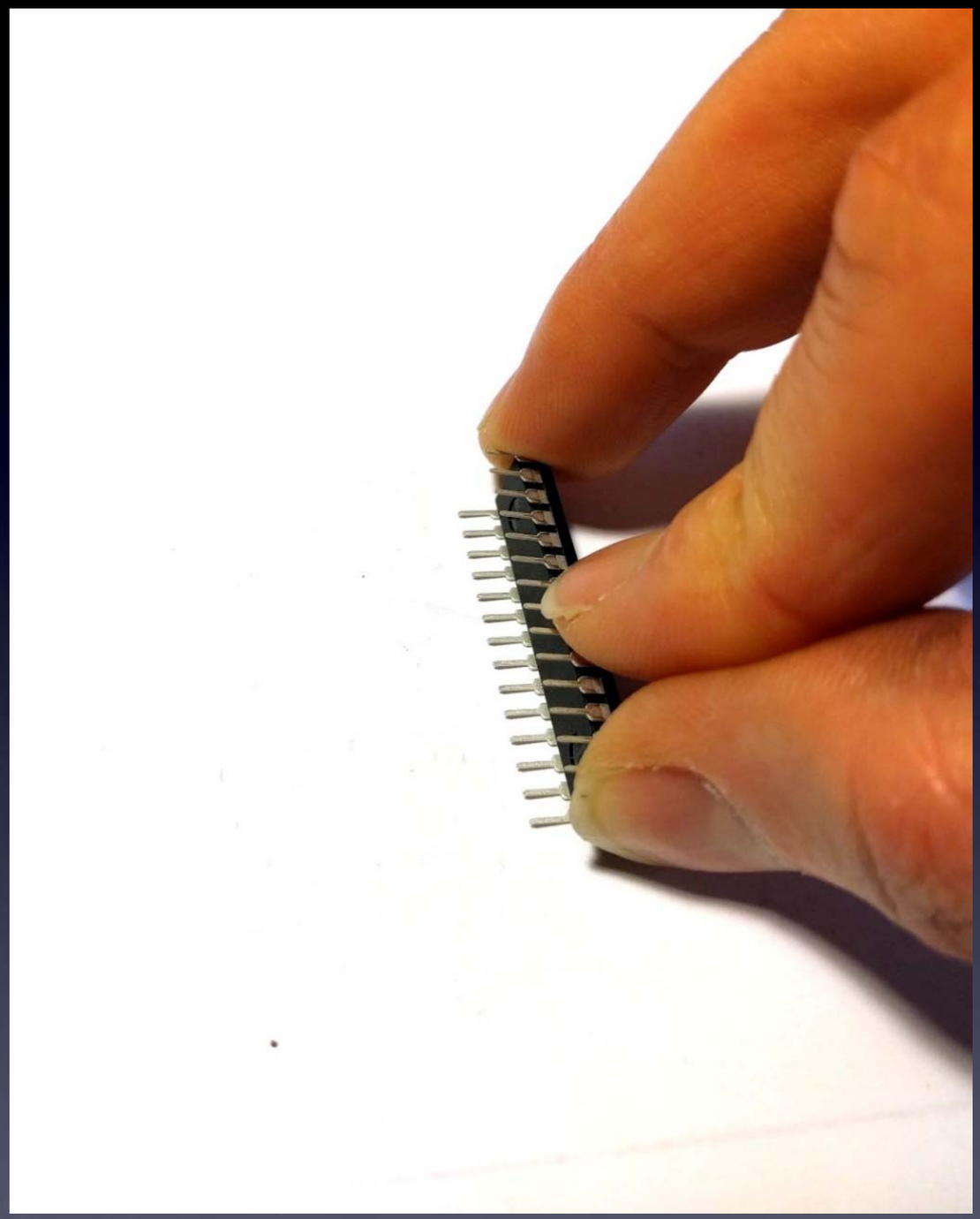
U1



When chips are new, their pins are bent out.

Note: Your kit's U1 chip may or may not have its pins already bent straight and parallel.
If not, you need to bend them, as shown in the next picture.

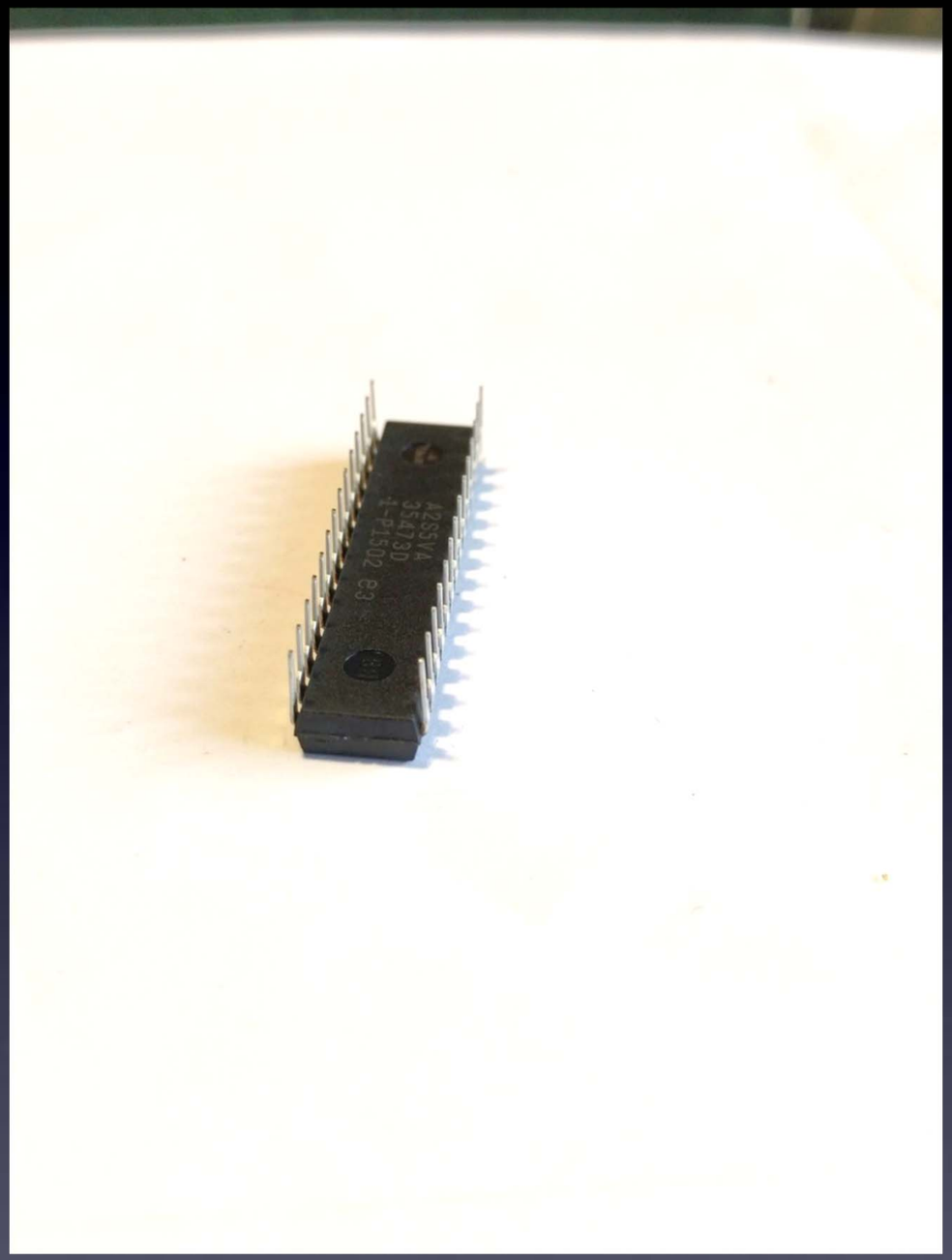
U1: microcontroller



Note: Your kit's U1 chip may or may not have its pins already bent straight and parallel.
If not, you need to bend them, as shown in this picture.

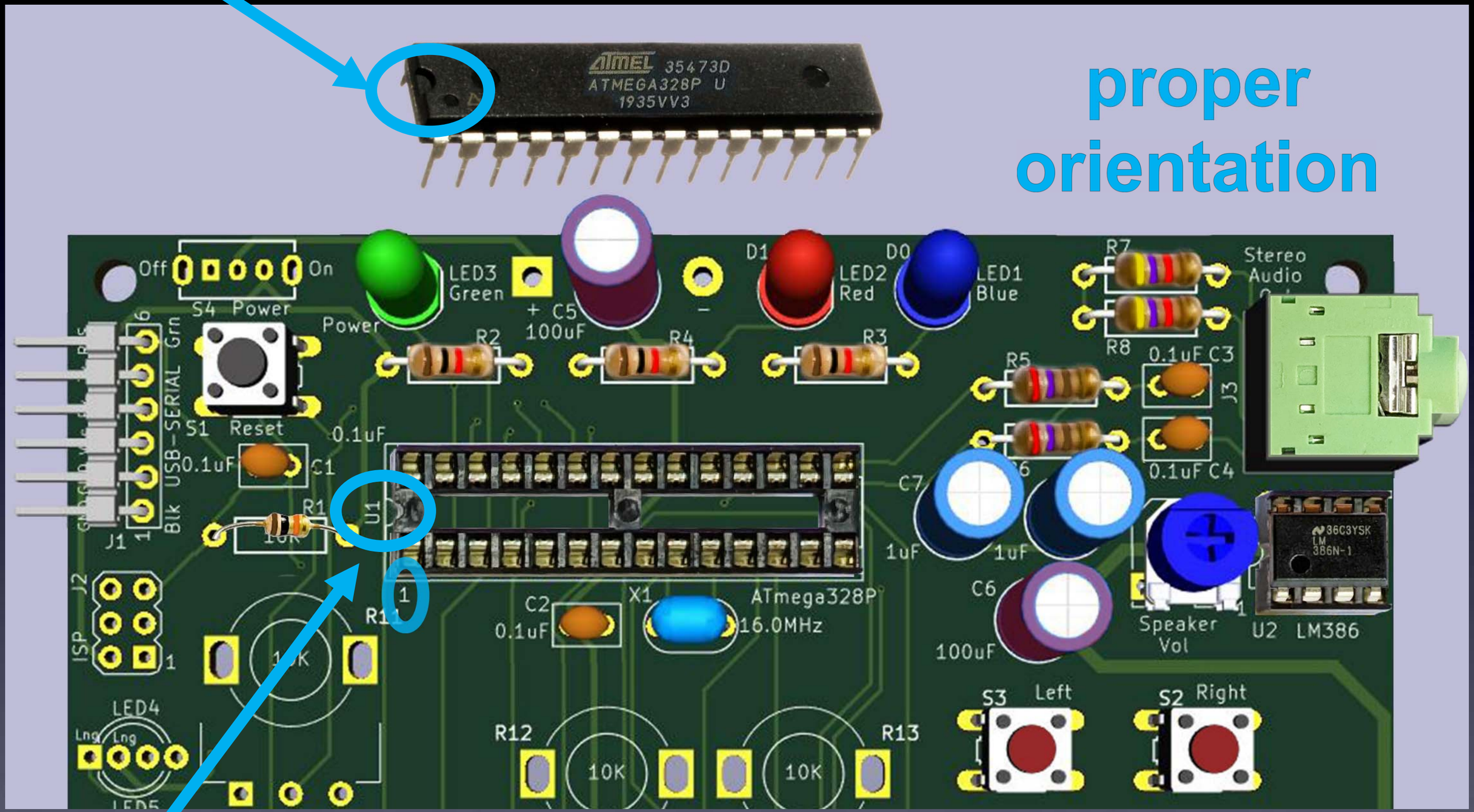
We need the pins bent straight and parallel.
Use your work table to (gently) bend the leads.

U1: microcontroller

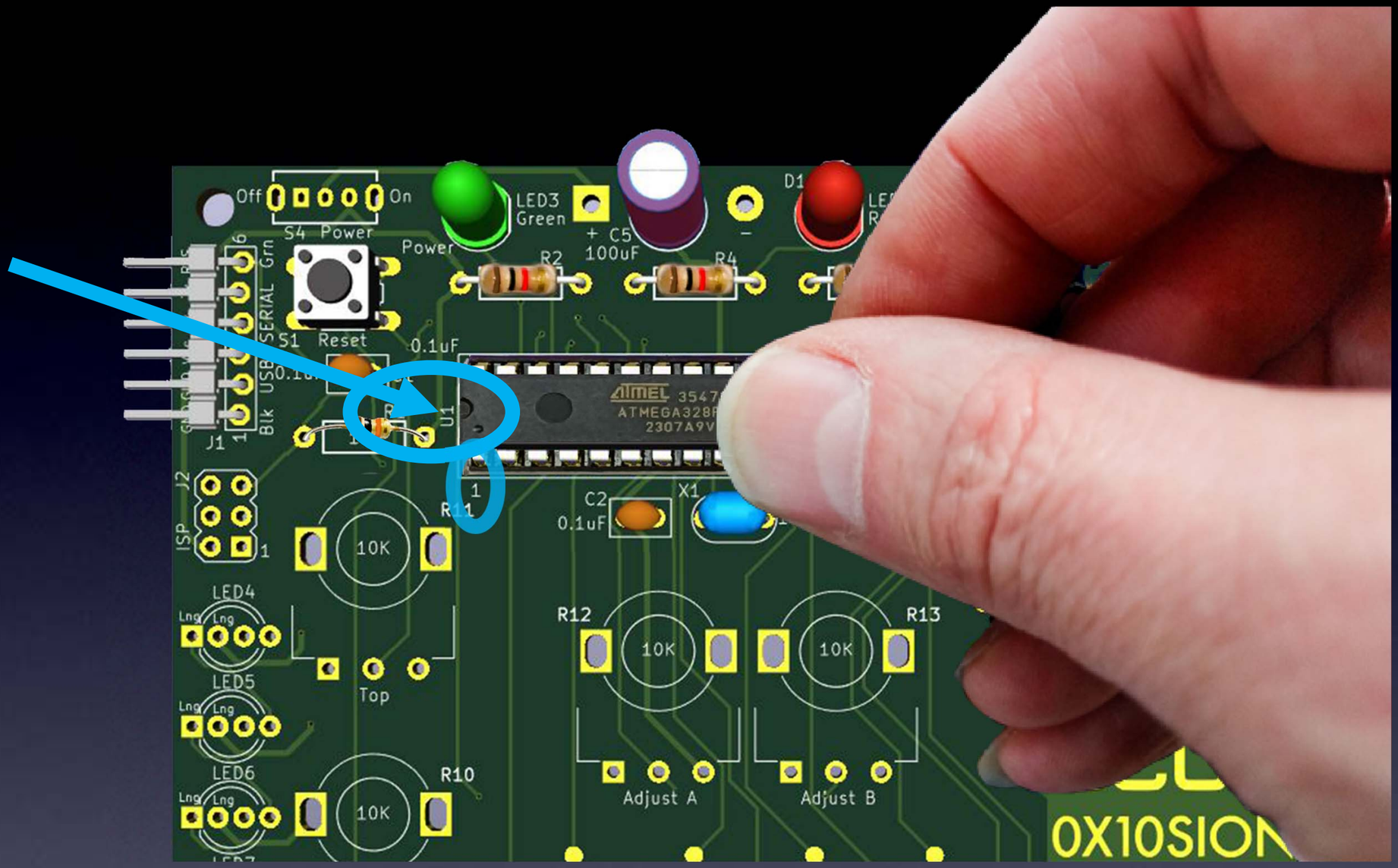


These pins must be straight and parallel

U1: microcontroller



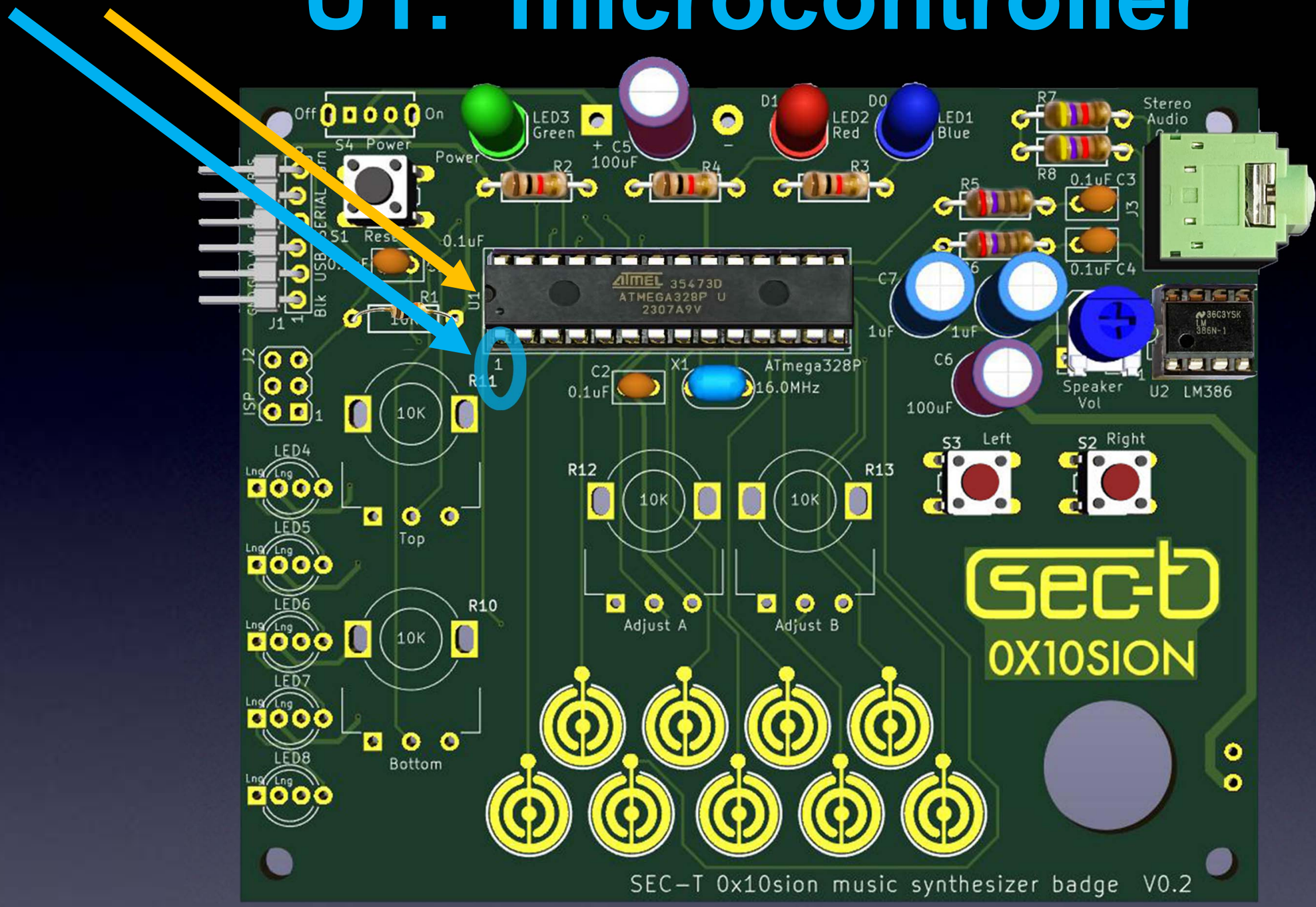
U1: microcontroller



Make sure all 28 pins rest in their holes in the socket
→ with the proper orientation

(This is actually way easier with 2 thumbs.)

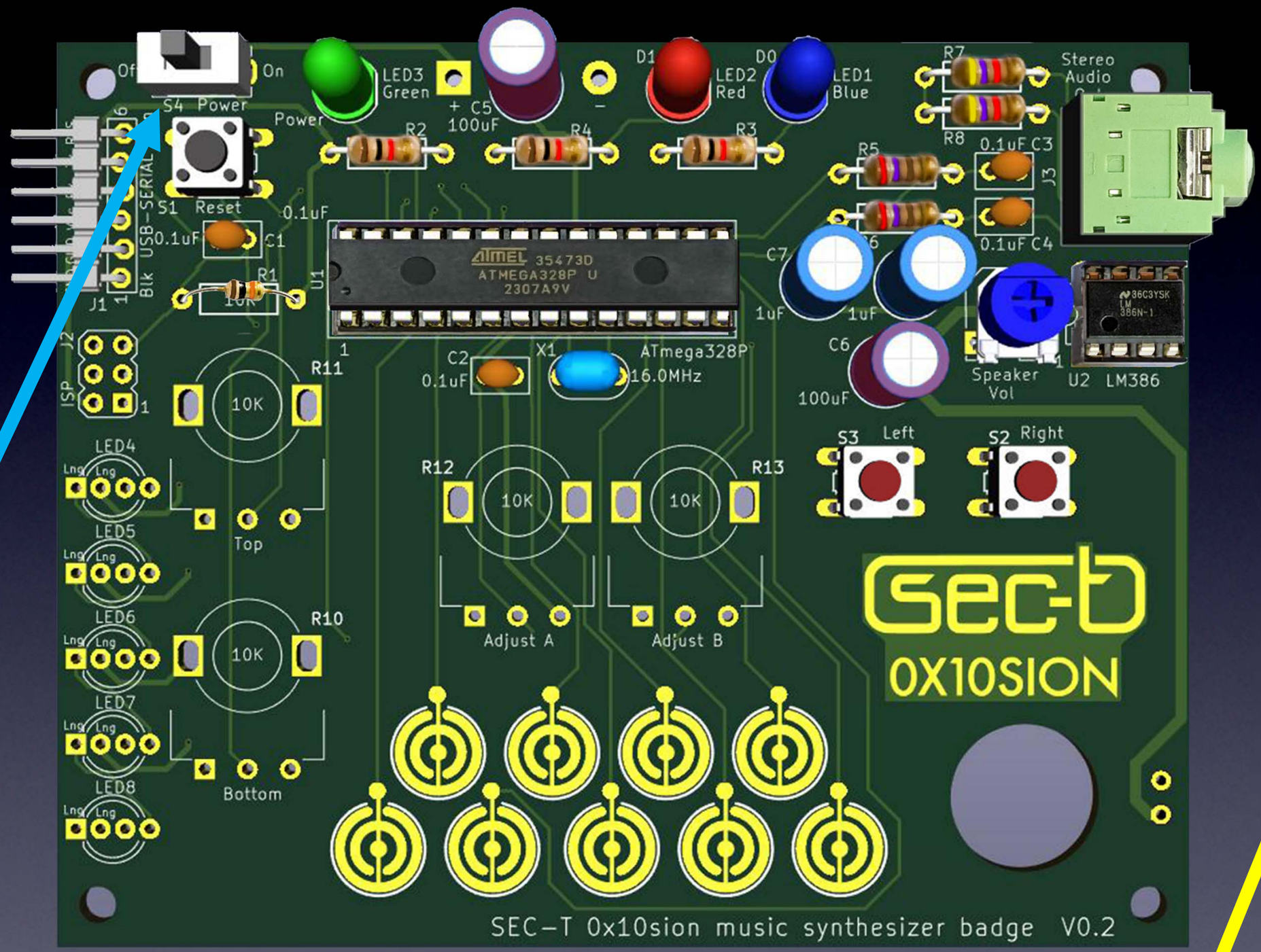
U1: microcontroller



Inspect all pins, and be sure each went into its hole in the socket – not bent.

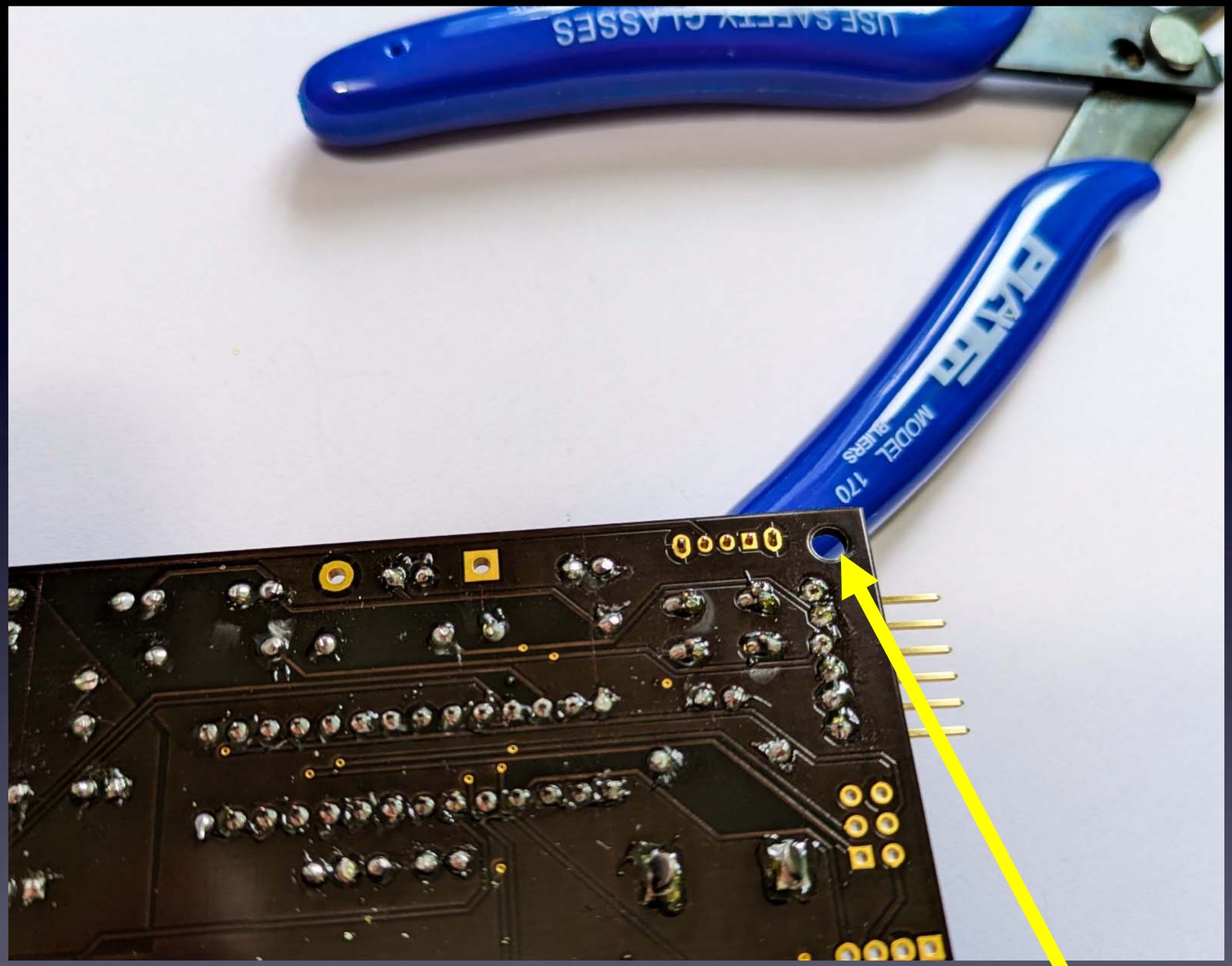
If any pins are bent, (gently) pry out chip, straighten pins, and insert again.

J4: Power switch



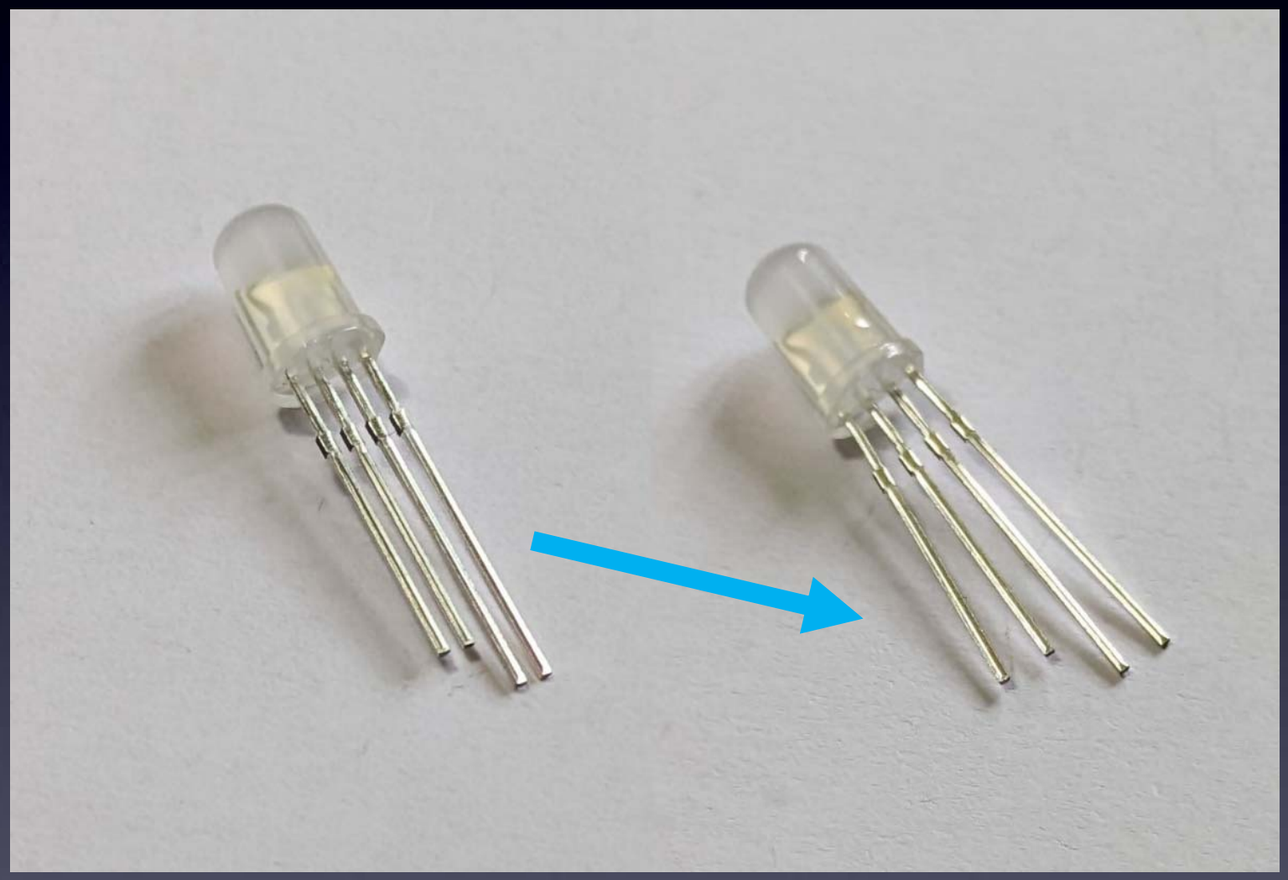
→ See next photo for how to solder this part…

J4: Power switch



Resting on the wire cutters, S4 is held in place here, so you can solder all 5 leads.

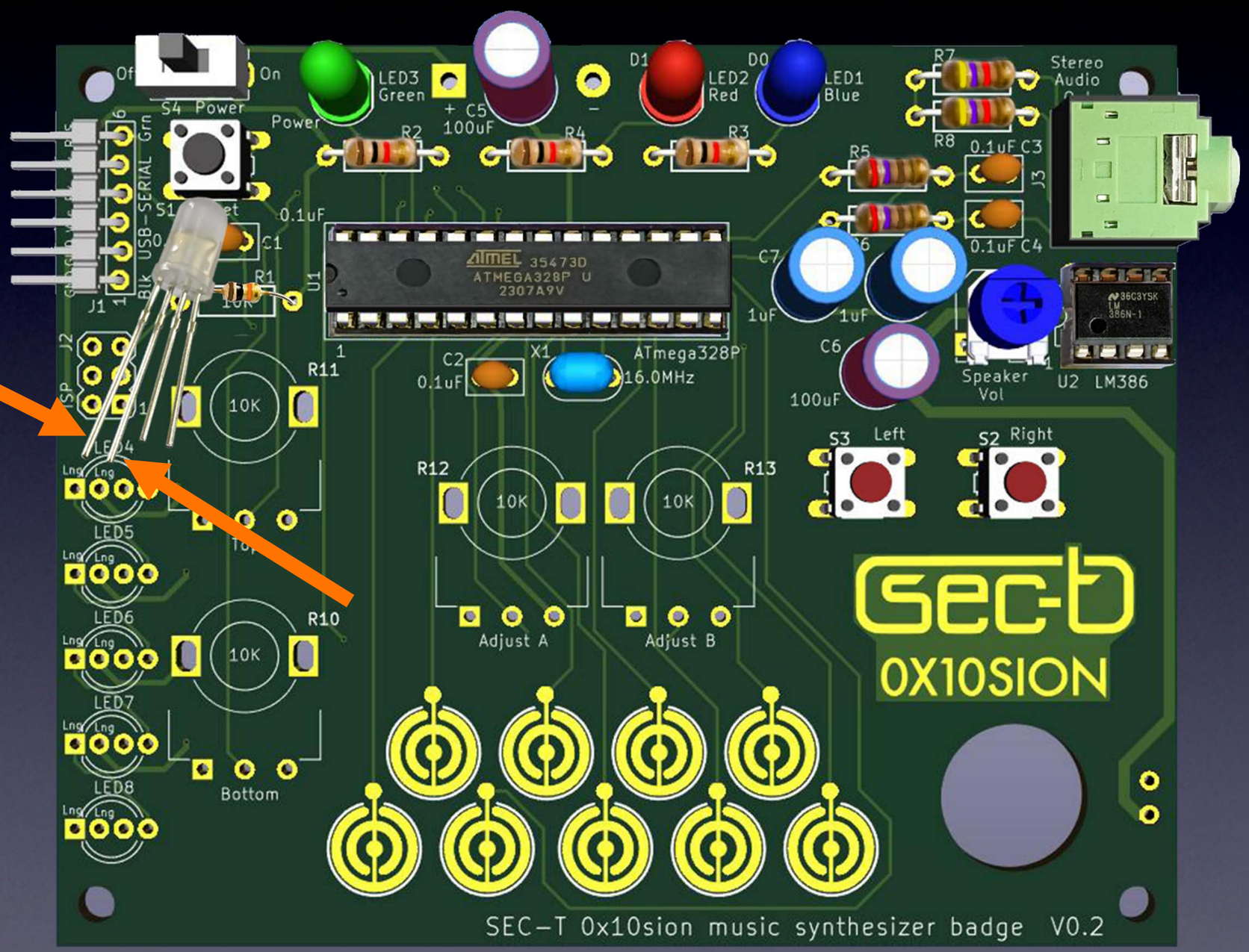
LED4, LED5, LED6, LED7, LED8: RGB LEDs



Bend the leads out a little so they fit into their pads.

LED4, LED5, LED6, LED7, LED8: RGB LEDs

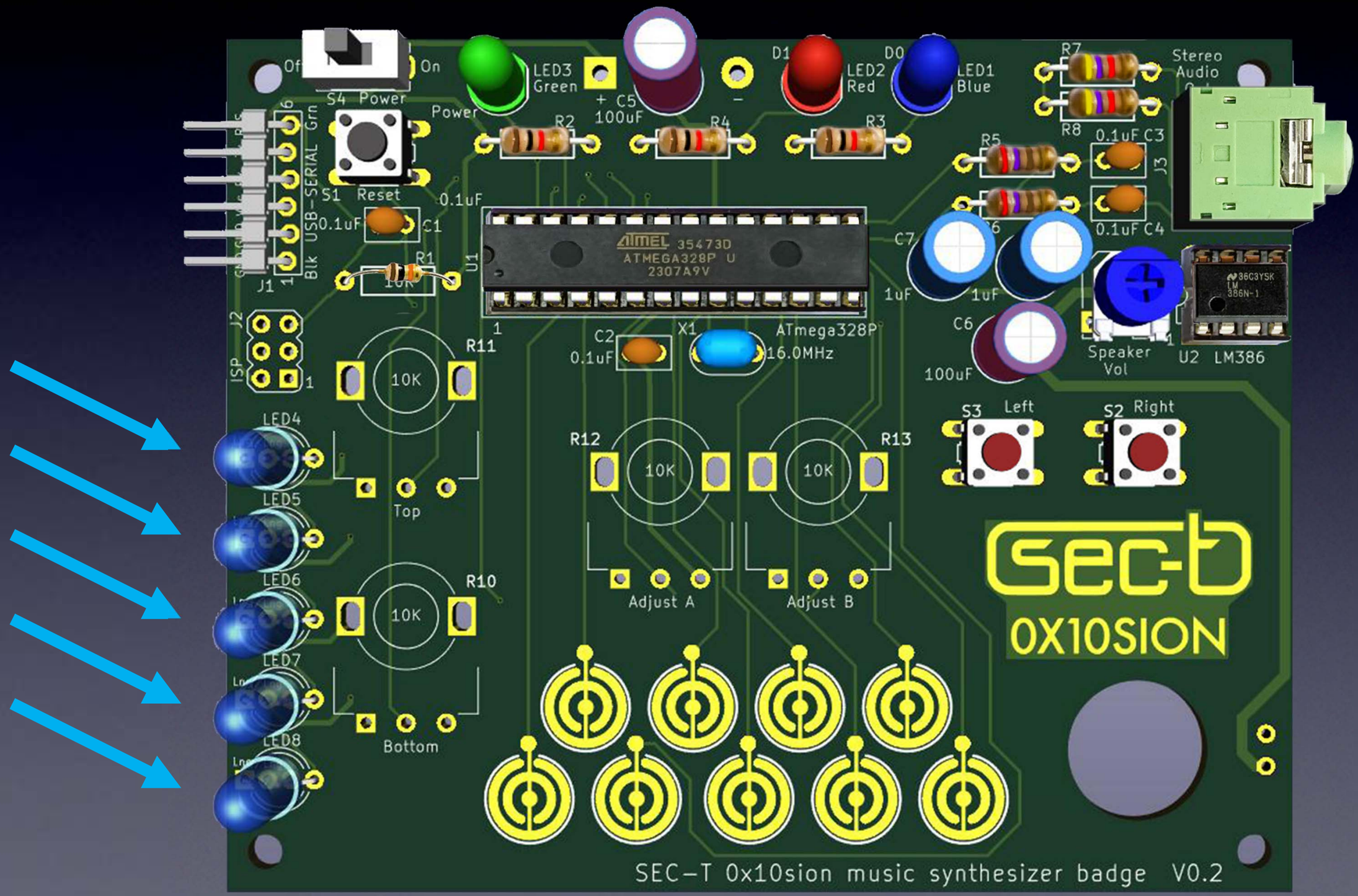
Note the orientation of the 2 long leads



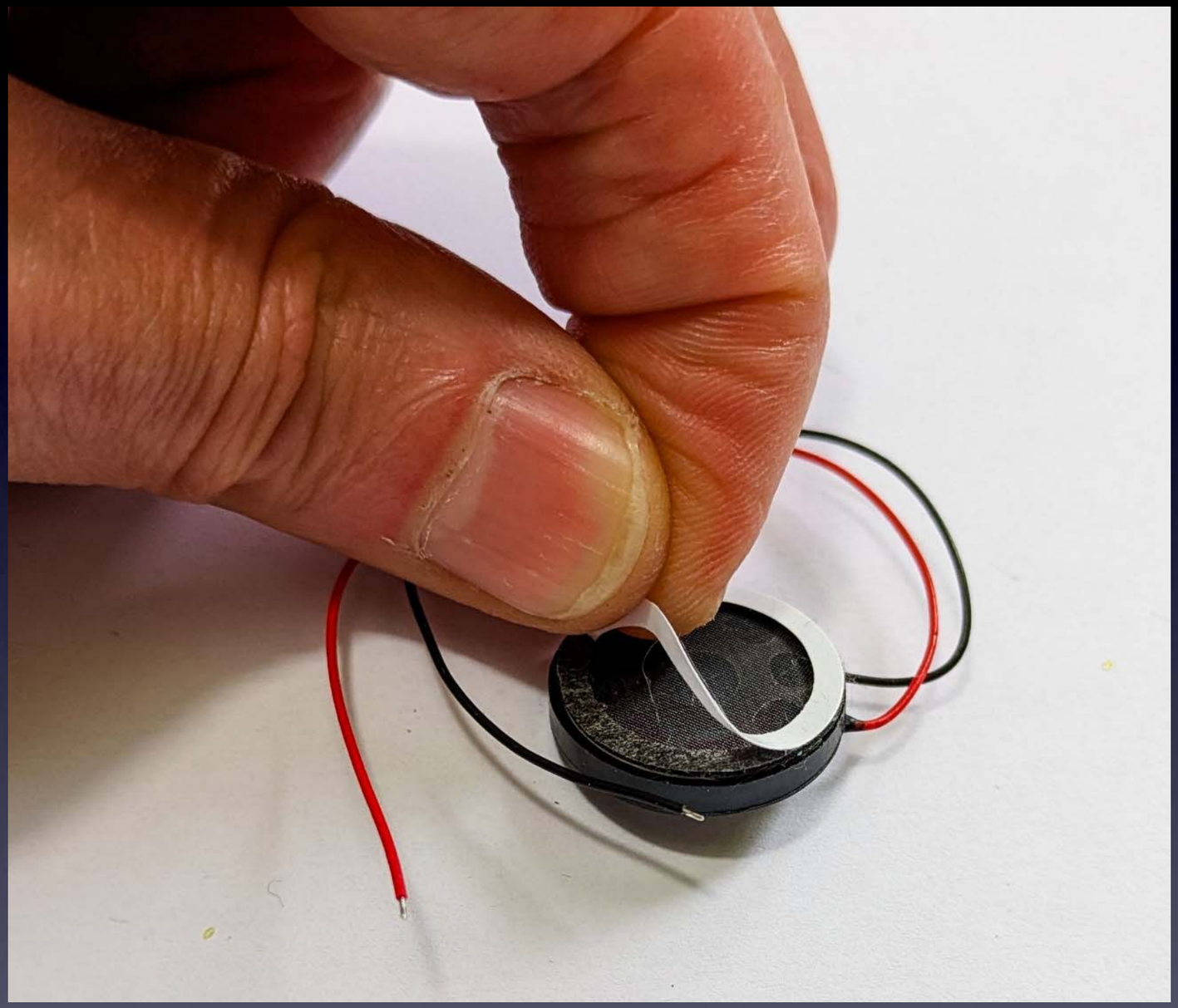
LED4, LED5, LED6, LED7, LED8: RGB LEDs

These LEDs will be inserted about this far into the board

LED4, LED5, LED6, LED7, LED8: RGB LEDs

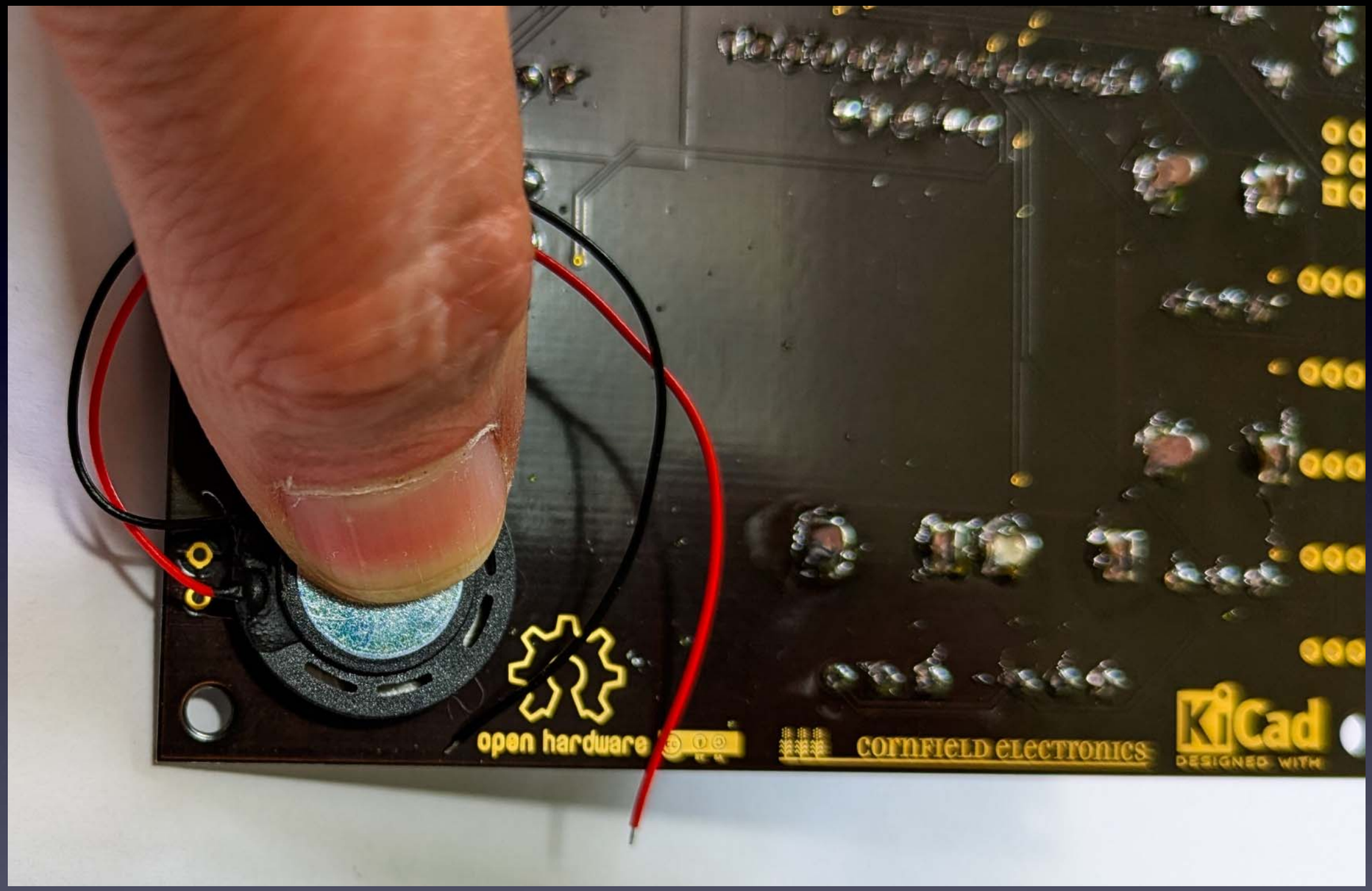


Speaker



Pull off the protective covering to reveal the adhesive

Speaker

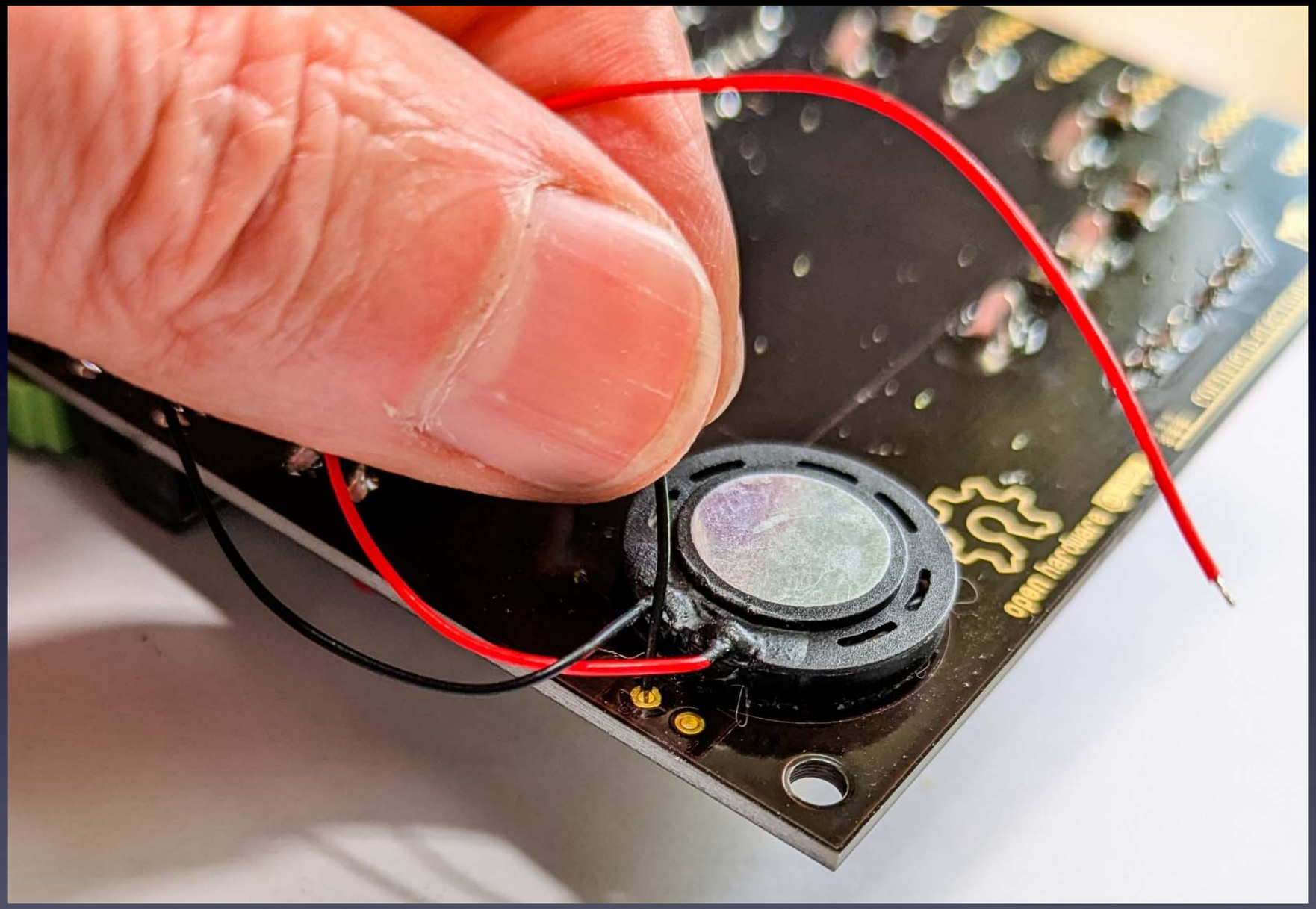


Carefully center the speaker over its hole, and then push

Speaker

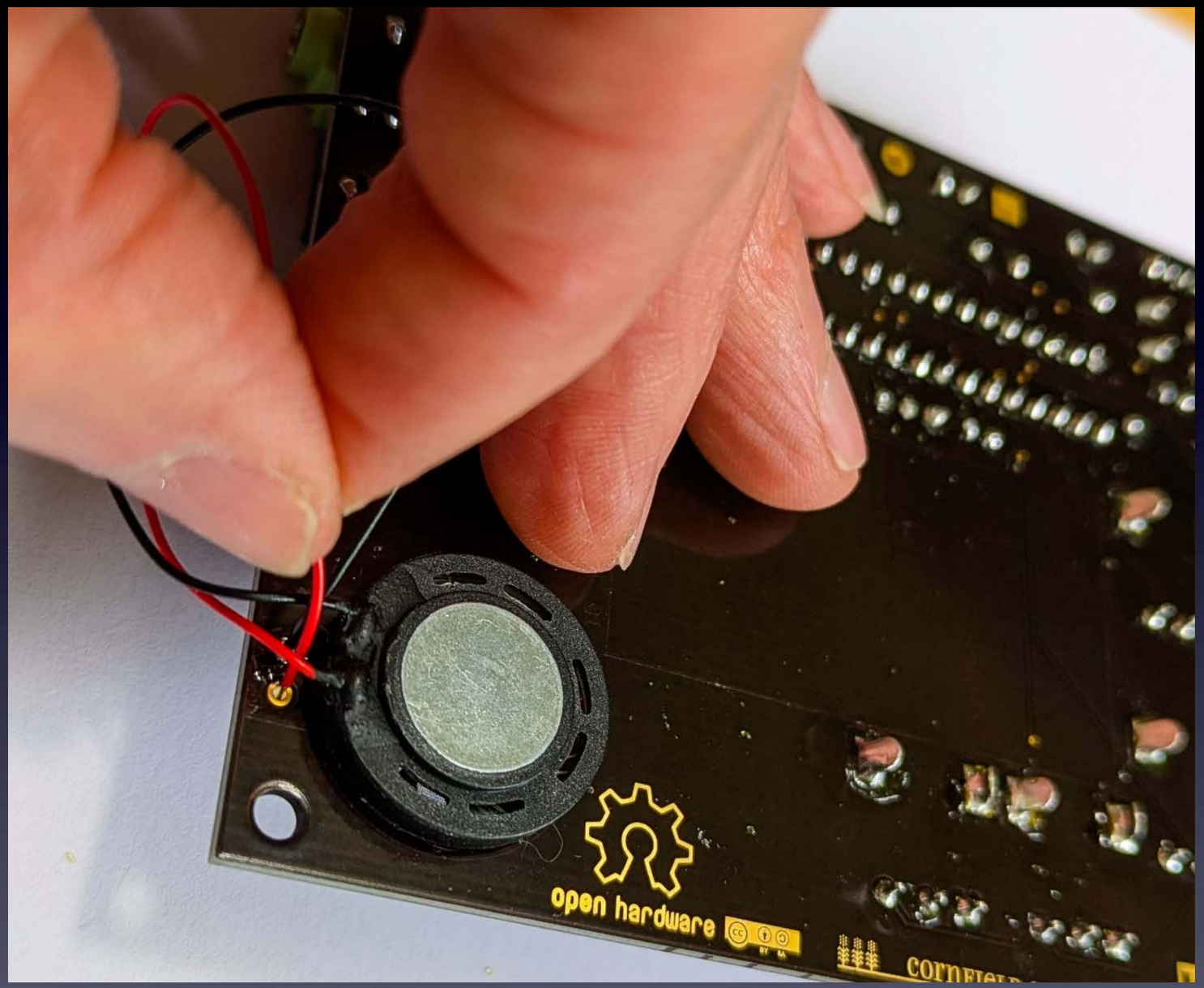
Here you can see the speaker centered under its hole

Speaker



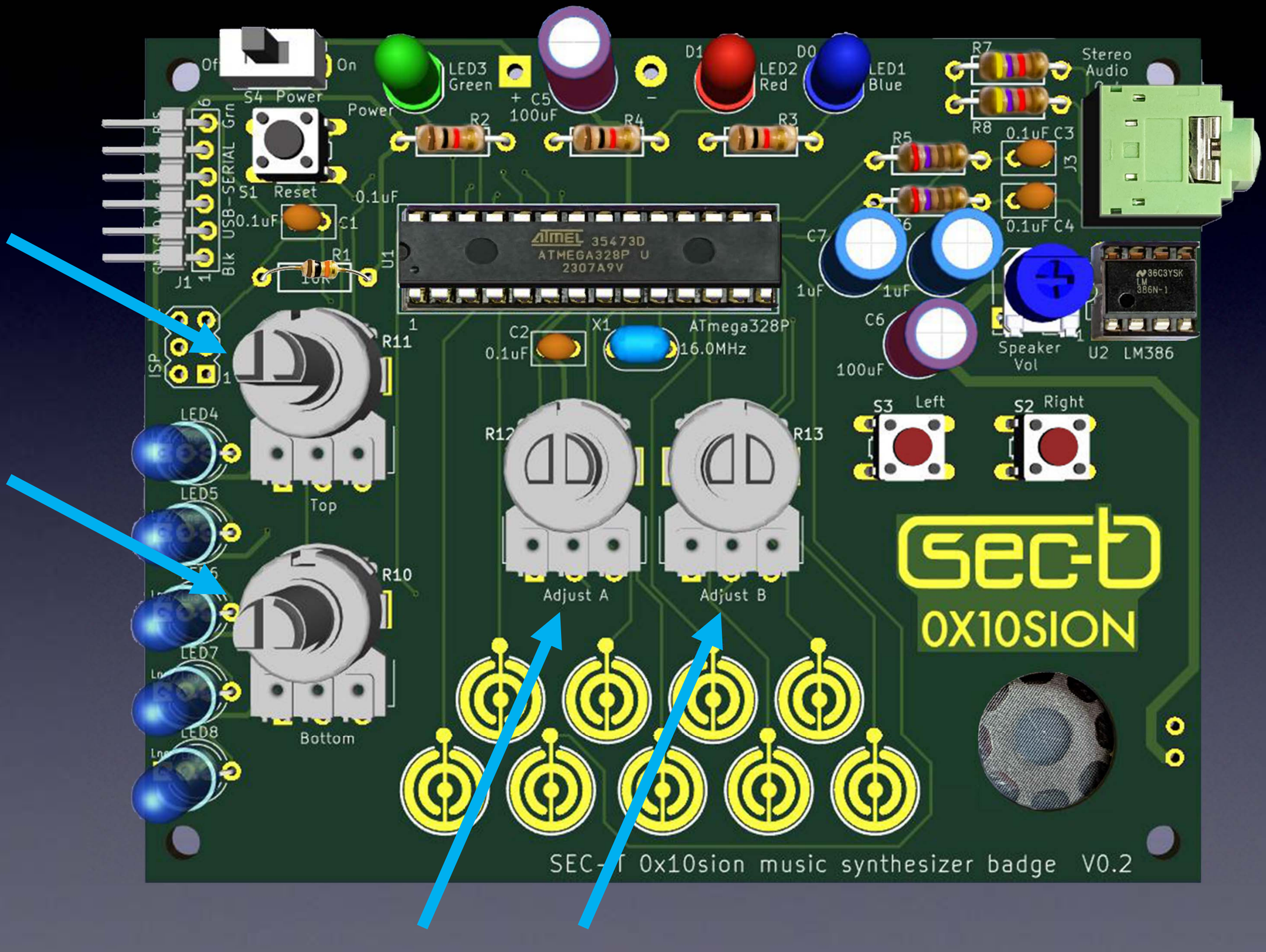
Place the black wire into its pad, and solder it

Speaker

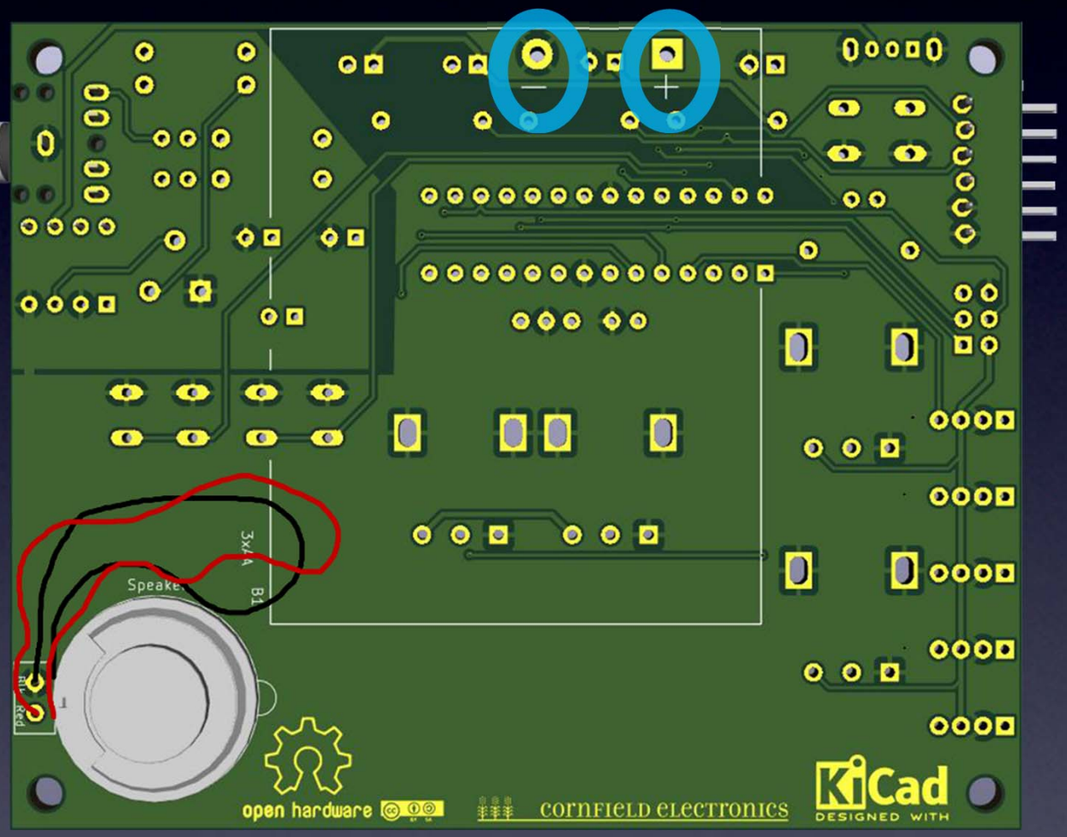


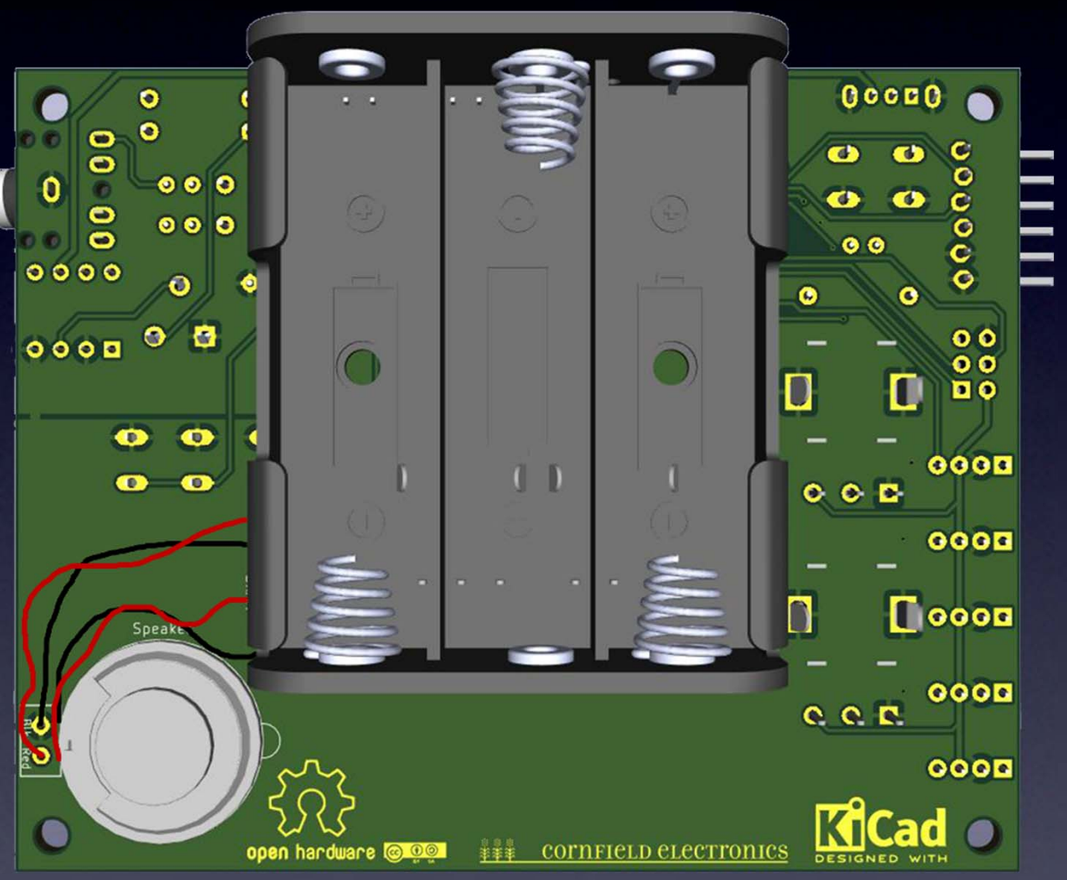
Place the red wire into its pad, and solder it

R10, R11, R12, R13: potentiometers



Battery pack



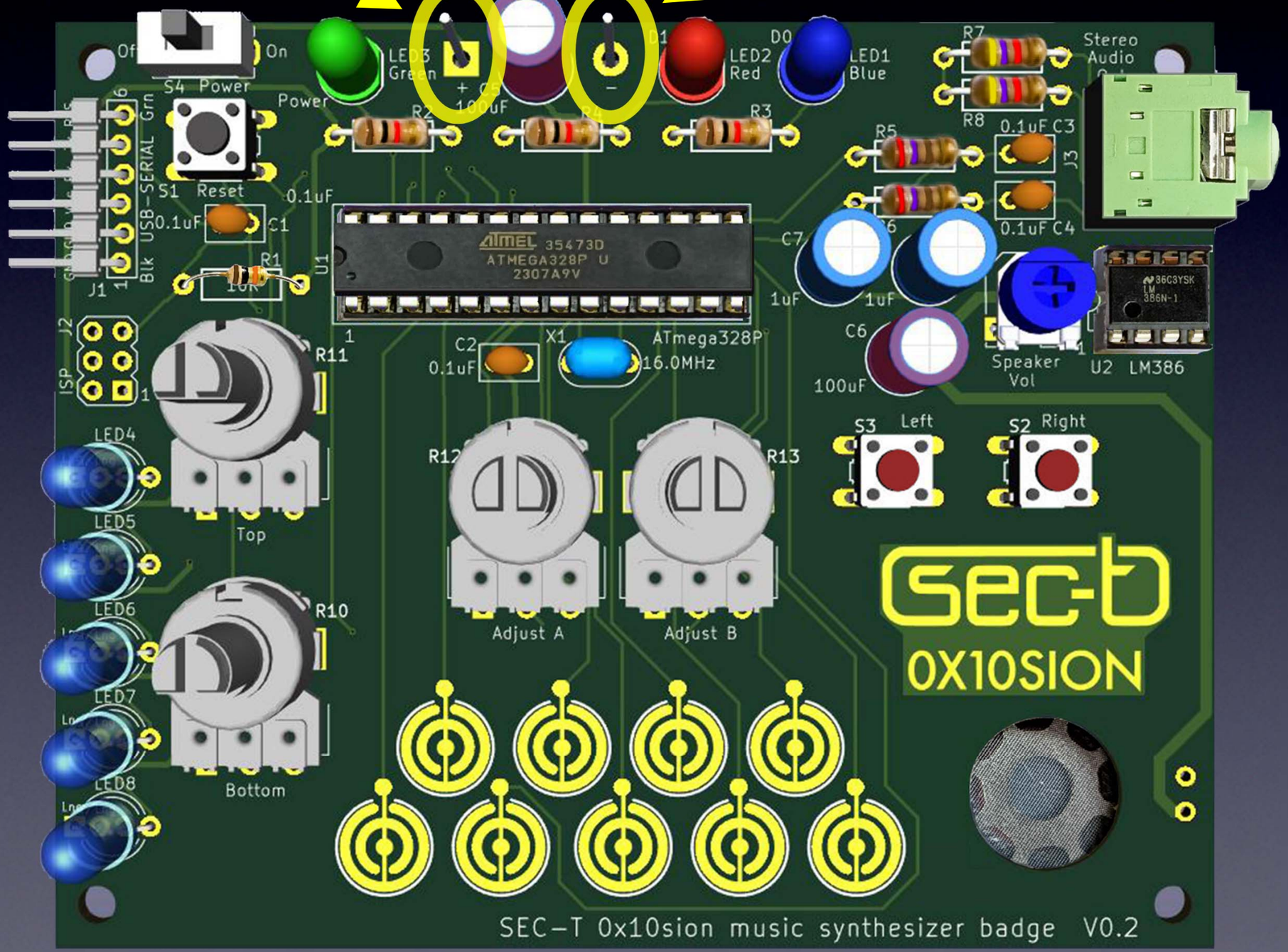


Place the Battery pack into place, with its leads in its “-” and “+” pads and solder the leads on the front

DO ***NOT*** CUT THE “-” and “+” leads !

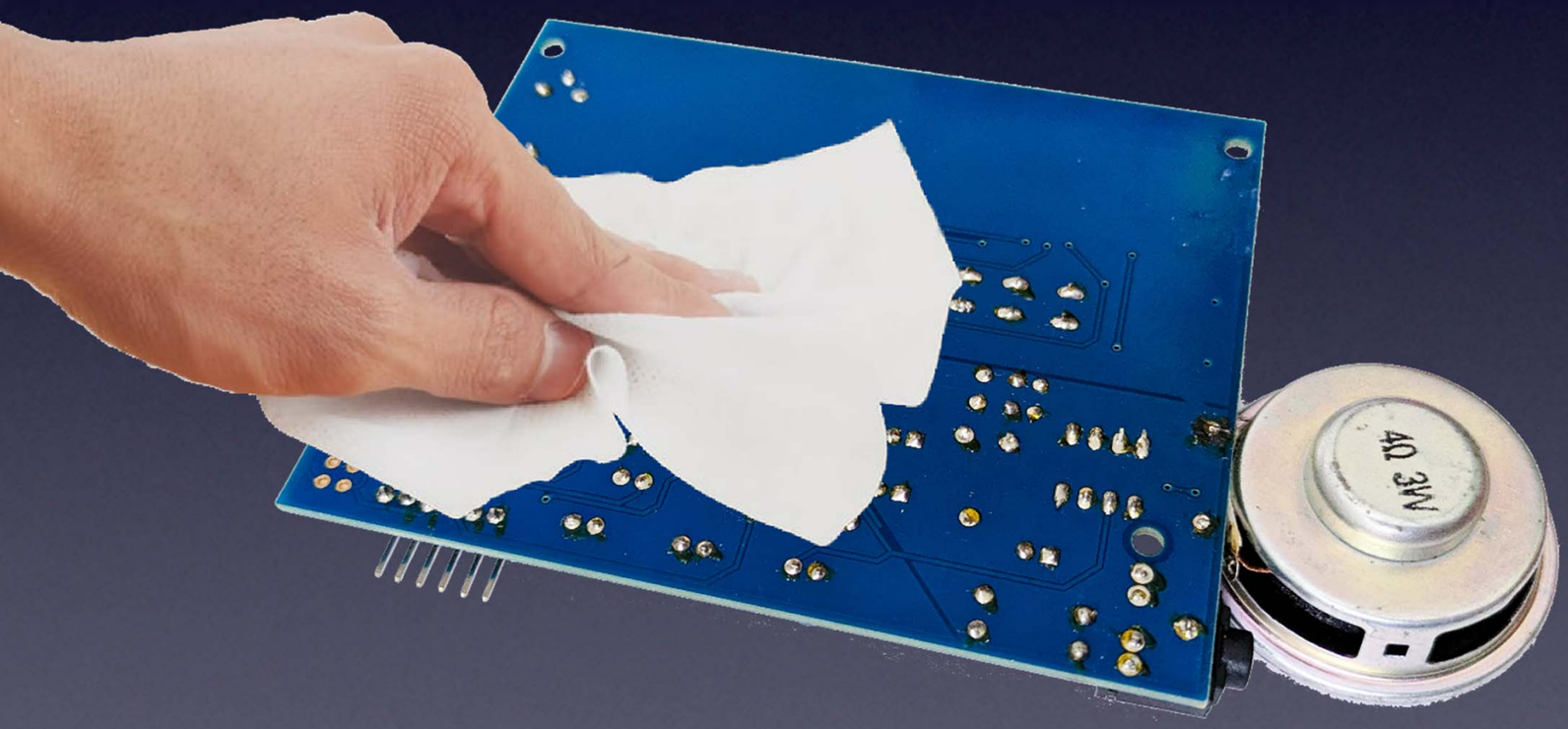
Battery pack

DO ***NOT*** CUT THE “-” and “+” leads !



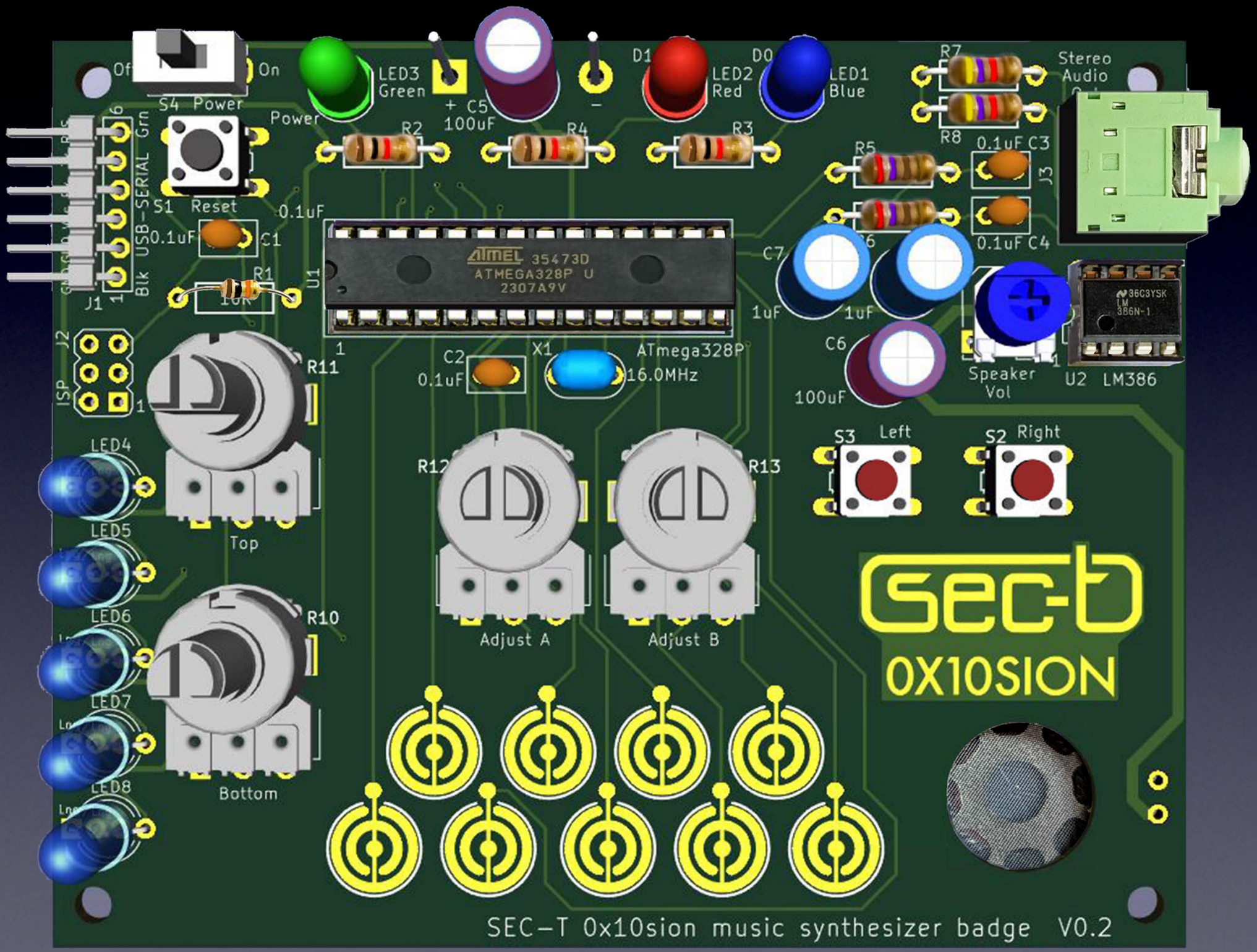
If you used any *flux paste for re-working problems*

The bottom of the PCB will be sticky from the flux

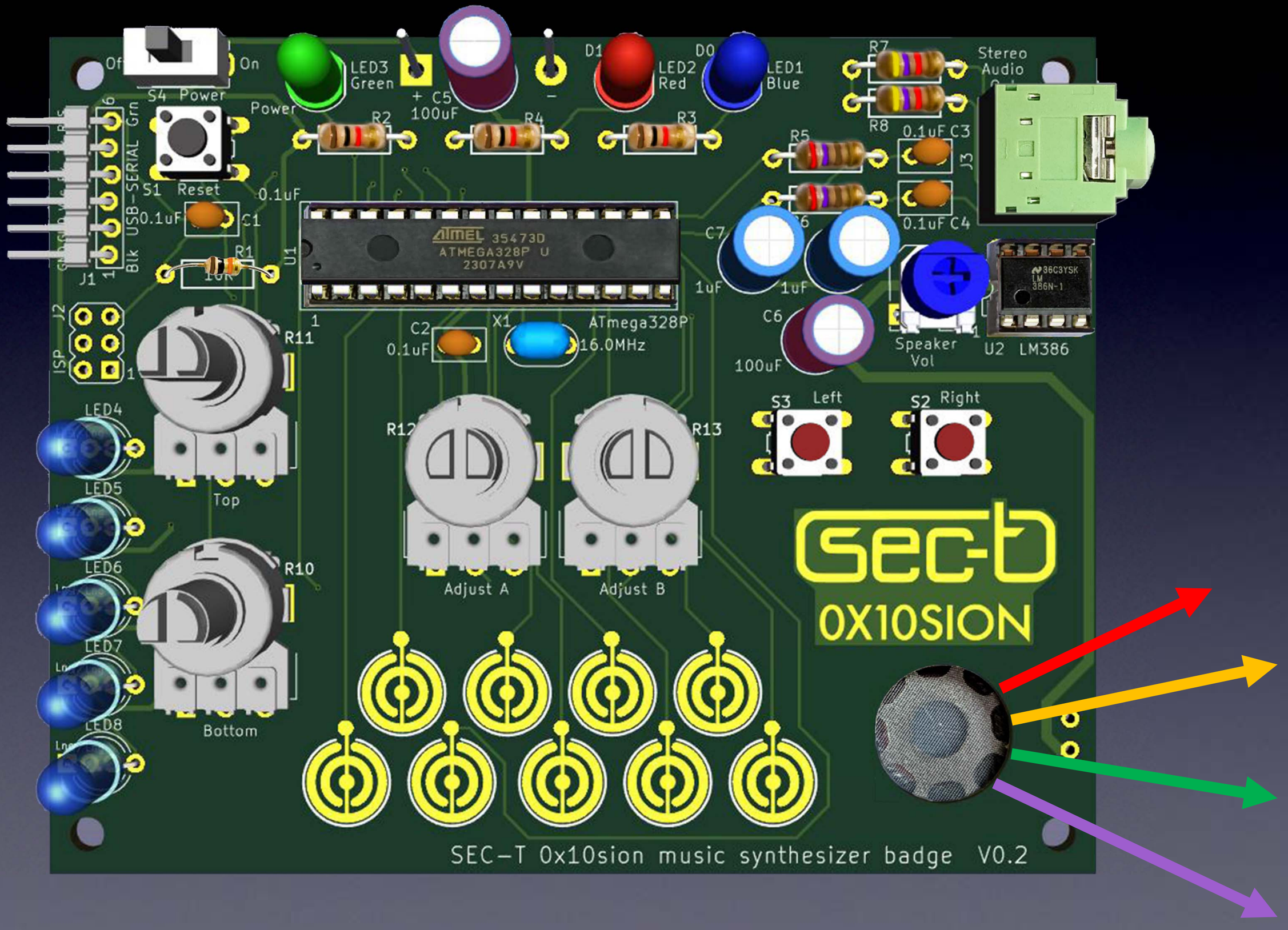


You can clean it with a cloth wet with Isopropyl Alcohol

Done!


S4 Power
S1 Reset
LED3 Green
LED2 Red
LED1 Blue
+ C5 100uF
R2
R4
R3
R7
R8
R5
0.1uF C3
0.1uF C4
Stereo Audio
J3
J1
Blk USB-SERIAL Grn
0.1uF C1
R1
U1
ATMEL 35473D
ATMEGA328P U
2307A9V
C7
1uF
C6
100uF
Speaker Vol
U2 LM386
J2
ISP
R11
C2 0.1uF
X1
ATmega328P
16.0MHz
S3 Left
S2 Right
LED4
LED5
LED6
LED7
LED8
Top
Bottom
R10
R12
R13
Adjust A
Adjust B
SEC-T
0X10SION
SEC-T 0x10sion music synthesizer badge V0.2

Let’s make some noise!



Please Remember:

to

Wash your hands

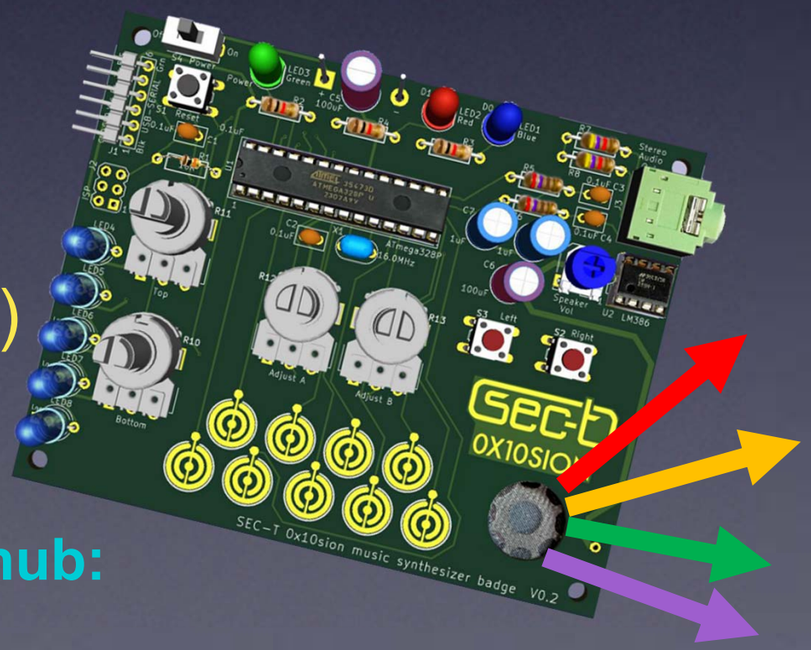
after soldering

Let's make noise!

Your SEC-T 0x10sion Synth Badge comes pre-programmed with a really cool synthesizer, called "Quadrant".

"Quadrant" is like a looper, but with the ability to warp each time around the loop.

- the left and right red buttons change octaves
- double-click the right red button and then hit a bottom-row touch-key to change to different preset – there are 5 very different presets
- the Bottom knob controls the looping rate
- the Top knob controls warping (detuning) each time through the loop
- the Adjust A knob controls the wah speed
- the Adjust B knob controls the wah depth
- Try playing with these and see!
- Long-pressing the left and right red buttons changes other parameters that the knobs can control (more info on the project's Github)
- Experiment, and see what sounds you get!



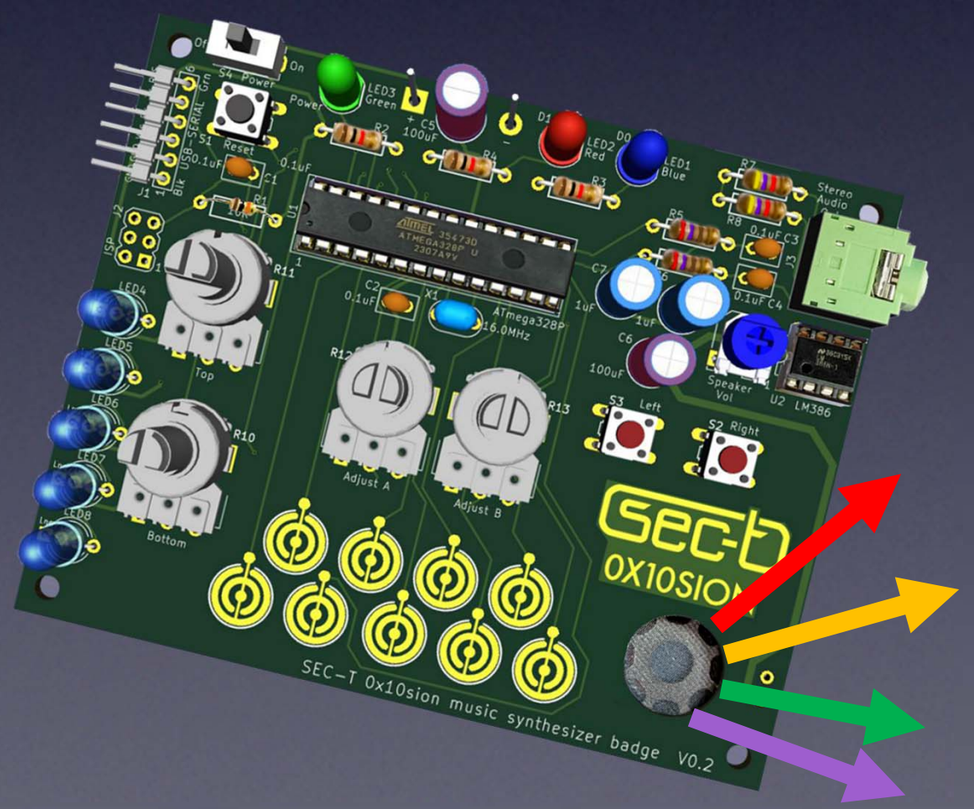
Complete documentation is on the project's Github:
https://github.com/SEC-T/badge-2024

Let’s make noise!

Your SEC-T 0x10sion Synth Badge comes pre-programmed with a really cool synthesizer, called “Quadrant”.

If you are happy playing with “Quadrant” then no need to re-program your SEC-T Synth Badge.

But if you want to program other synths into your SEC-T Synth Badge, the next pages show you how…

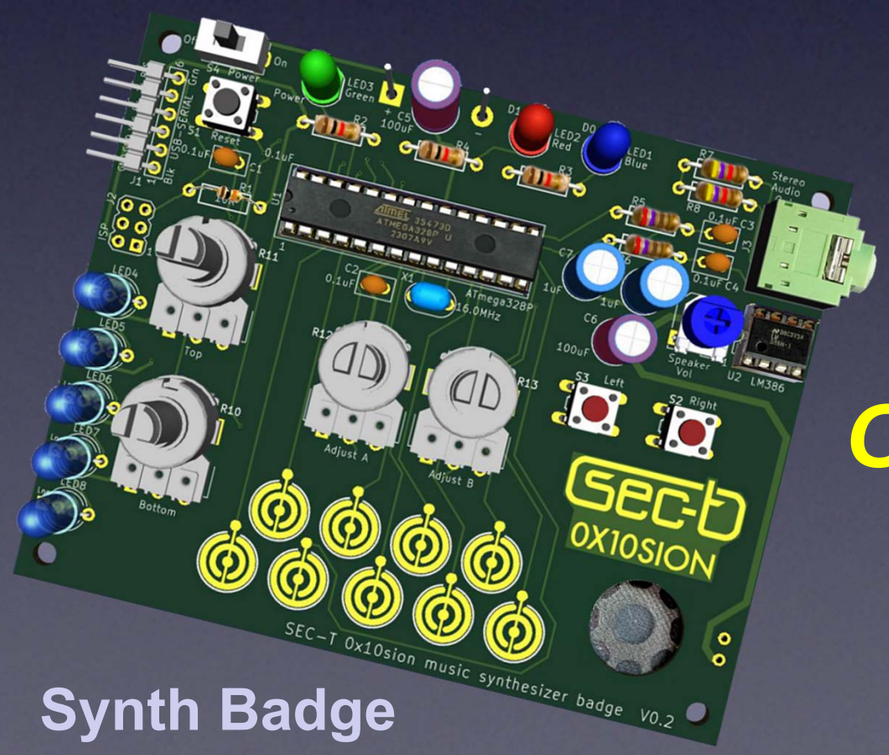


Re-programming the Synth Badge

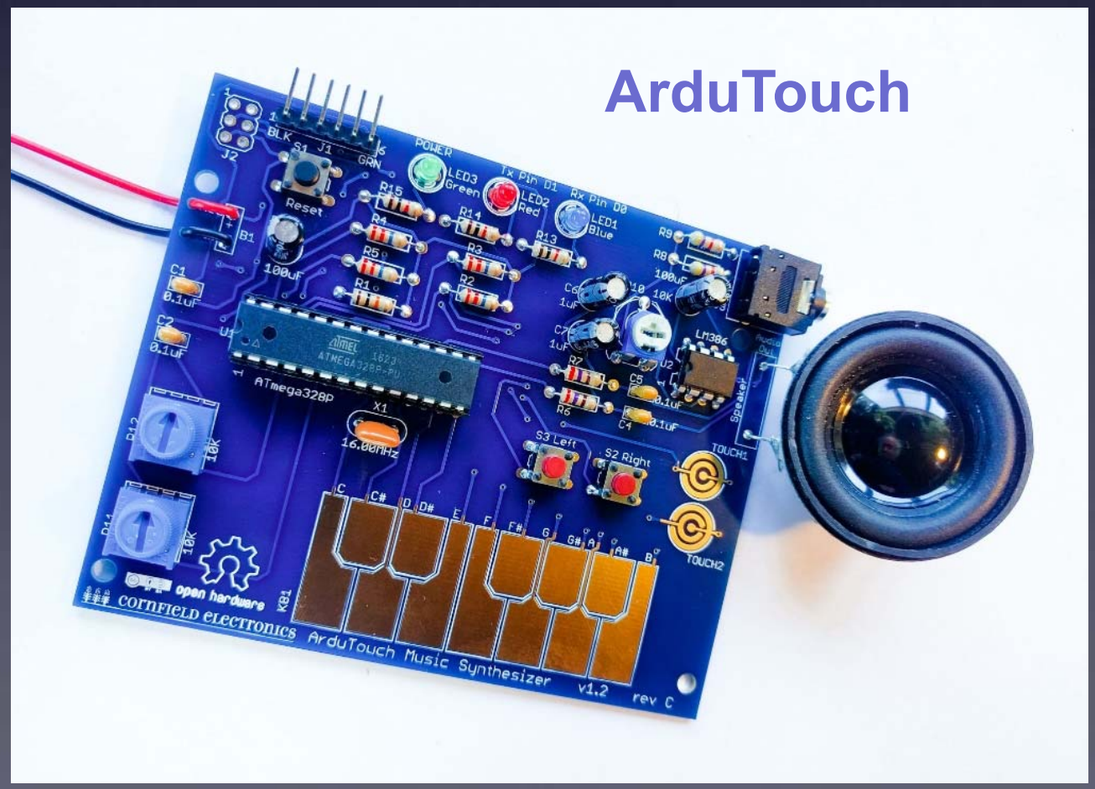
The SEC-T 0x10sion Music Synthesizer Badge
was created from the ArduTouch Music Synthesizer project

The following pages show how to re-program the ArduTouch synthesizer

Re-Programming the Synth Badge board
is the same as
re-programming the ArduTouch board



Or

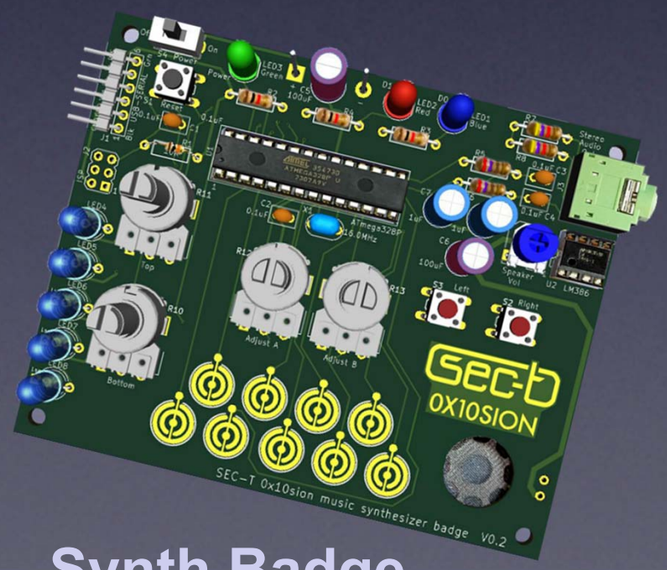


Re-programming the Synth Badge

We have written several way cool synthesizers for the ArduTouch!
Each is unique, and each way different than the others.

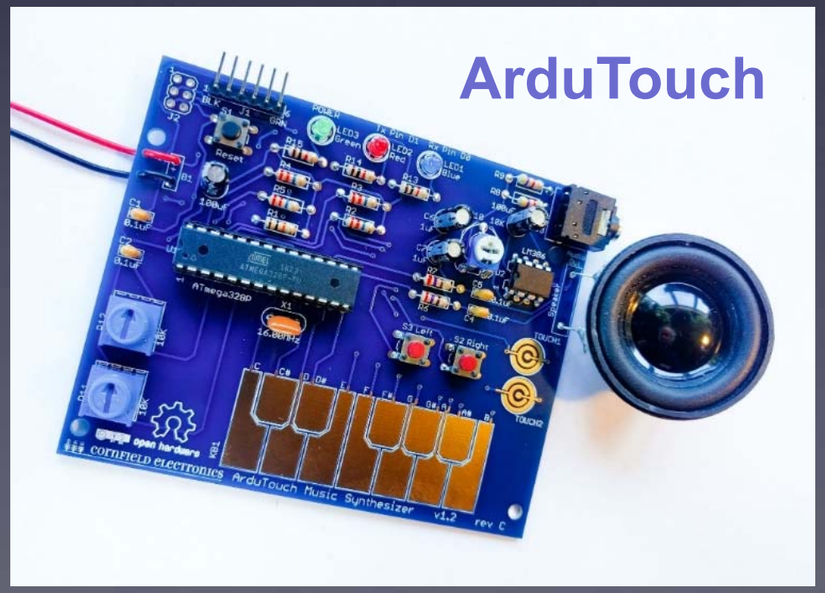
<u>To program in a new synth in your Synth Badge, you will need</u>:

- the Arduino software
  <http://arduino.cc>
- a USB-Serial adapter cable (such as an FTDI, or equivalent)
  with its driver installed
- the SEC-T 0x10sion Synth Badge Arduino library
  <https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library>
- An ArduTouch synth sketch
  <http://cornfieldelectronics.com/cfe/projects.php#ardutouch>



Synth Badge

Or



Re-programming the Synth Badge

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  <https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library>
- An ArduTouch synth sketch
  <http://cornfieldelectronics.com/cfe/projects.php#ardutouch>

The following pages show you how to do the above, in detail.

Arduino

Arduino is a very powerful tool!
But it is very easy to use.
It was designed for total beginners to use successfully.

I won't give a complete tutorial here – just some basics.
For more info, there are many good Arduino tutorials online.
A good place to start is:
<https://www.arduino.cc/en/Tutorial/HomePage>

Arduino

First:
Download and install the Arduino software
< http://arduino.cc >

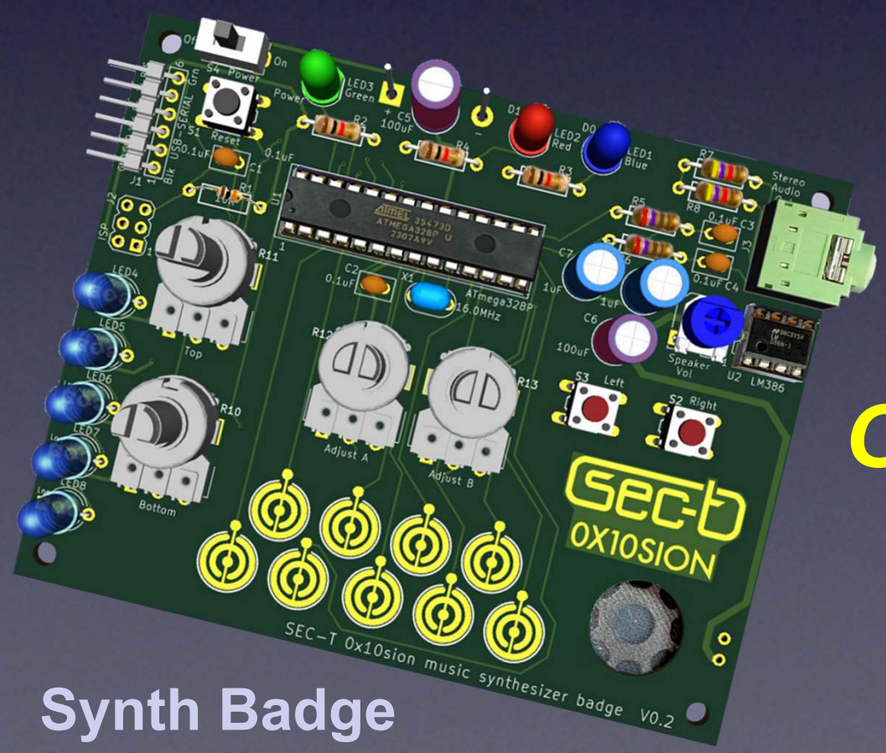
Any version is OK

Re-programming the Synth Badge

Second:

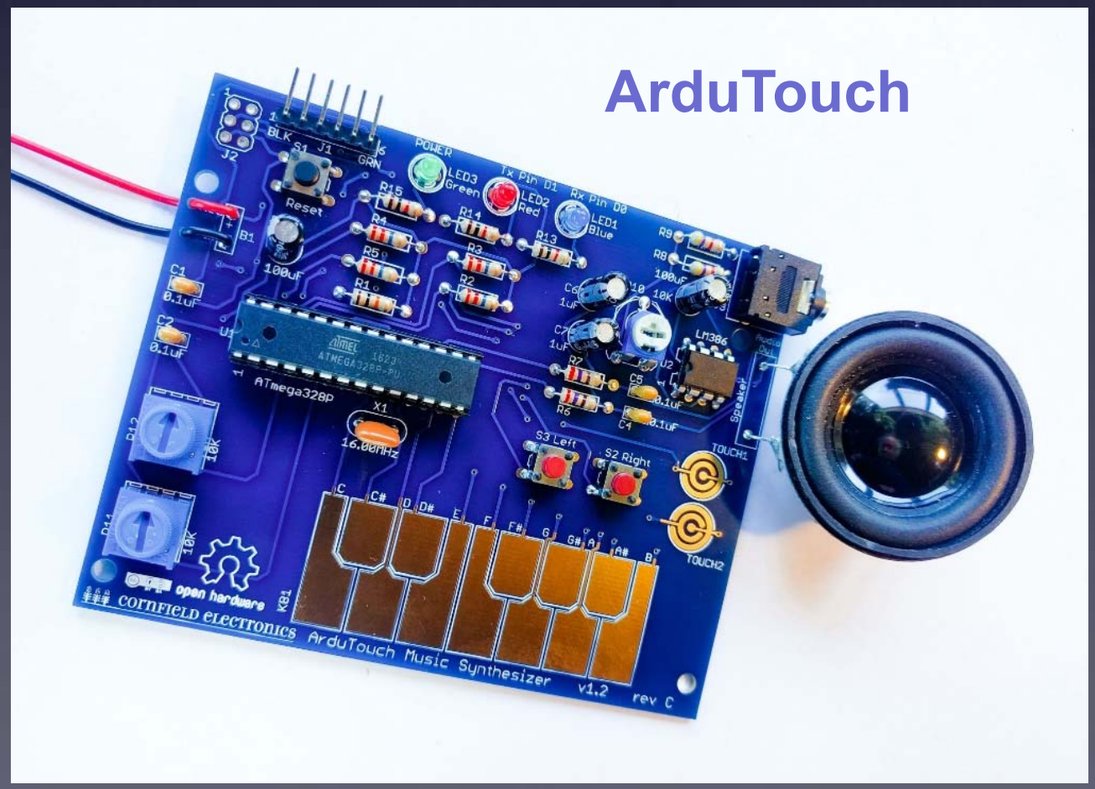
Download and install the SEC-T 0x10sion ArduTouch Arduino library
<https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library>

(details on this soon)



Synth Badge

Or



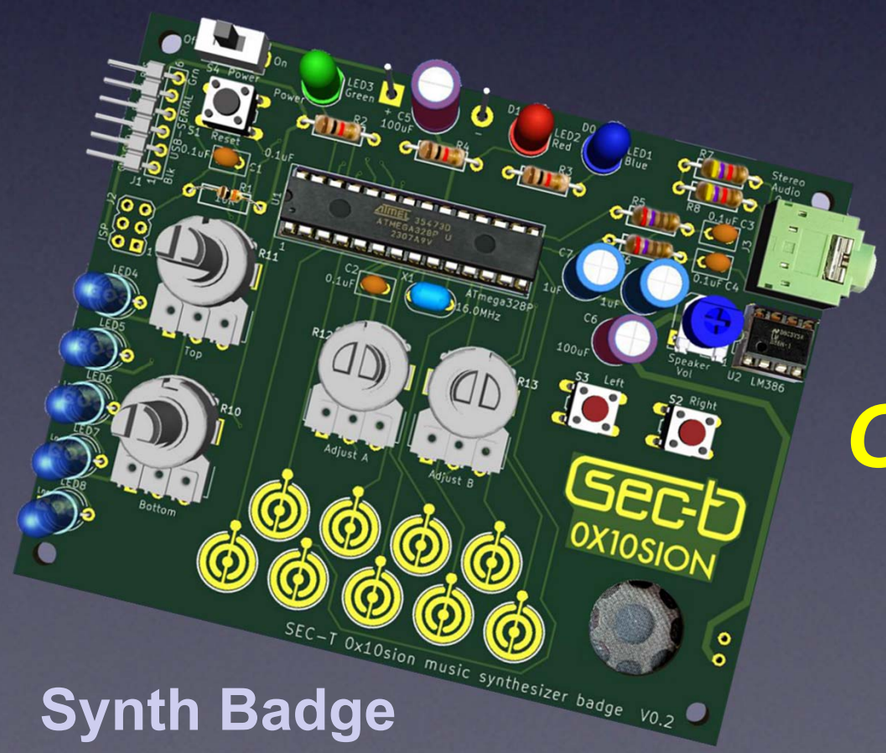
Re-programming the Synth Badge

Third:

Download ArduTouch synth sketches
<http://cornfieldelectronics.com/cfe/projects.php#ardutouch>

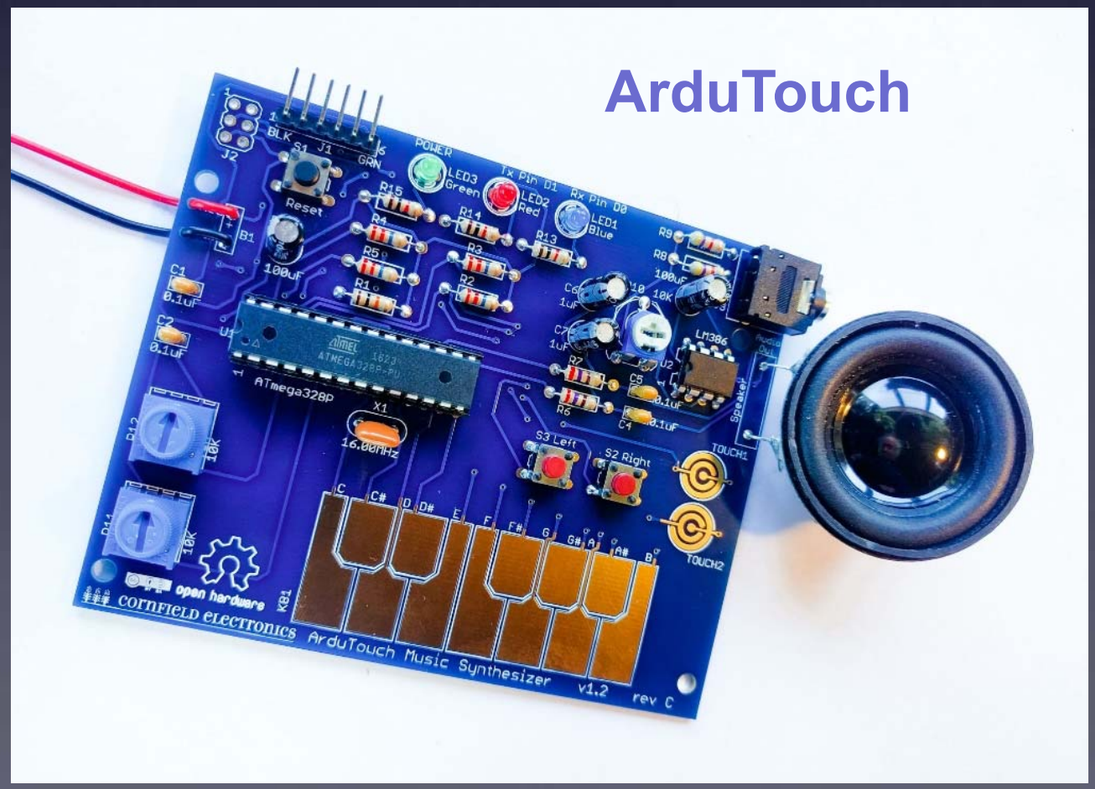
Store them on your computer anywhere you like.

(details on this soon)



Synth Badge

Or

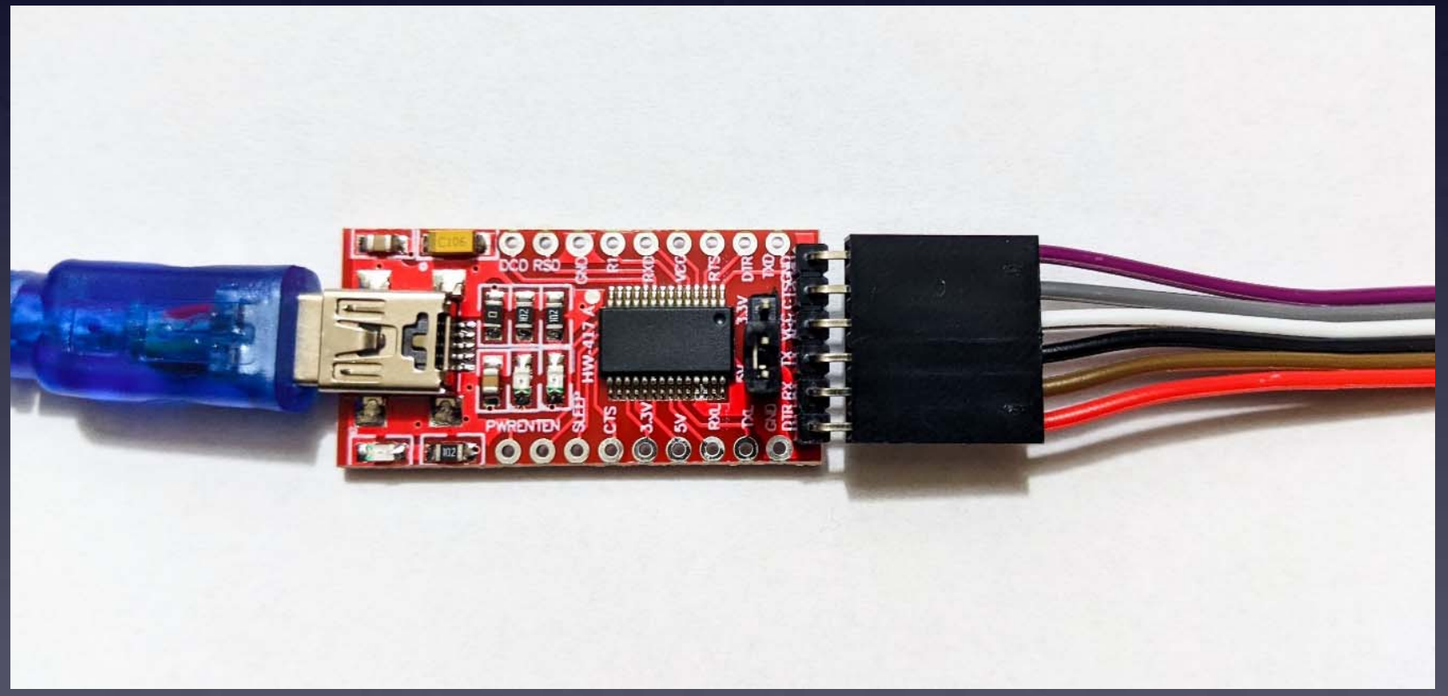


Connecting your Synth Badge to your computer

USB-Serial adapter cable

Ones available from Cornfield Electronics look like this:

<https://cornfieldelectronics.com/cfe/products/buy.php?productId=usbcable>

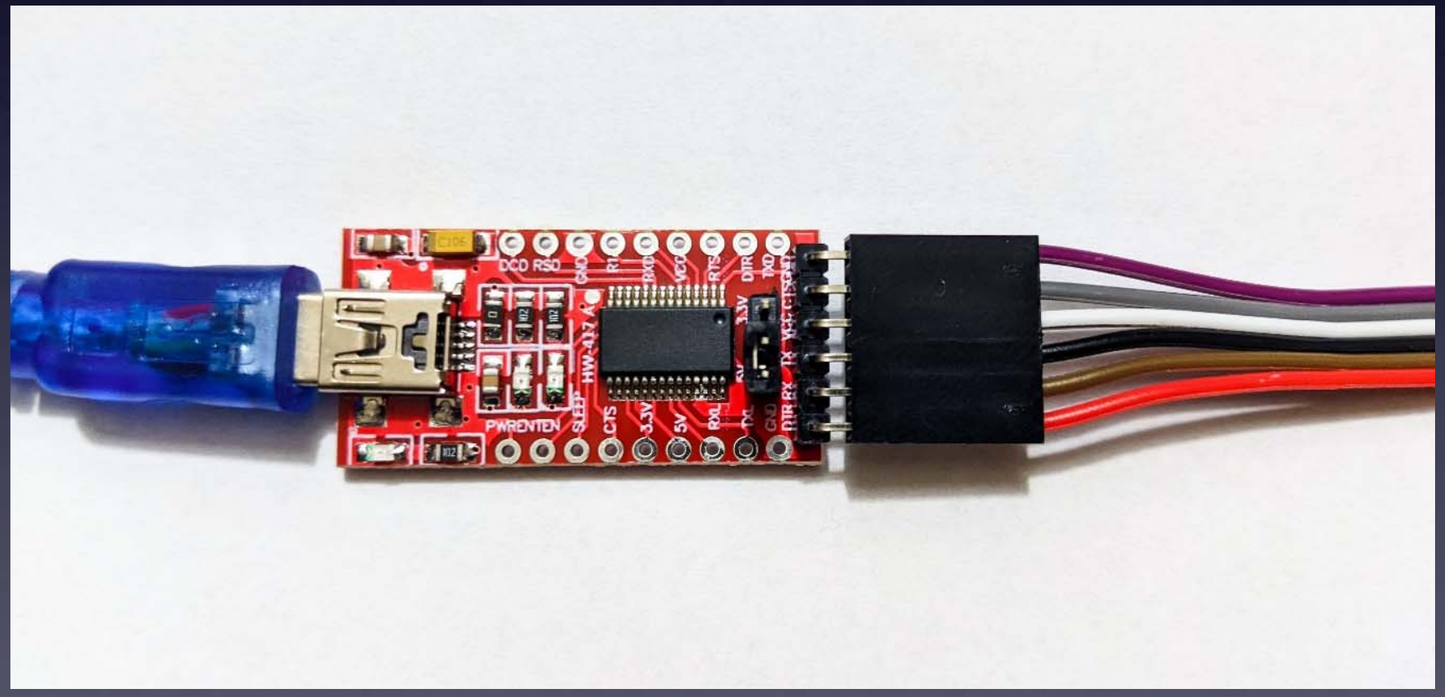


Connecting your Synth Badge to your computer

USB-Serial adapter cable

Ones available from Cornfield Electronics look like this:

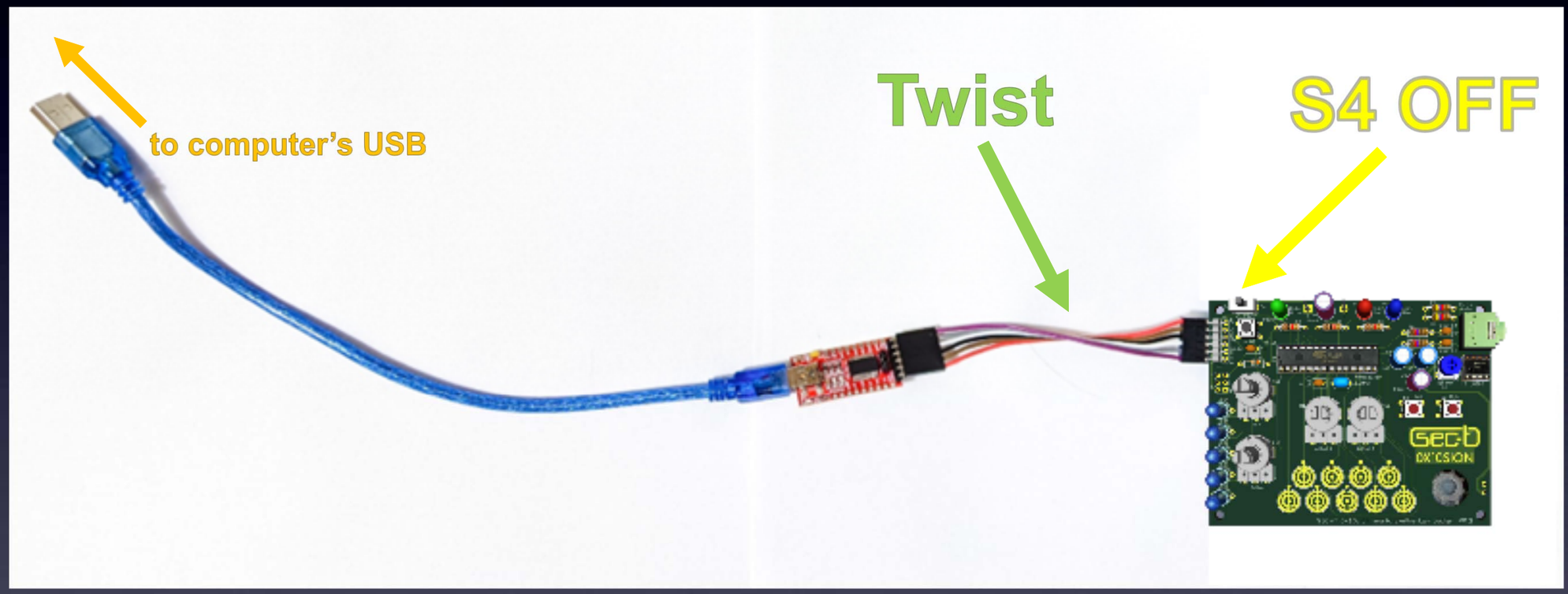
<https://cornfieldelectronics.com/cfe/products/buy.php?productId=usbcable>



You may need to download and install a driver for your Operating System (Windows, MacOS, or Linux):

<https://ftdichip.com/drivers/vcp-drivers/>

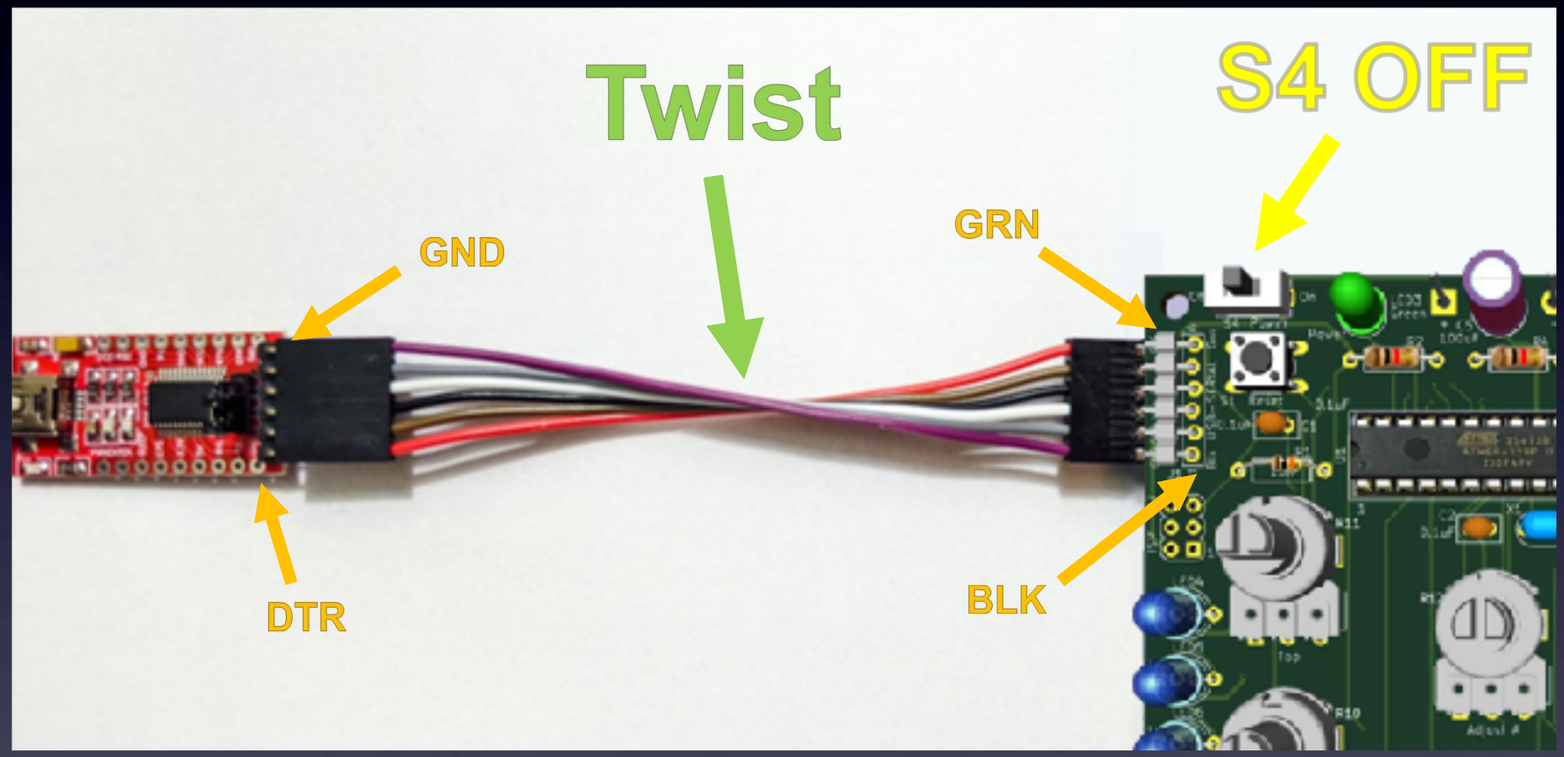
Connecting your Synth Badge to your computer



<u>IMPORTANT</u>:
Make sure the Power switch S4 on your Synth Badge is OFF

Connecting your Synth Badge to your computer

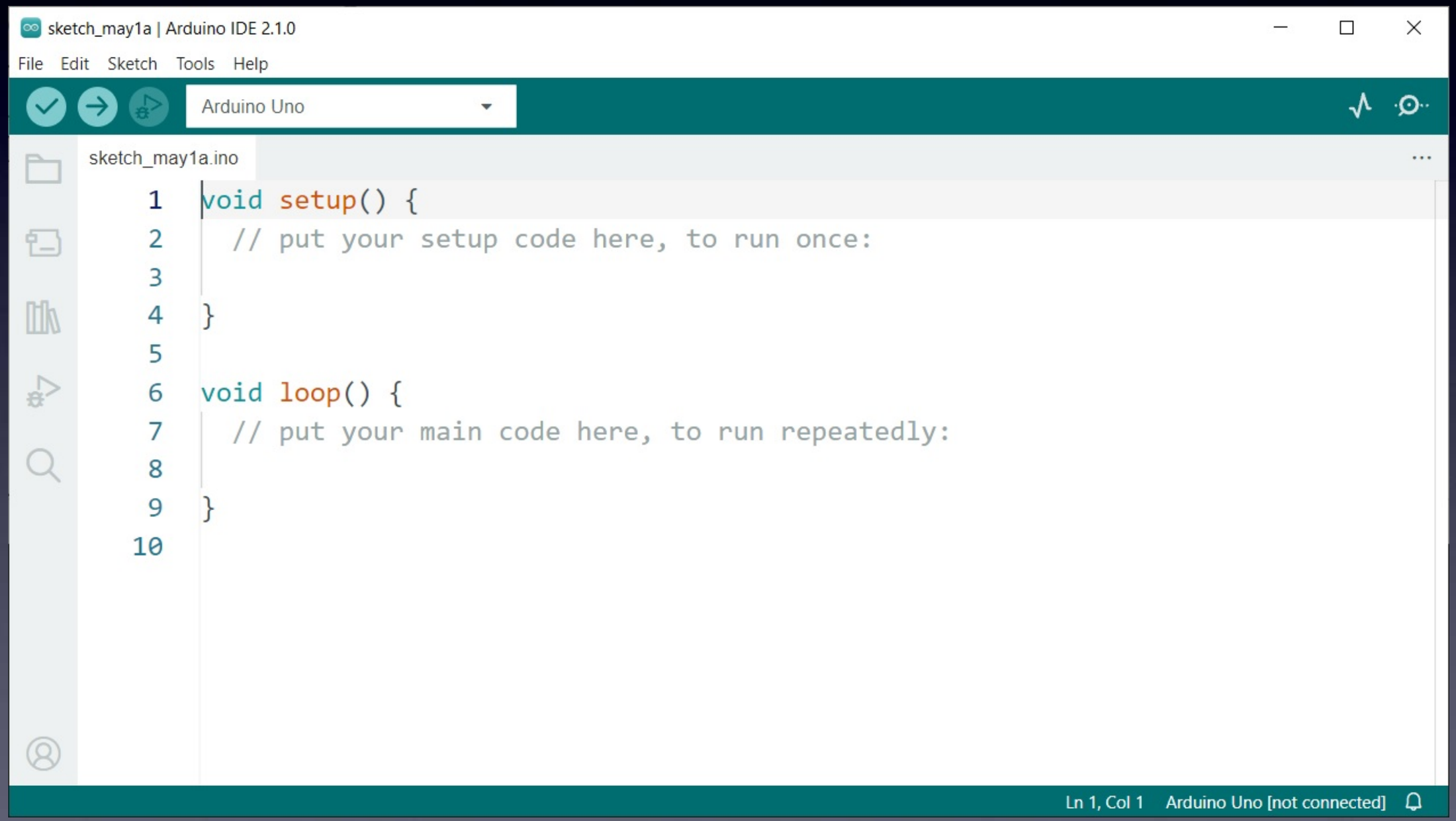
This shows a few more details:



<u>IMPORTANT</u>:
Make sure the Power switch S4 on your Synth Badge is OFF

Arduino

After you download and install the Arduino software start it, and you will see a screen that looks like this:

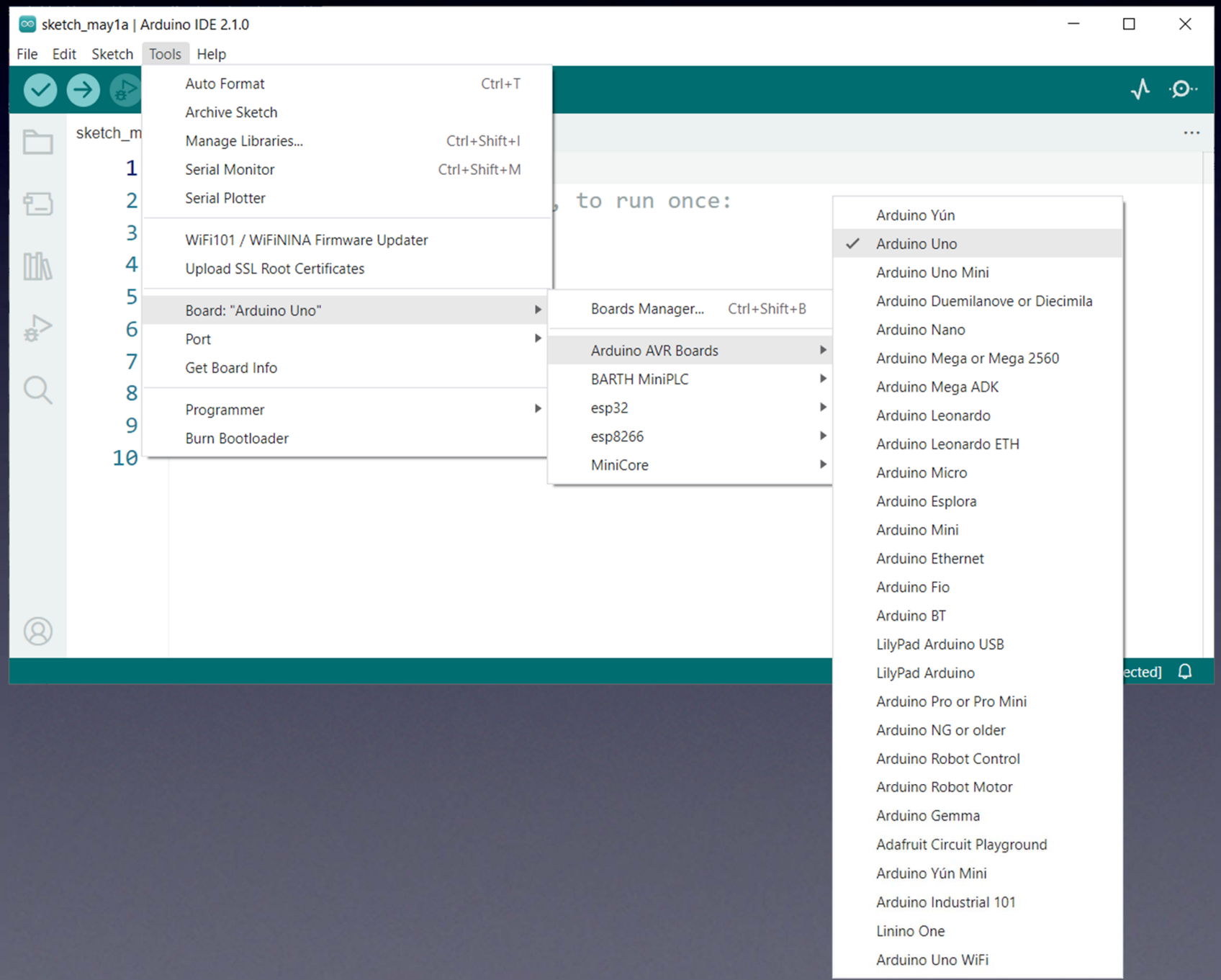


Arduino

The first time you start your Arduino software you need to do three things to set things up

(1)
Choose “Uno” as the Board

(Your Synth Badge board acts just like an Arduino Uno board)

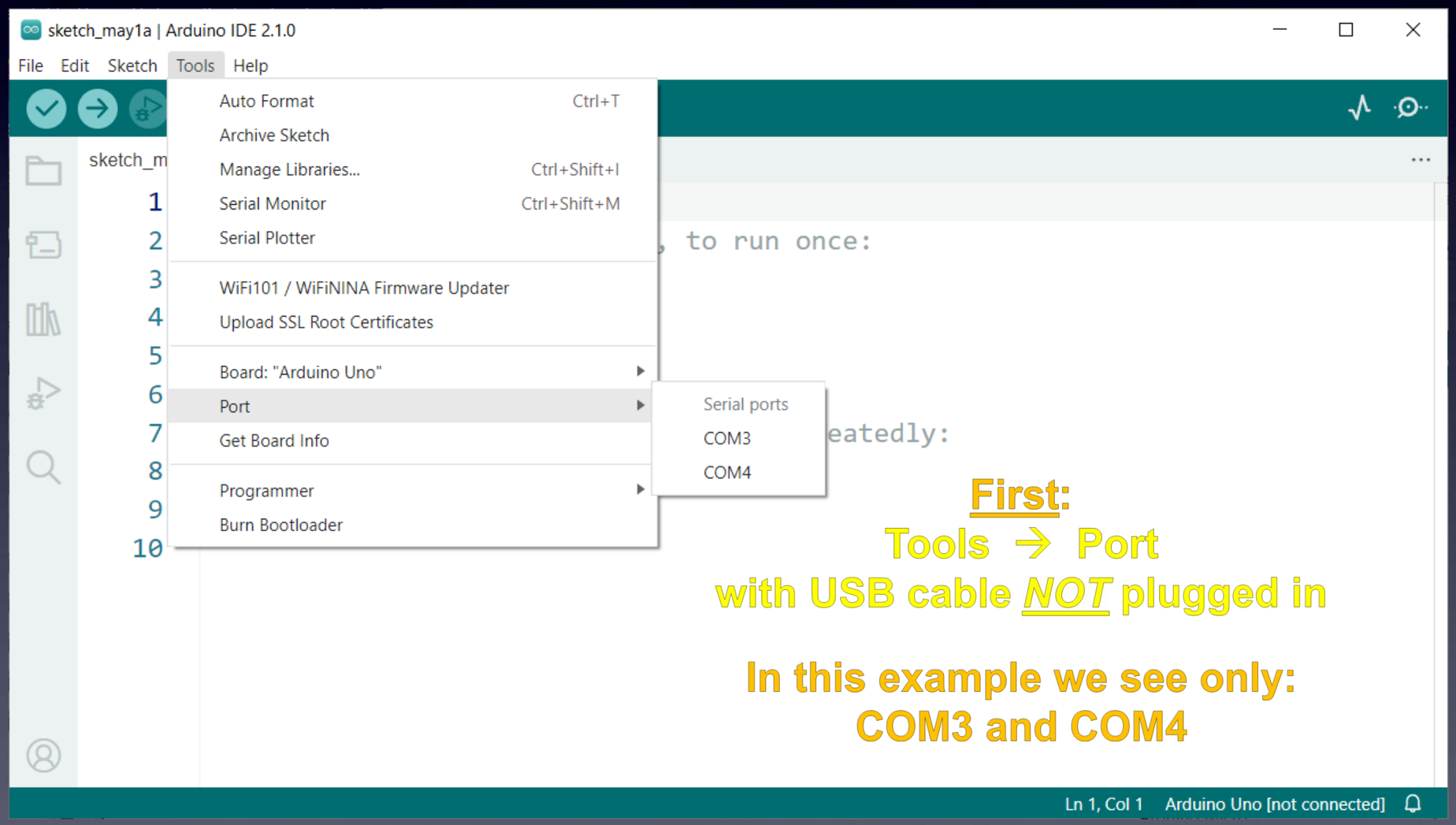


Arduino

The first time you start your Arduino software you need to do three things to set things up

with USB cable *NOT* plugged in:

(2) Choose the Port (this will be different depending on your Operating System)



First:
Tools → Port
with USB cable *NOT* plugged in

In this example we see only:
COM3 and COM4

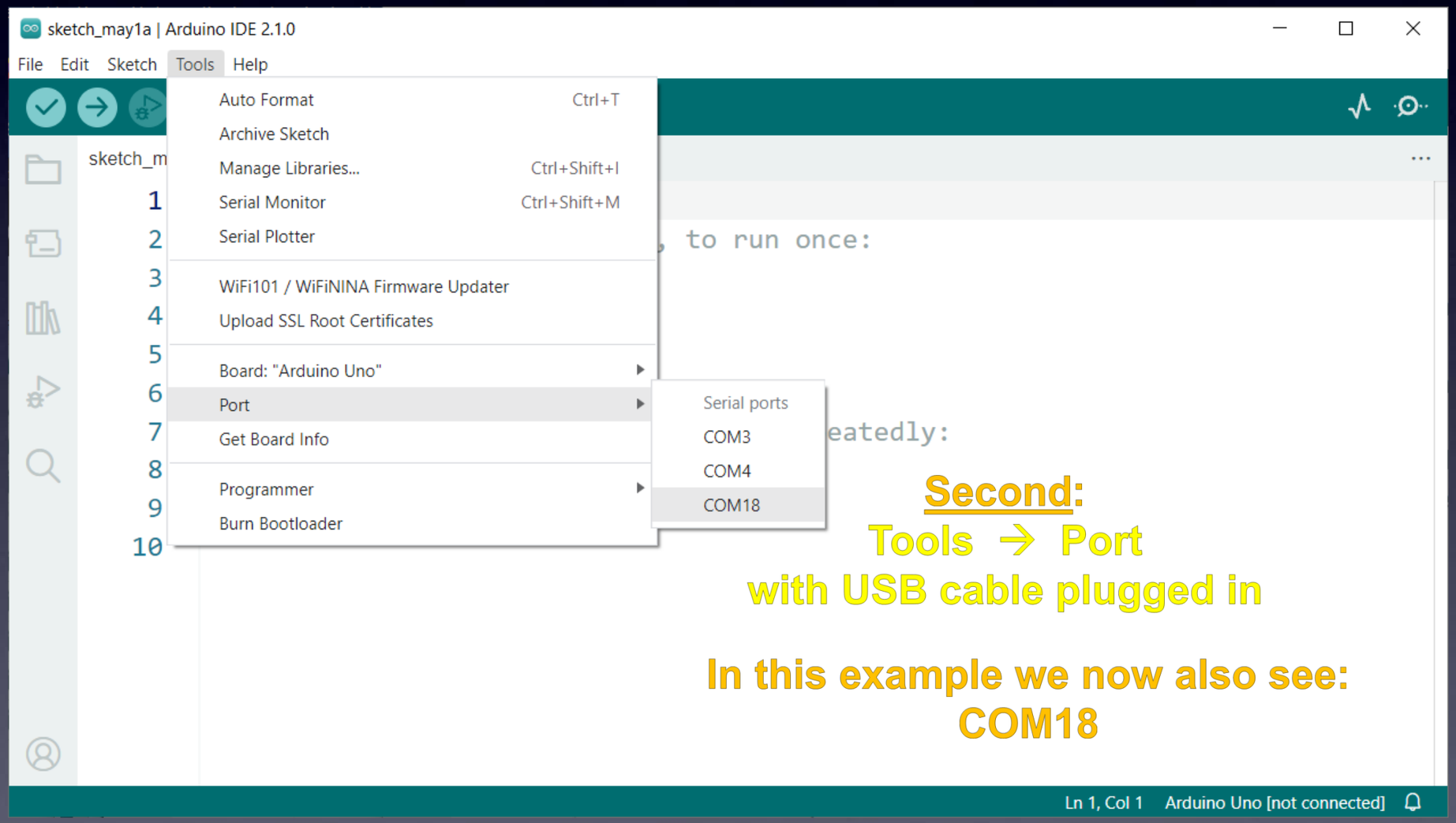
Arduino

The first time you start your Arduino software you need to do three things to set things up

with USB cable plugged in (with driver installed):

(2) Choose the Port (this will be different depending on your Operating System)

(After installing the driver for your USB-Serial cable and plugging it in your operating system will see a serial port and it appears here.)



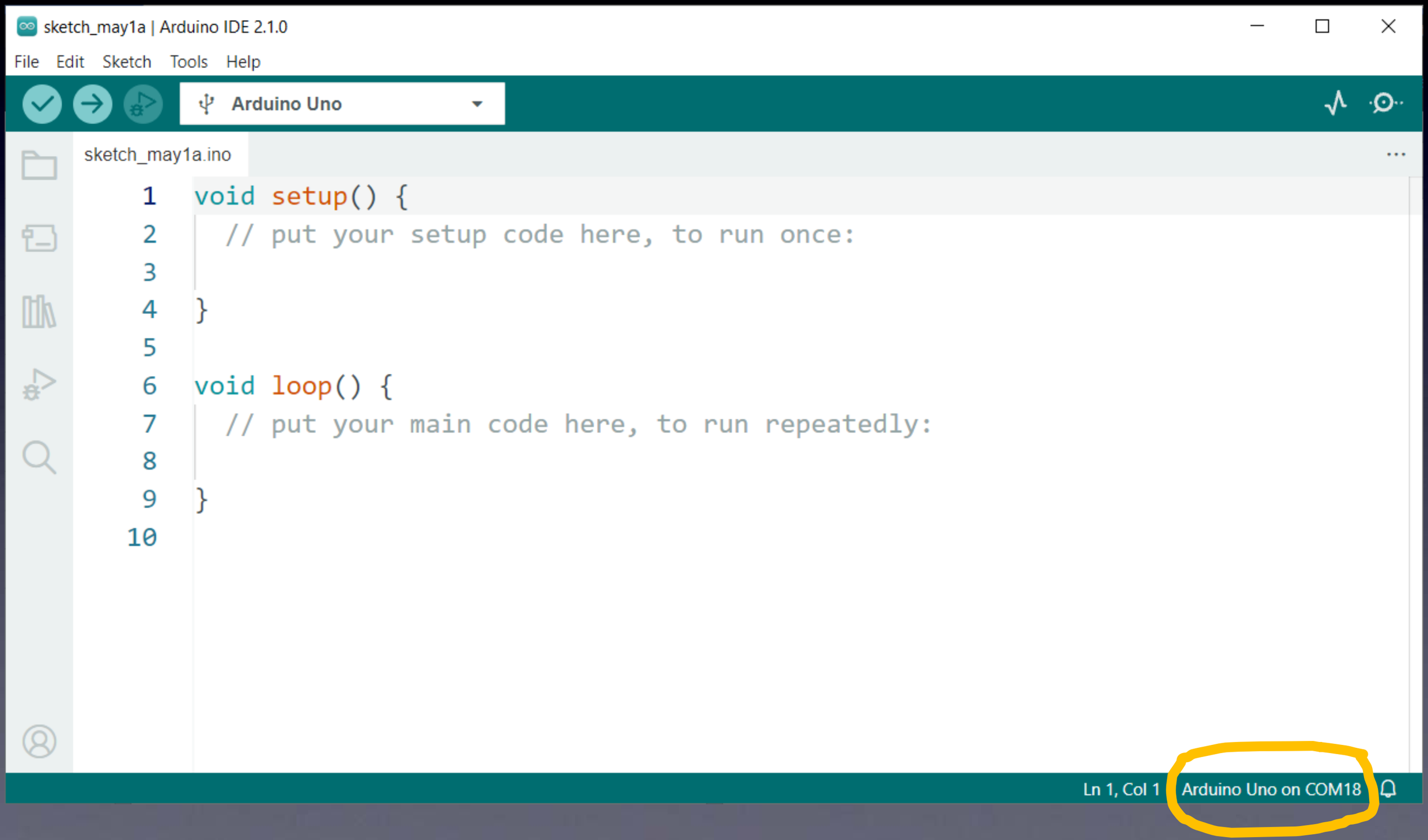
Second:
Tools → Port
with USB cable plugged in

In this example we now also see: COM18

Choose the new port: In this example: COM18

Arduino

Your Arduino software is almost ready

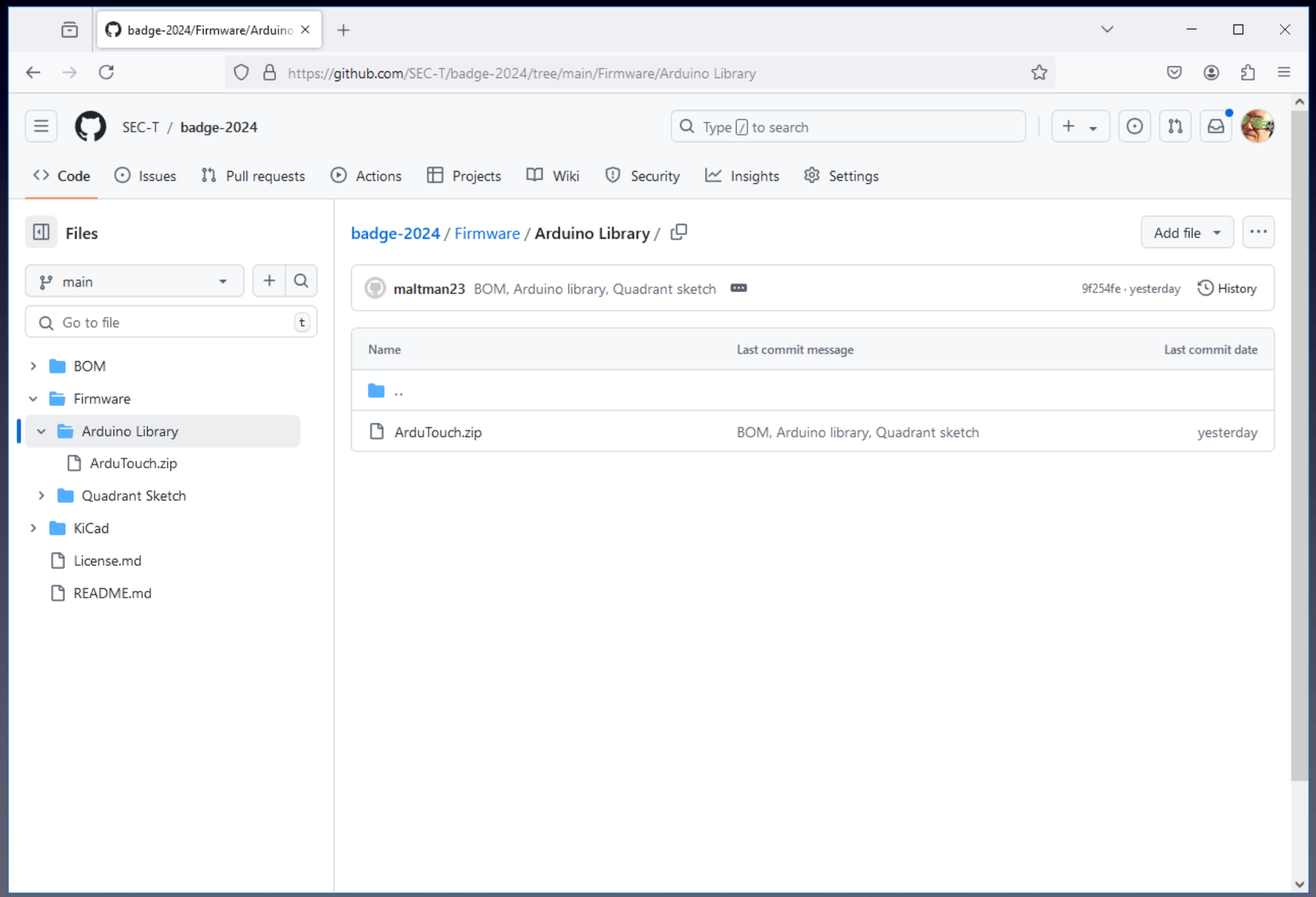


Arduino

The first time you start your Arduino software you need to do three things to set things up

(3) Install the SEC-T 0x10sion Synth Badge library

https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library



Arduino

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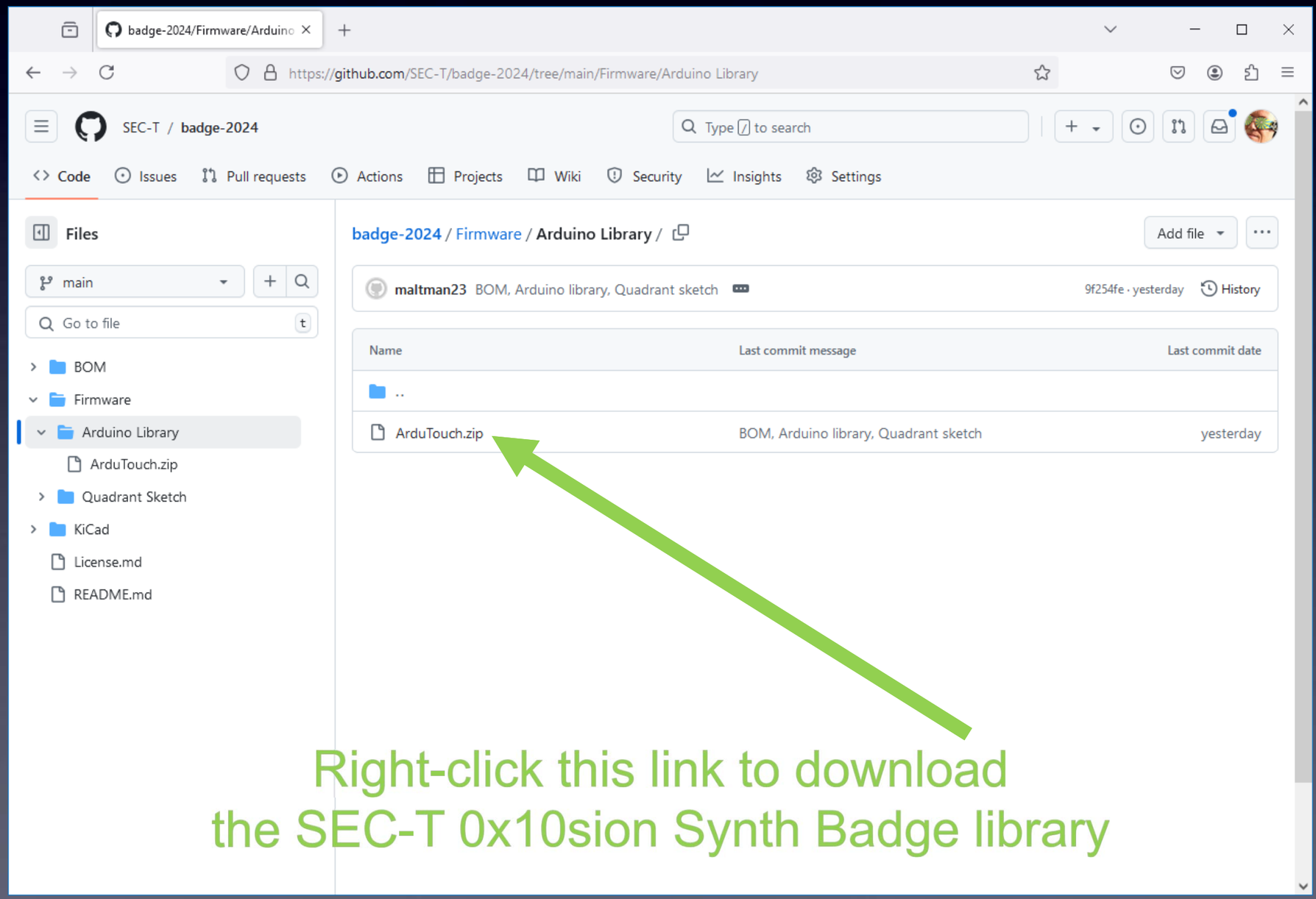
https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library

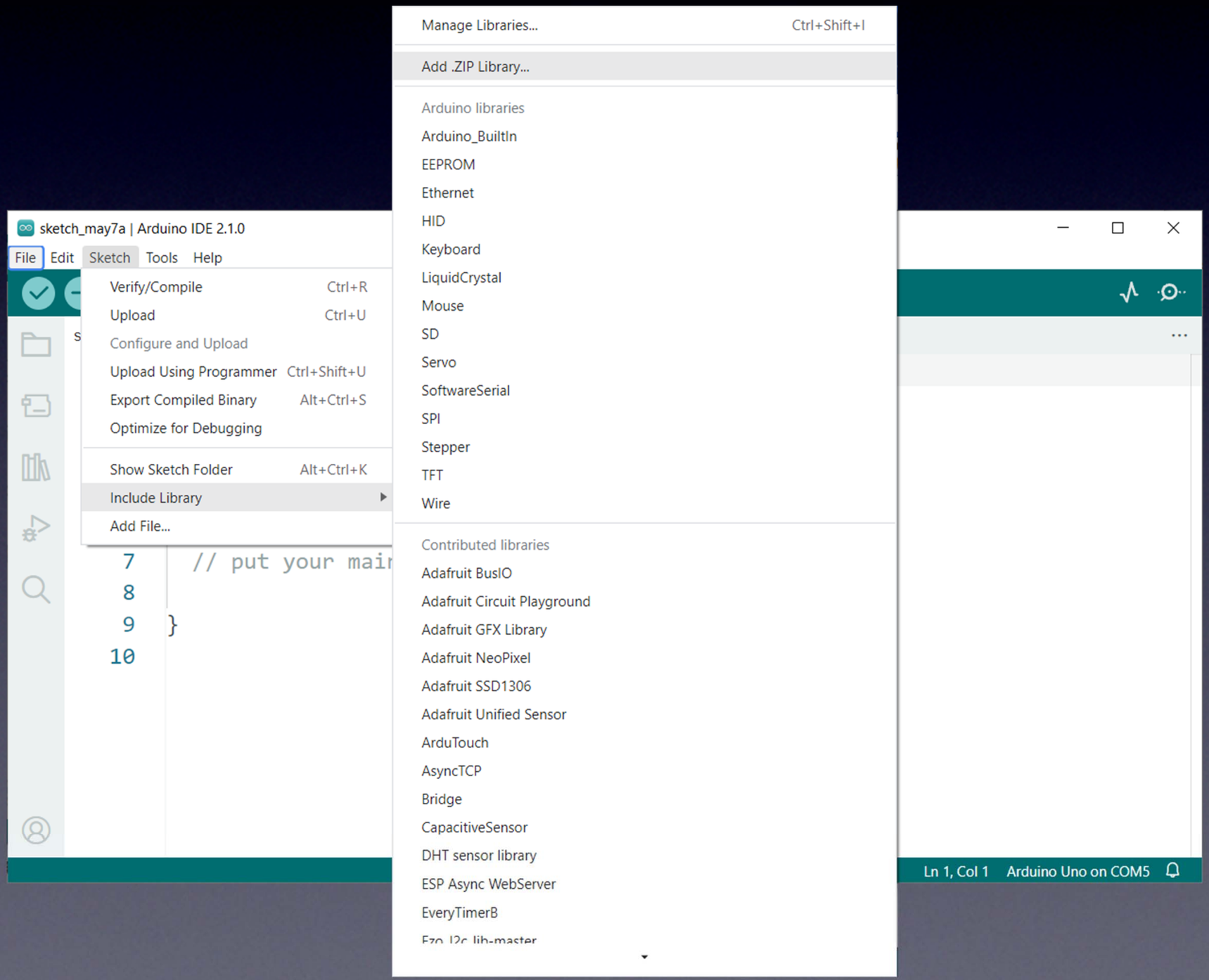


Right-click this link to download the SEC-T 0x10sion Synth Badge library

Arduino

The first time you start your Arduino software you need to do three things to set things up

(3) Install the SEC-T 0x10sion Synth Badge library



Arduino

The first time you start your Arduino software
you need to do three things to set things up

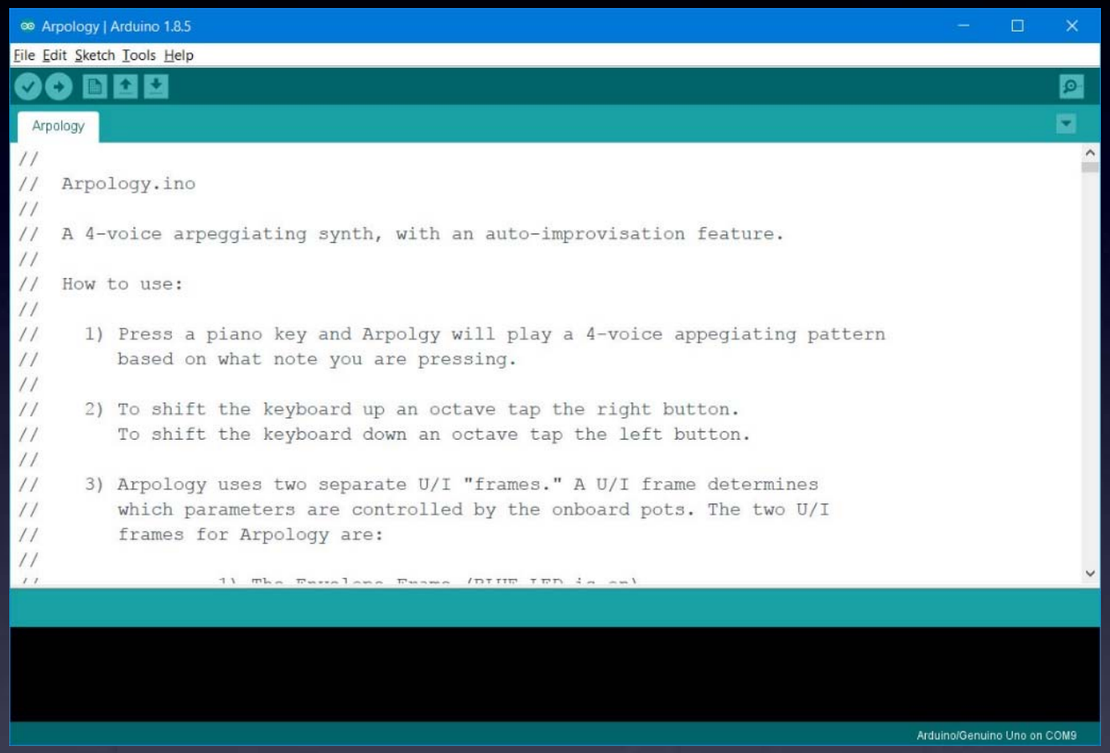
Your Arduino software is now ready

to program a synth sketch
into your SEC-T 0x10sion Synth Badge !

Arduino

Designed for non-geeky artists

Download a new Synth Badge synth “sketch”



“Sketch” :

an Arduino program

Arduino

Download a new Synth Badge synth “sketch”

Arduino

Download a new Synth Badge synth “sketch”

Arduino

Download a new Synth Badge synth “sketch”

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DO-IT-YOURSELF PROJECTS

by Mitch Altman, and friends.

Last modified: 5-Oct-2022

You Can Make Cool Things With Electronics!

The projects on this page were all created for total beginners, with no experience,
so everyone can complete them successfully at my workshops, or at home, or anywhere!

All you need is:
a desire, a handful of parts, a soldering iron (with stand and sponge), a wire-cutter, a wire-stripper, solder, and an afternoon.

Here is a really nice tutorial on how to solder -- for total beginners!
Soldering Tutorial for total beginners

Open Hardware!

Everything on this page (and everything I do) is free and open source!
(That's *free* as in *freedom*.)
(But everything here is free to download -- and that is *free* as in *beer*.)
If you have any questions on anything, please feel free to email me:
ch AT CornfieldElectronics DOT com

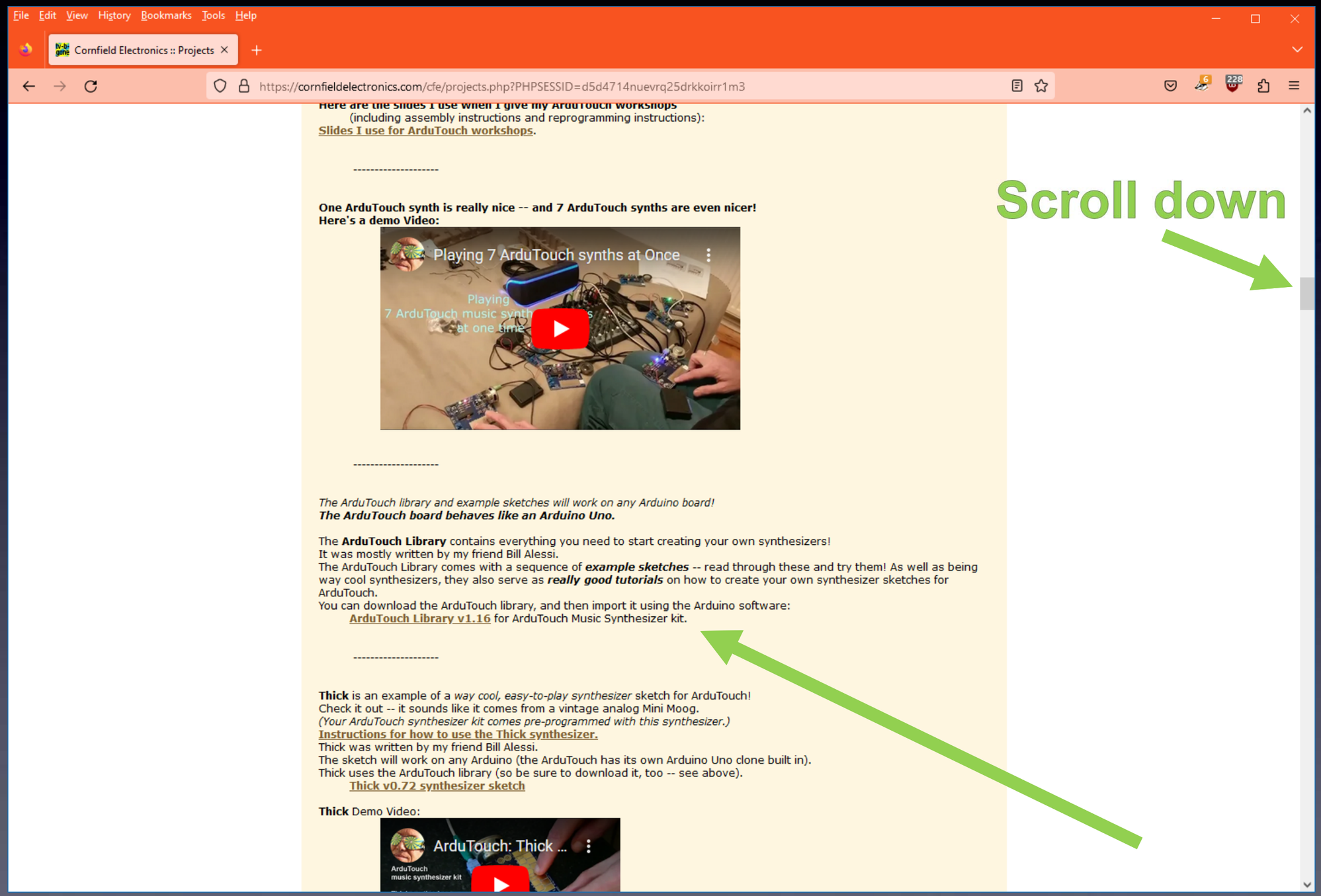
Transferring data from cornfieldelectronics.com...

Arduino

Download a new Synth Badge synth “sketch”

Arduino

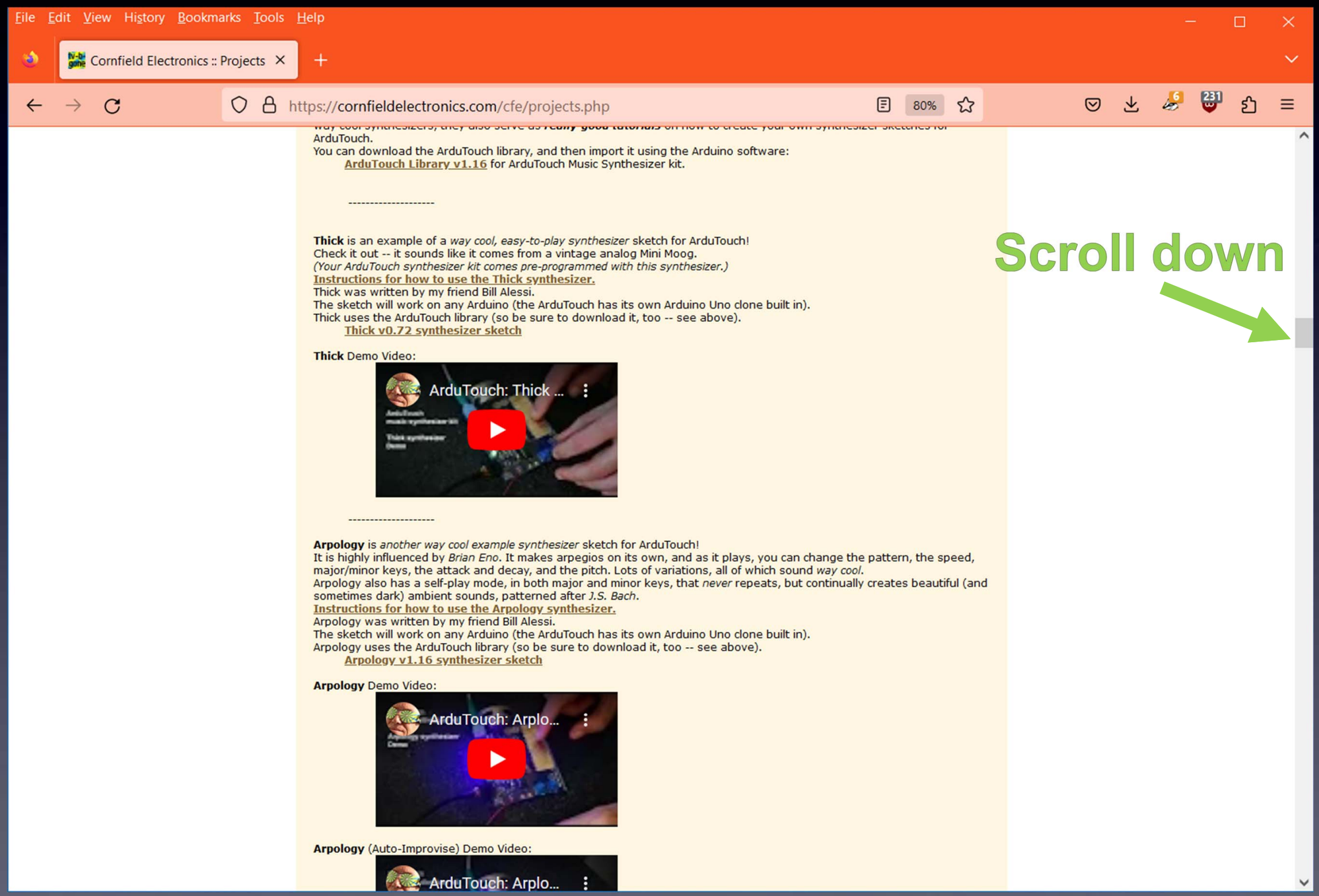
Download a new Synth Badge synth “sketch”



NOTE: *Do NOT use this ArduTouch library for the SEC-T Synth Badge*

Arduino

Download a new Synth Badge synth “sketch”



Scroll down

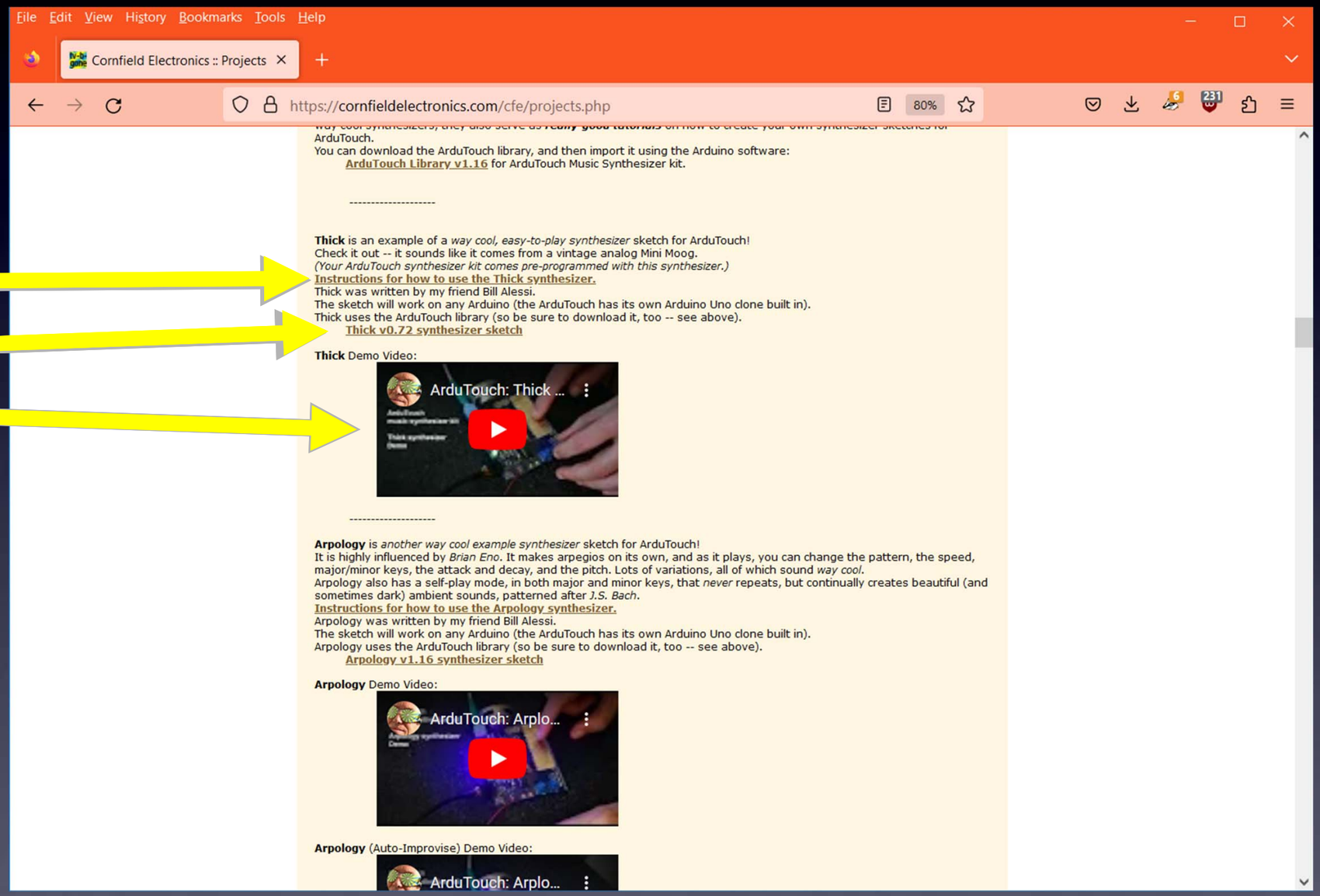
Here you will see several ArduTouch synth sketches

Arduino

Download a new Synth Badge synth “sketch”

Each ArduTouch synth sketch has:

- Instructions
- Sketch download
- Demo video(s)



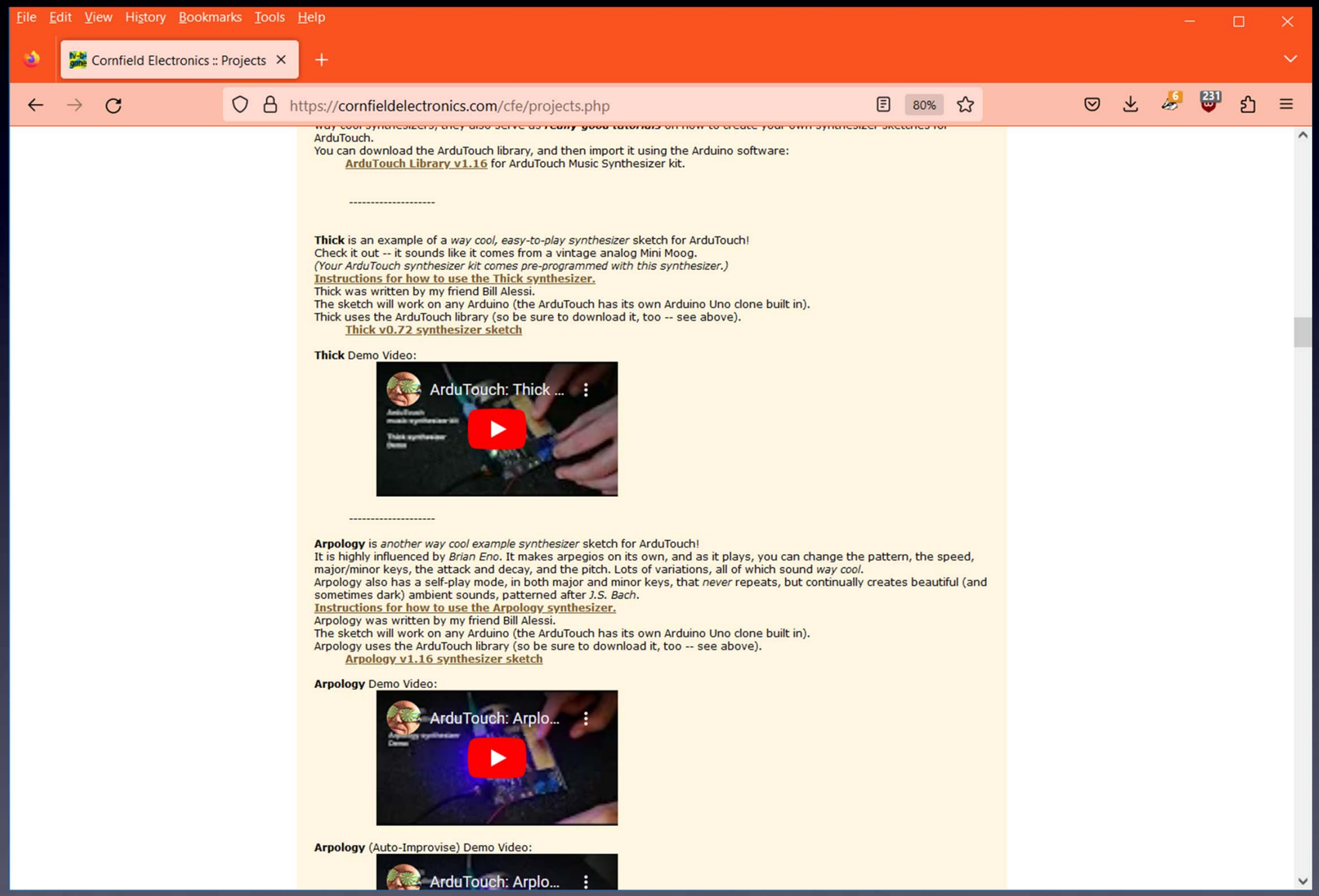
Here you will see several ArduTouch synth sketches

Arduino

Download a new Synth Badge synth “sketch”

NOTE:
These sketches do <u>NOT</u> have functionality for:

- RGB LEDs
- knobs A and B



Here you will see several ArduTouch synth sketches

Arduino

Download a new Synth Badge synth “sketch”

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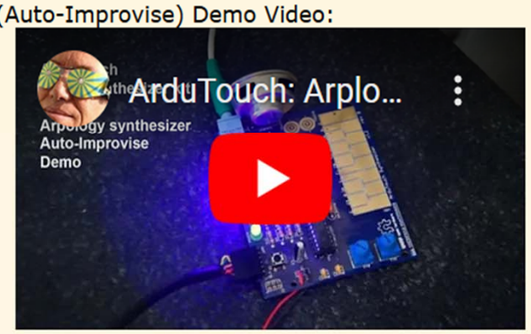
https://cornfieldelectronics.com/cfe/projects.php 120%

Arpology is *another way cool example synthesizer* sketch for ArduTouch!
It is highly influenced by *Brian Eno*. It makes arpegios on its own, and as it plays, you can change the pattern, the speed, major/minor keys, the attack and decay, and the pitch. Lots of variations, all of which sound *way cool*.
Arpology also has a self-play mode, in both major and minor keys, that *never* repeats, but continually creates beautiful (and sometimes dark) ambient sounds, patterned after *J.S. Bach*.
Instructions for how to use the Arpology synthesizer.
Arpology was written by my friend Bill Alessi.
The sketch will work on any Arduino (the ArduTouch has its own Arduino Uno clone built in).
Arpology uses the ArduTouch library (so be sure to download it, too -- see above).
Arpology v1.16 synthesizer sketch

Arpology Demo Video:

Arpology (Auto-Improvise) Demo Video:

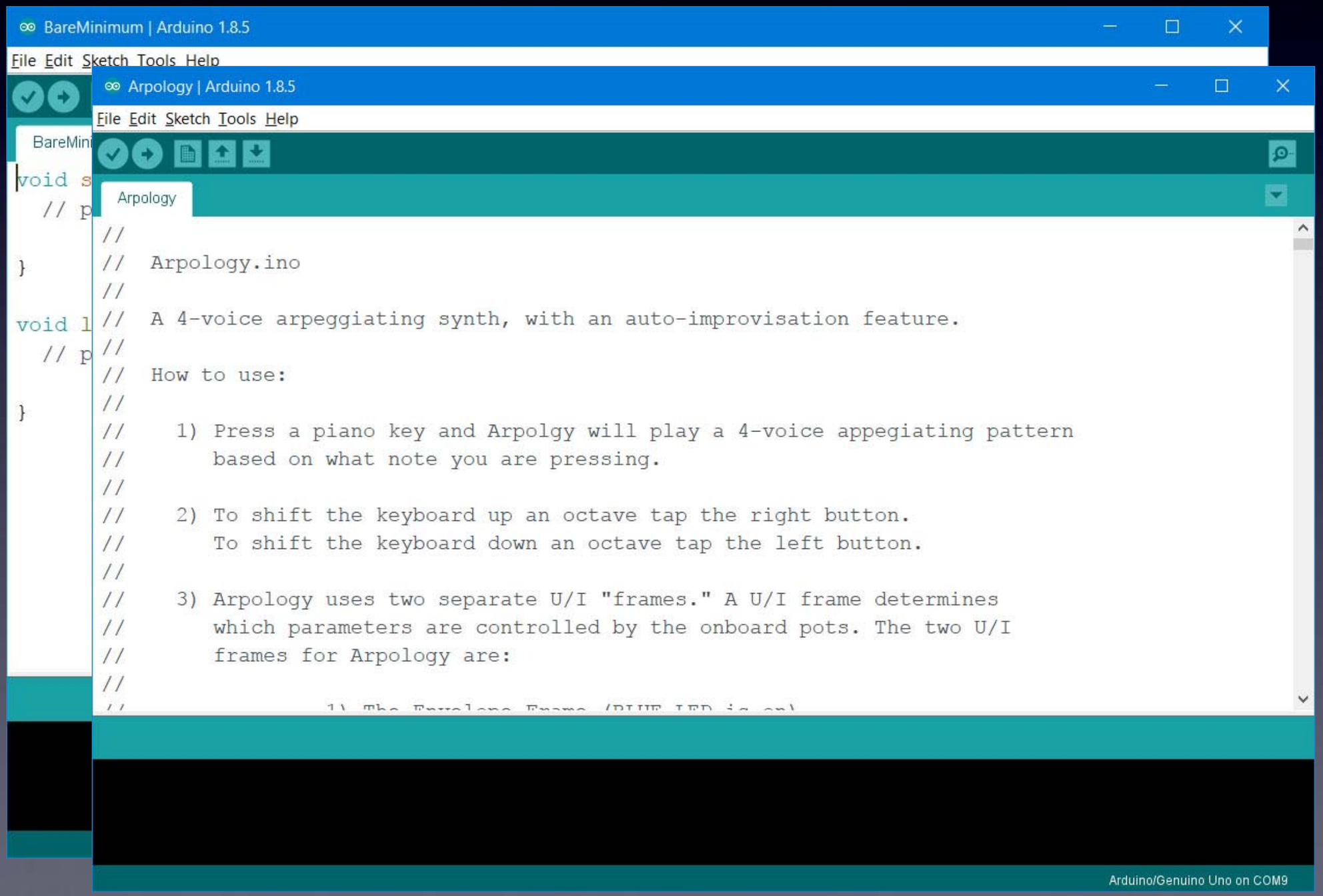


Click the sketch link to download the sketch

Arduino

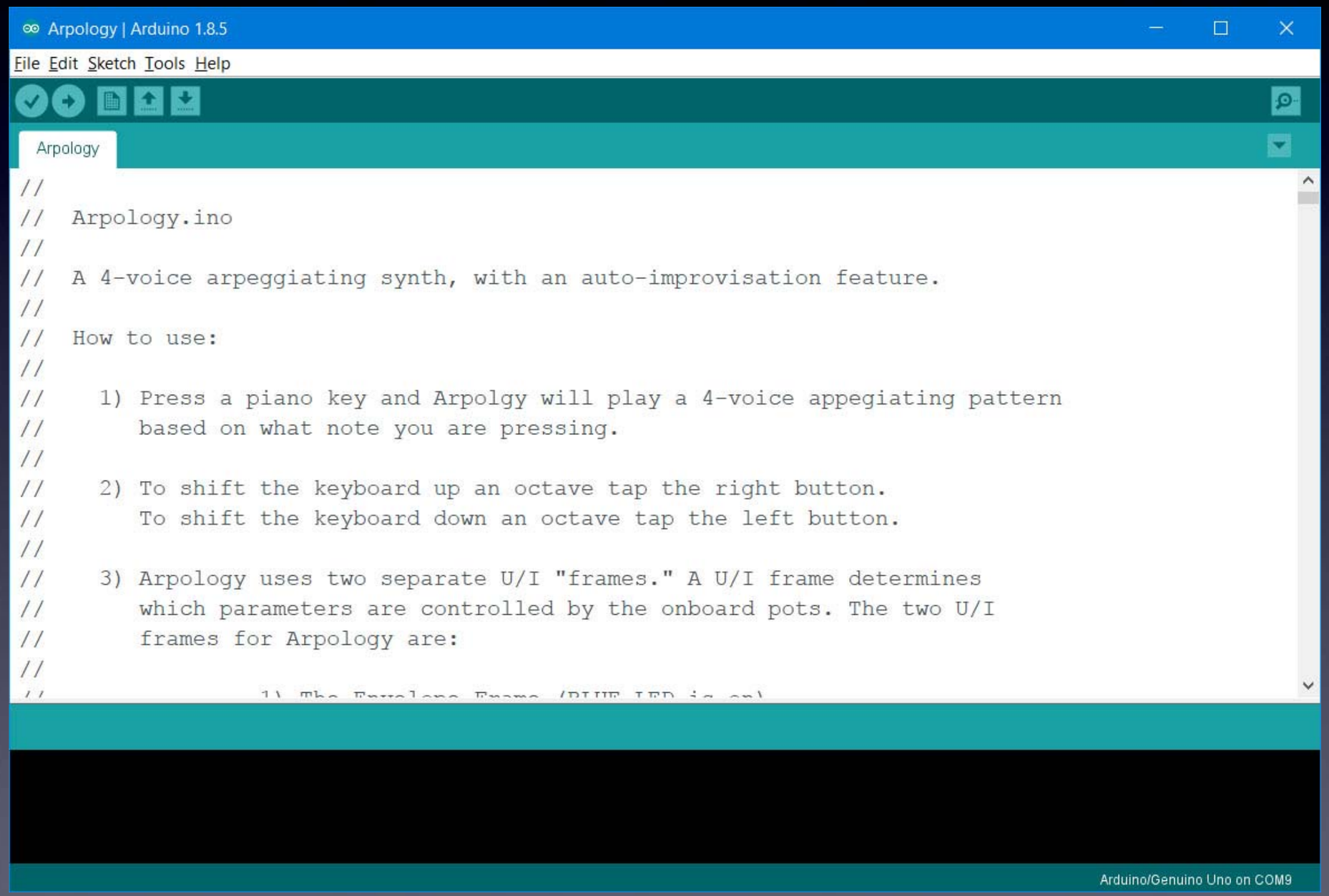
You can open the ArduTouch / Synth Badge synth sketch:
File → Open…

(I opened “Arpology here)



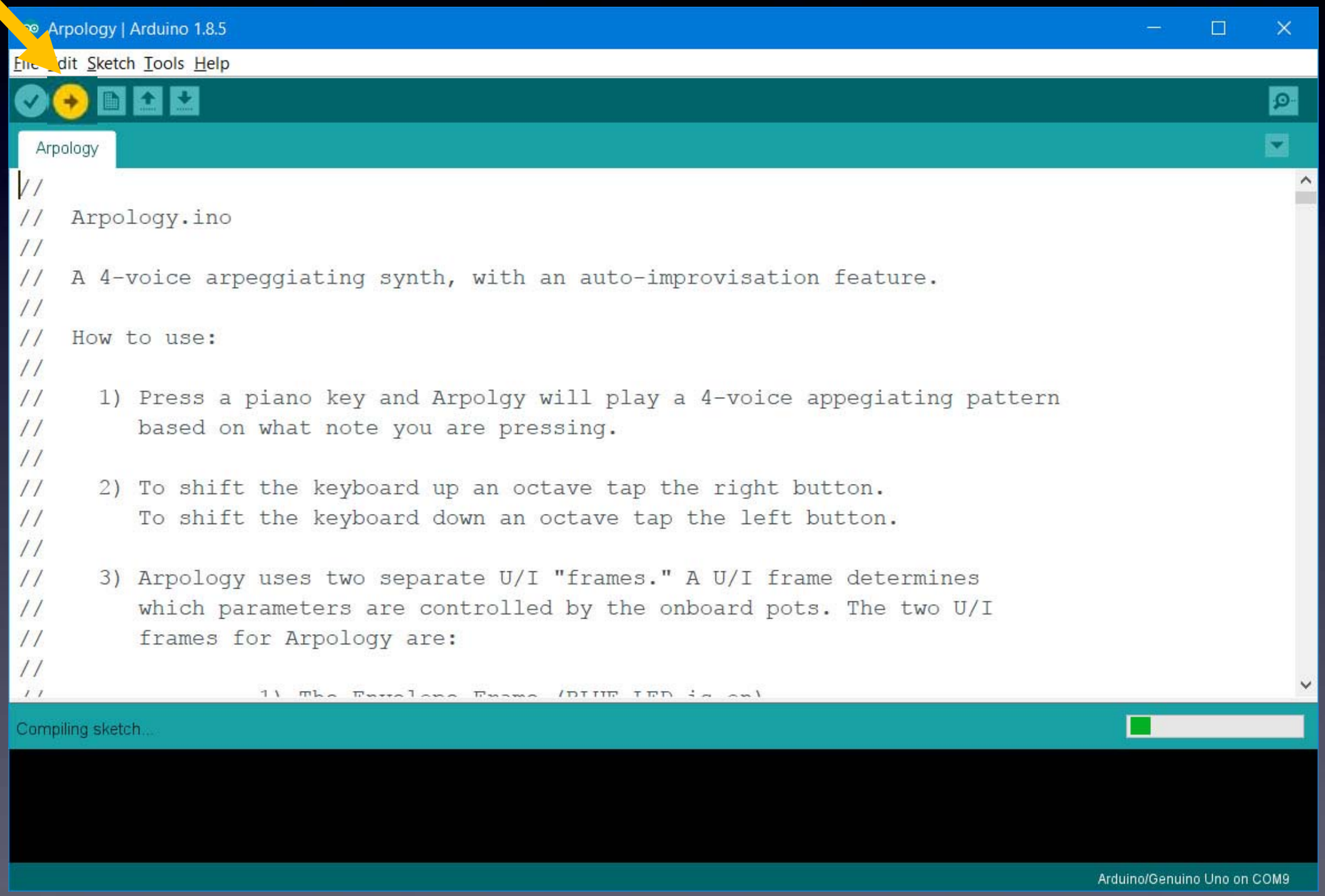
Arduino

You can now program your Synth Badge with a new synth sketch !



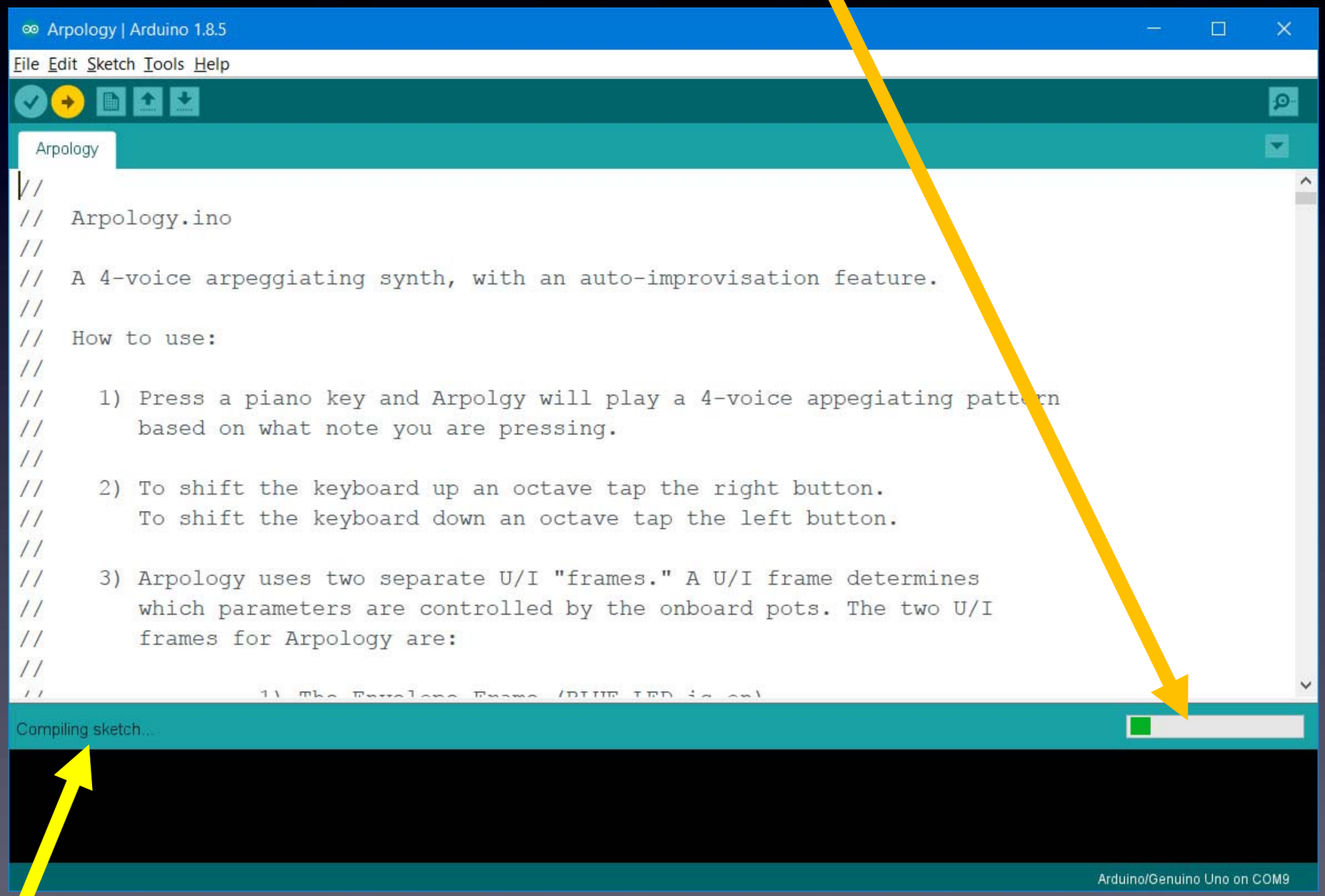
Arduino

With the USB-Serial cable connected to your Synth Badge board press the Upload button



Arduino

While uploading, you will see a progress bar…



…and when it's completed successfully, it says: "Upload done"

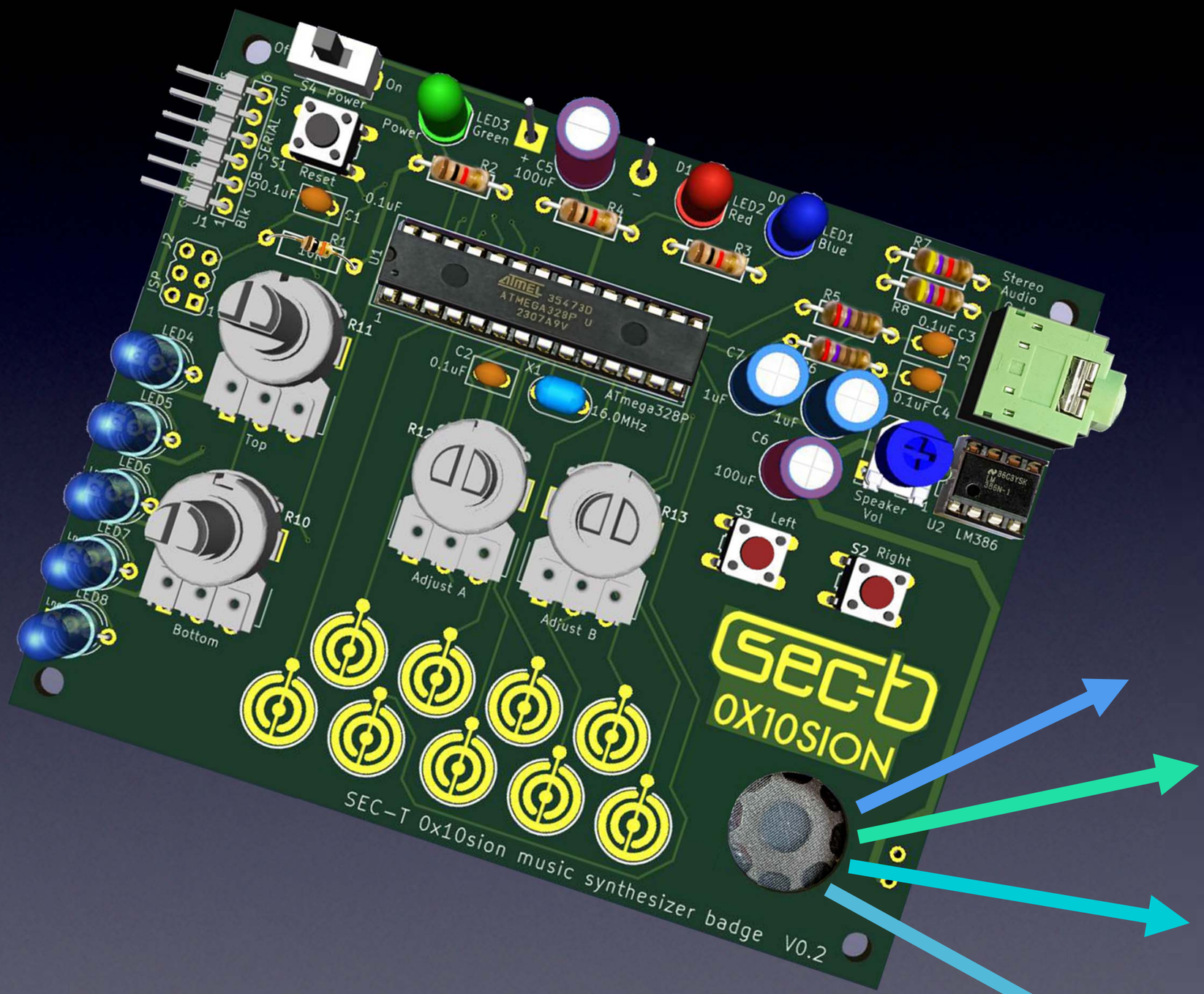
ArduTouch

Disconnect your SEC-T 0x10sion Synth Badge board from the USB-Serial cable,

turn on the Power switch S4,

And…

Let’s make new noise!



Please Remember:

to

Wash your hands

after soldering

12-Sep-2025

SEC-T 0x10sion Music Synthesizer Badge

Assembly Instructions

(with re-programming how-to)

V 0.2

open source hardware

0X10SION

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